

**UNIVERSITI TEKNOLOGI MARA**

**CANCERVIVE: DATA  
VISUALIZATION OF CANCER  
SURVIVAL AND TREATMENT  
RECOMMENDATION.**

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**BACHELOR OF INFORMATION  
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Thesis submitted in fulfillment  
of the requirements for the degree of  
**Bachelor of Information Technology (Hons.)**

**School of Computing  
College of Computing Informatics and Mathematics**

**August 2023**



## AUTHOR'S DECLARATION

I declare that the work in this thesis was carried out in accordance with the regulations of Universiti Teknologi MARA. It is original and is the results of my own work, unless otherwise indicated or acknowledged as referenced work. This thesis has not been submitted to any other academic institution or non-academic institution for any degree or qualification.

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## ABSTRACT

Cancer is the second leading cause of death worldwide, after cardiovascular disease; by 2020, nearly 10 million people will have died as a result of cancer. Cancer is a disease that causes certain cells in the body to grow uncontrollably and spread to other parts of the body. However, not all cancer patients will die from cancer. With early detection and treatment advancements, many cancer patients can survive and live long, healthy lives after being diagnosed. Treatment's success is one of the factors that influence patients survive. To get a good treatment outcome, the oncologist plays an important role in planning the best treatment and providing information about the patient's current condition. The cancer survival rate also helps the patient in treatment planning. Knowing a patient's survival percentage can assist the patient and their oncologist determine the best course of treatment and set and setting reasonable expectations for the patient and their family. Oncologists frequently use previous cases to compare with current ones when determining cancer survival rates and treatment plans. The problems doctors and oncologists often face is difficulty in accessing and analysing cancer-related data, which can lead to a lack of valuable insights. Thus, this project aims to assist doctors by the system in analysing cancer survival rate data, predicting patients' survival rates, and providing treatment recommendations by using machine learning predictive model. To complete the system, an Agile methodology with an OSEMN model was used, with a focus on doctors and medical staff, particularly those in the oncology field, such as oncologists. Usability testing also found the system features working and easy to use. The system will help the doctors and oncologists by visualize the survival rate for better understanding. As well as the system help by predicting cancer survival rate and recommending the treatment as oncologists' guideline to plan the treatment.

**Keywords:** Cancer, Survival rates, machine learning, Predicting survival rate, Big Data, treatment recommendation, Prediction

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## LIST OF ABBREVIATIONS

### Abbreviations

|                |                            |
|----------------|----------------------------|
| <b>WHO</b>     | World Health organization  |
| <b>BDA</b>     | Big Data Analytics         |
| <b>BD</b>      | Big Data                   |
| <b>CT Scan</b> | Computed Tomography Scan   |
| <b>MRI</b>     | Magnetic Resonance Imaging |
| <b>US</b>      | United States              |
| <b>DNA</b>     | Deoxyribonucleic Acid      |
| <b>ML</b>      | Machine Learning           |
| <b>BI</b>      | Business Intelligence      |
| <b>AI</b>      | Artificial Intelligence    |
| <b>RSs</b>     | Recommender Systems        |



# CHAPTER ONE

## INTRODUCTION

### 1.1 Research Background

According to the World Health Organization (2018), Cancer is the second leading cause of worldwide after cardiovascular disease. Cancer develops from transforming normal cells into tumor cells in a multi-stage process that usually goes from a pre-cancerous lesion to a malignant tumor. Some cancers have the potential to spread to other tissues. There will be 19.30 million new cancer cases worldwide in 2020 and almost 10.0 million cancer deaths (Ferlay al., 2021). The primary cause of cancer death, representing 23% of all cancer deaths, was lung cancer. Cancers of the colon and rectum (9%), pancreatic (8%), female breast (7%), prostate (5%), and liver and intrahepatic bile duct (5%) were the other leading causes of mortality from cancer. Other malignancies accounted for fewer than 5% of all cancer fatalities (Centers for Disease control and prevention, 2020). Figure 1.1 shows statistics for the year 2020 in the United States of fatalities caused by 6 major types of cancer.

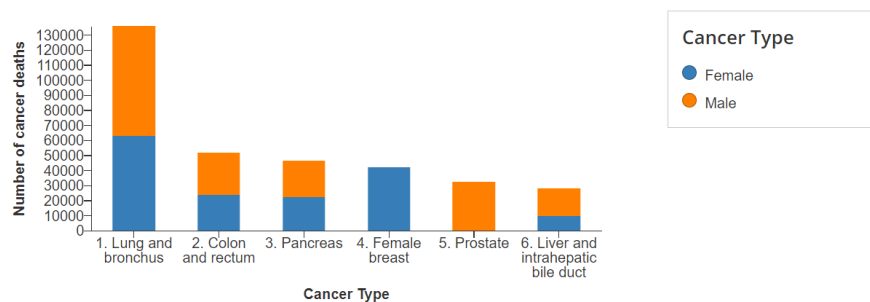


Figure 1.1: The statistics for the year 2020 in the United States (Source: Centers for Disease and control and prevention, 2020)

However, not all cancer patients will die from cancer. With early detection and treatment advancements, many cancer patients can survive and live long, healthy lives after being diagnosed (American Cancer Society, 2020). Many factors influence each patient's outcome, including the type of cancer, the cancer stage at the time of diagnosis, the patient's overall condition, and the treatment's success (Clin,2018). Cancer survival is the percentage of individuals living a specific amount of time after a cancer diagnosis. It assesses how people are doing well after being diagnosed with cancer. The 5-year survival rate, the percentage of persons who survive five years after being diagnosed

with cancer, is the most commonly used survival metric. Various survival metrics include 1-year and 10-year survival rates (Spencer et al., 2019). The cancer survival rate helps the patient in treatment planning. Knowing a patient's survival percentage can assist the patient and their doctor or oncologist in determining the best course of treatment and setting reasonable expectations for the patient and their family. This is because survival rates estimate the possibility of a patient surviving for a set period after being diagnosed with cancer.

Based on survival data, a doctor or oncologist can evaluate the efficacy of various treatment choices and provide suggestions to the patient. This can assist the patient and their family in making the best decisions about their care and setting realistic expectations for treatment outcomes Neris et al. (2019). Hui et al. (2018) state that oncologists are medical specialists that specialize in the treatment of cancer patients. They collaborate closely with other specialists, such as surgeons and radiation therapists, to develop and administer cancer treatment strategies. This includes alleviating cancer-related symptoms, dealing with the adverse effects of treatment, and providing insight into the outcome of the disease.

## **1.2 Problem Statement**

The main problem is doctors or oncologists have difficulties accessing the latest cancer data because there is a limitation for them to getting the latest update about worldwide cancer data including survival and treatment cases data where they can access it, this also includes in-country cancer data. According to Taylor et al. (2021), there is difficulty with accessing information regarding the disease and issues with restricted visitation rights, and Rieke et al. (2020) also stated that public medical data is typically task- or disease-specific, it is frequently provided with varied degrees of license limitations, limiting its use. This problem leads to a lack of data insight on cancer-related data. It is important for a doctor to know the latest data on cancer because the field of oncology is always expanding with new research and treatment improvements. In addition, doctors use a strategy to predict a patient's survival rate is to look at information from similar patients, such as those with the same type and stage of cancer (Sauer et al., 2019). However, according to Nunez et al. (2023), survival rates have traditionally been calculated retrospectively and classified by only a few factors, primarily cancer site, and histology. Despite their familiarity with these probabilities,

oncologists frequently struggle to accurately predict an individual patient's future survival because personal factors such as age are difficult to account for. By knowing most recent treatment data allows a doctor to stay current on the most successful treatment and diagnostic methods, allowing them to make informed decisions about their patients' care. Furthermore, the latest data can assist doctors in identifying areas where additional study is required and make more accurate prognoses for their patients. Besides, having access to the most recent data allows them to understand better the patterns in cancer in their region, which is critical for planning and providing the best possible care for their patients. Knowing cancer survival is important for both cancer patients and doctors because knowing a patient's cancer survival rate can help doctors set realistic goals and expectations for the patient and their family, improving patient outcomes and satisfaction as well as assist a doctor in making more accurate prognoses for their patients, allowing them and their family to plan for the future (Spencer et. al,2019). Nunez et al. (2023) also stated that accurately predicting a patient's survival can improve cancer care and treatment. It shows that cancer survival rate and treatment have a significant relationship.

### **1.3 Project Aim**

This project aims to develop a dashboard for Cancervive: Data visualization of cancer survival and treatment recommendation. The purpose of Cancervive: Data visualization of cancer survival and treatment recommendation dashboard is to show how survival rates change depending on cancer type, stage, gender, and age factors. This dashboard also provides treatment recommendations as a guideline for doctors or oncologists to make treatment decisions for their patients.

### **1.4 Objectives**

- i. To identify the system requirement for CANCERVIVE: Data Visualization of Cancer Survival and Treatment Recommendation dashboard.
- ii. To design a dashboard for CANCERVIVE: Data Visualization of Cancer Survival and Treatment Recommendation.

- iii. To develop a dashboard for CANCERVIVE: Data visualization of cancer survival and treatment recommendation.

## **1.5 Project Scope**

The data contains information about cancer survival and treatment in the United Kingdom based on cancer patient factors, including age, gender, cancer type, and the year since diagnosis.

Features in CANCERVIVE: Data visualization of cancer survival and treatment recommendation. The user can filter cancer survival data visualization based on cancer type, stage, gender, and age and generate predictions about patients' cancer survival rates and treatment recommendations by filtering based on cancer patients' characteristics input. The user will enter information such as cancer type, stage, age, and gender. Furthermore, the dashboard allows users to observe the cancer survival trend analysis. The primary stakeholder for the CANCERVIVE dashboard project is the doctor or oncologist, as the dashboard can help them decide to plan the treatment for the patients. The secondary user is Cancer patients, who will receive the results and help them in decision-making.

In this project, two datasets were used: cancer survival rate datasets and treatment datasets. The datasets obtained for the project are from The National Cancer Registration and Analysis Service (NCRAS) and the Office for National Statistics. Public Health England manages the National Cancer Registration and Analysis Service (NCRAS). It is in charge of cancer registration in England to help with cancer epidemiology, public health, service monitoring, and research in the United Kingdom. The Office for National Data (ONS) is the largest independent producer of official statistics in the United Kingdom, as well as the country's recognised national statistical institute.

## **1.6 Project Significance**

The outcome of this project is for a doctor or oncologist to predict cancer survival and suggest the best treatment for their patient. It enables them to get updated on the latest survival trends and understand and make informed decisions about the best course of treatment for their patients. The suggestion of cancer treatment can be the outline for oncologists to plan the treatment. Furthermore, survival rates allow oncologists to explain potential treatment outcomes to patients and their families, allowing them to make the best decisions about their care.

## **1.7 Summary**

This chapter explain the outline of the dashboard CANCERVIVE: Data visualisation of cancer survival rate and treatment recommendation. The introduction chapter of this project starts with a research background and problem statement, which outline the main issue that the system aims to address. This chapter is followed by the project aim, which explains the purpose of the dashboard, project objectives, project scope that explains stakeholders and features of the system, and project significance which highlights the benefits the system will provide to the stakeholders.

## **CHAPTER TWO**

### **LITERATURE REVIEW**

#### **2.1 Introduction**

This chapter discusses the literature review on data visualisation of cancer survival and treatment recommendation cancer treatment. It includes all related topics that may help this project's study and provide a better understanding in the project's development. In this chapter will cover explanations of cancer, cancer survival and big data.

#### **2.2 Cancer survival**

According to Islam et.al (2018), the term "survival" refers to the amount of time a person diagnosed with an illness is predicted to live. Survival is related to the stage of the disease at the time of diagnosis, with better survival the sooner the disease is discovered and treated (National Institute for Health and Care Excellence,2020). Several statistics are used to predict cancer patient survival (Islam et.al, 2018). Moreover, another strategy that doctors use to predict a patient's survival is to look at information from similar patients, such as those with the same type and stage of cancer (Sauer et al, 2019). Cancer survival is classified into three phases, as described by Mullan (1985) as cited in Neris et al. (2019). The first phase is called the acute phase, which starts with a cancer diagnosis and involves treatment. The second phase is known as the extended phase and is characterised by remission of the disease and the need for periodic check-ups. The final phase is called the permanent phase, also called the "cure" phase, in which there is no evidence of the active disease for at least five years. Cancer survivors who reach this stage are expected to live without the disease for the rest of their lives.

According to Neris et al. (2019), knowing cancer survival rates is important for several reasons, as it provides an idea of the prognosis of the disease, meaning how likely it is to progress and what the chances of recovery are. It also helps patients, and their families make informed decisions about their treatment options and plan for the future. Because surviving cancer can have both physical and psychological impacts on

patients and their families., the physical effects of cancer treatments can include long-term issues such as sexual dysfunction, bowel problems, and incontinence. Patients and their families may also worry about the possibility of a cancer relapse. In addition to physical effects, surviving cancer can also have psychological impacts, such as changes in self-esteem and body image, which can affect social interactions (Neris et al. ,2019).

According to Islam et.al (2018), Survival can be with or without evidence of cancer, so terms like "disease-free survival" or "recurrence-free survival" are often used to describe the results of treatment. This requires careful monitoring and analysis to understand the disease's course and the treatments' impact. The results are often shown as graphs that compare different treatments and their effects over time for different stages of the disease.

### **2.2.1 Cancer survival rate**

According to Spencer et al. (2019), survival rate refers to the percentage of patients still alive after a certain number of years following treatment. Islam et al. (2018) also stated that the survival rate is considered an indicator of success, but it should take into account the duration of monitoring, typically five years, and A 5-year survival rate is often considered a sign of a cure.

The relative or net survival rate is a way to compare the survival of people with cancer to the survival of a similar population without cancer, considering factors such as age, gender, and race (Islam et al., 2018). This comparison is usually done over 5 years and provides an understanding of the impact of cancer diagnosis and treatment on survival, and net survival is the most commonly used method to determine public cancer survival statistics (Spencer et al., 2019). According to Anderson (2018), this information provides valuable insights for patients, their physicians, and survivors about the prognosis of the disease and the effectiveness of treatments at different stages of survivorship.

#### **i. Cancer prognosis**

The term prognosis refers to the probability of achieving a desired outcome. Age, health status, stage at diagnosis, tumour stage, cell trait, treatment, and response to treatment are all factors that can affect cancer prognosis (Islam et al., 2018).

ii. Cancer survival factor

Factor that determine the survival of cancer are age, sex, cancer type and cancer stage. Below is the explanation of cancer survival factors.

i) Cancer type

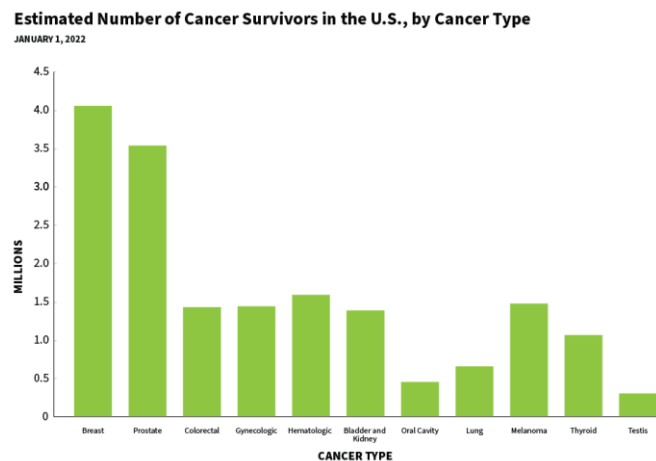


Figure 2.1 cancer survivors by cancer type  
(Source: National Cancer institute)

One factor that affects cancer survival rates is the type of cancer. Different types of cancers have varying growth rates, locations, and responses to treatment, which can significantly impact the success of treatment and the individual's overall survival (Miller et al., 2022). Figure 2.1 displays a bar graph showing the number of cancer survivors according to the type of cancer, and as seen in the figure, there are higher survivors of breast cancer compared to testicular cancer.



## ii) Age

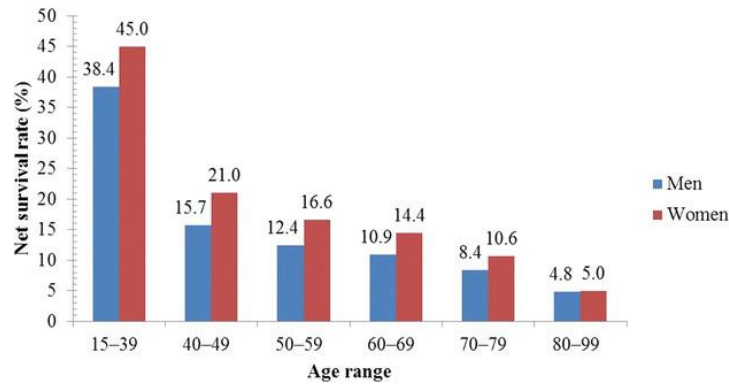


Figure 2.2: cancer survivors by cancer type

(Source: [https://www.researchgate.net/figure/Five-year-net-survival-rate-of-lung-cancer-patients-by-age-in-the-United-Kingdom-UK\\_fig1\\_305722874](https://www.researchgate.net/figure/Five-year-net-survival-rate-of-lung-cancer-patients-by-age-in-the-United-Kingdom-UK_fig1_305722874))

Age is another factor that can affect cancer survival rates, with older individuals generally having a lower survival rate than younger individuals. Pilleron et al. (2021) found that younger patients diagnosed with early-stage lung cancer had higher survival rates than older patients, likely due to differences in treatment received and outcomes. A study by Berkman (2020) also found that the long-term survival rates of ovarian cancer patients were higher for those diagnosed at younger ages, with the highest survival rates among those diagnosed at ages 15 to 19 or 20 to 29. The 25-year survival rate was 74% for patients aged 15 to 19 and 72% for those diagnosed at ages 20 to 29, compared to 64% for those aged 30 to 39. This proves that younger age has greater survival in cancer,

### iii) Gender

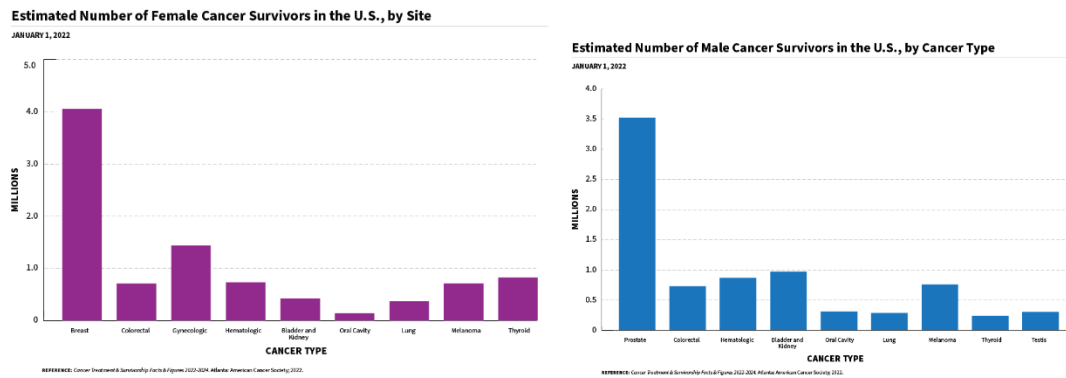


Figure 2.3: shows number of cancer survivors male and female based on cancer type.  
(Source: National Cancer institute)

Gender can also influence cancer survival rates. According to research, certain types of cancer, such as lung cancer, are more aggressive in men than in women, resulting in lower survival rates in men. This is because men smoke more than women, resulting in higher rates of incidence and mortality in men (Groot et al, 2018). Furthermore, Mergen et al. (2019) discovered that when women have a disease, they go to the doctor sooner than men. However, both men and women were diagnosed with the disease at a similar stage and had the same incidence of metastatic disease at the time of diagnosis. This means that an earlier stage of disease at diagnosis is not a likely explanation for the difference in survival between men and women.

#### iv) Cancer stage

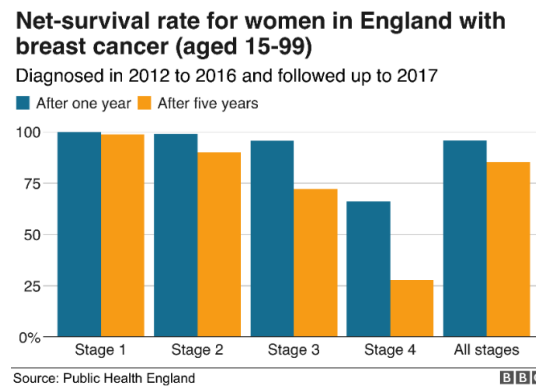


Figure 2.4: cancer survival rate by age  
(Source: <https://www.bbc.com/news/health-46986800>)

According to the National cancer institute (2021), the stage of cancer refers to the extent to which cancer has spread from its original location to other parts of the body. If a cancer is detected at an early stage, before it has spread, the chances of survival are higher than cancer that has advanced or metastasized to other body parts. Figure 2.4 shows that stage 1 has a higher survival rate than stage 4.

#### 2.2.2 Cancer treatment plan.

A treatment plan, according to Newton et al. (2017), is a comprehensive guide that outlines the course of action for a patient's therapy. It includes information on the goal of treatment, the schedule and dosage of medications or procedures, how to monitor progress, what to avoid, and how to manage symptoms. It also details a follow-up plan to ensure that the patient and care team stay on track throughout the treatment journey. Having a well-defined treatment plan helps patients and their care team stay organized and provides a sense of autonomy and control over the process. Popovtzer & Eisbruch (2021) has found that having a good plan for radiation treatment is important for it to be successful. This plan involves determining the complete extent of the cancer, including the histology of the tumour and areas where it may have spread, as well as any nearby tissues that may be affected by the treatment. Other factors, such as the stage of the cancer and the location of the tumour, must also be taken into consideration when developing the plan to ensure the best possible outcome. Jo (2021) stated the treatment plan is highly personalised for the patient and should consider the risk or benefit ratio of the various options, the patient's performance status, the patient's willingness to

consent to treatment, the availability of a treatment facility, financial constraints, and the impact of the treatment on the patient's quality of life.

## 1.8 Cancer

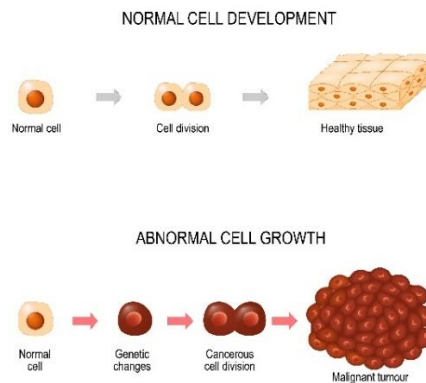


Figure 2.5: normal and abnormal cancer cell development

(Source: National Cancer Foundation: <https://www.cancerfoundation.org.au/what-is-cancer-.html>)

According to Hunter (2020), cancer is a major cause of death worldwide, ranking second as the leading cause of death. In the United States, more than 1.7 million new cases of cancer are estimated to be diagnosed in 2018, making it the primary cause of death. Among men, the most frequently diagnosed cancers are prostate, lung, colon and rectum, and bladder. The most commonly diagnosed cancers for women are breast, lung, colon and rectum, and uterus. Holt (2022) found that cancer develops when genetic abnormalities that disrupt normal cell growth develop. This can occur due to genetic factors to cancer, mutations in genes that transform normal cells into cancer cells, or a lack of function in genes that usually protect against cancer. Figure 2.5 shows different normal cell development and abnormal cell growth a cancer cell.

Cancer is a disease that occurs due to the accumulation of genetic mutations that cause abnormal cell growth and proliferation. These mutations can be inherited or arise spontaneously in a normal cell, turning it into a cancer cell. Genetic mutations can alter the metabolism of potentially carcinogenic compounds, activate oncogenes that promote cancer formation, or inactivate tumor suppressor genes that normally prevent uncontrolled cell division. As a result, cells can grow uncontrollably, invade surrounding tissues, and potentially spread to other parts of the body, leading to the development of cancer.

### 2.3.1 Cancer Trend

#### a) Worldwide

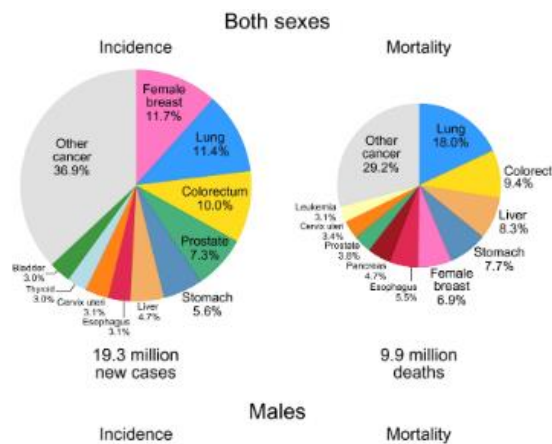


Figure 2.6: Cases and Deaths for the Top 10 Most Common Cancers in 2020  
(Source: <https://acsjournals.onlinelibrary.wiley.com/doi/10.3322/caac.21660>)

Worldwide, it's estimated that there were 19.3 million new cases of cancer and nearly 10 million deaths from cancer in 2020. The most commonly diagnosed type of cancer in women is breast cancer, which has surpassed lung cancer, with around 2.3 million new cases (11.7%). Lung cancer is still the leading cause of cancer death, with 1.8 million deaths (18%). Colorectal and liver cancers also cause a high number of deaths. The incidence of cancer is higher in developed countries than in developing countries, but the death rate from breast and cervical cancer is higher in developing countries. The number of cancer cases is expected to increase by 47% to 28.4 million in 2040 due to demographic changes and may increase even more due to the impact of globalization and a growing economy (Sung et.al,2021).

## b) Malaysia Trend

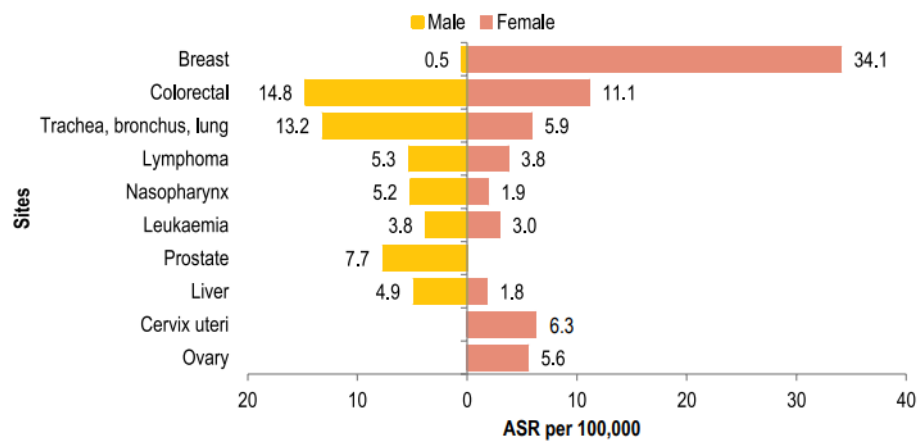


Figure 2.7: cancer cases in Malaysia 2012-2016

(Source: Malaysia National Cancer registry)

During the period of 2012-2016, a total of 115,238 new cancer cases were reported. Out of these, 44.7% were in males and 55.3% were in females. The average number of cancer cases per 100,000 people was 86.1 for males and 101.6 for females. In the last five-year period (2012-2016), the number of cancer cases increased by 2.3% in females and decreased by 0.8% in males when compared to the previous period (2007-2011). The ten most common types of cancer in males were colorectal, lung, prostate, lymphoma, nasopharynx, liver, leukaemia, stomach, non-melanoma skin, and bladder. In females, the ten most common cancers were breast, colorectal, cervix uteri, ovary, lung, corpus uteri, lymphoma, thyroid, leukaemia, and non-melanoma skin. However, the impact of different types of cancer varies greatly with age. For children (0-14 years old), the most common cancers are leukaemia and cancers in the central nervous system, which together account for over 50% of all cancers in boys and girls. In adolescents (15-24 years old), lymphoma is the most common type of cancer in both sexes. For men over 25 years old, the most common cancer is colorectal cancer, followed by lung cancer. In women aged 25-59, the most common cancers are breast cancer and colorectal cancer, with cervical cancer also being common. Among women over 70, colorectal cancer is slightly more common than breast cancer (Azizah et al. (2019).

### 2.3.2 Cancer Diagnosis

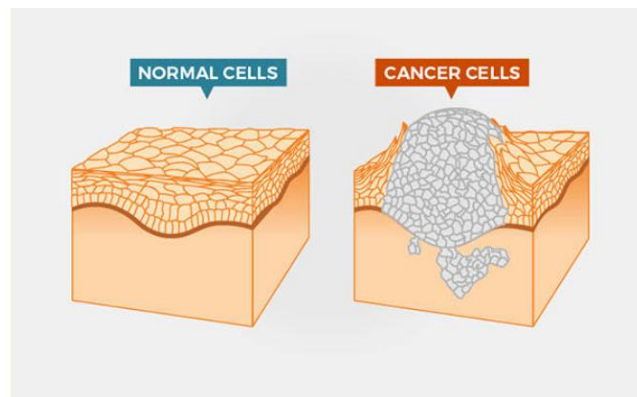


Figure 2.8: cancer cell illustration in body

(Source: National Cancer Institute: <https://www.cancer.gov/about-cancer/understanding/what-is-cancer>)

According to Holt (2020), most cancers form solid tumours (Figure 2.3.2) that can grow large enough to affect the function of important organs. Doctors often make the first diagnosis through a sample of the tumour cells or through a biopsy. There are special proteins that can help identify certain types of cancer. Holt (2022) also found, the TNM system is a commonly used way to describe the stage of a solid tumour based on its size (T), if it has spread to the lymph nodes (N), and if it has spread to other parts of the body (M). This helps doctors understand the patient's prognosis and predict their chances of recovery. The way that cancer spreads depend on how it affects the tissue around it and how easy it is for the cancer cells to move through the body. This can also depend on the location of the cancer and the presence of lymphatic vessels in that area. CT scans and MRI are often used to get a better understanding of the extent of the cancer. Additionally, Lewis et.al found that patients are diagnosed with cancer in a variety of contexts, including screening, through their physician, as an emergency admission, or during a routine hospital outpatient or inpatient session. Many professionals work together to care for cancer patients, and management decisions are made at multidisciplinary team meetings (usually tumour site-specific), with input from surgeons, oncologists, radiologists, pathologists, palliative care physicians, clinical nurse specialists, and allied health professionals.

### 2.3.3 Cancer Factor

There are several risk factors that may increase a person's chances of developing cancer: family history and genetics, lifestyle, environment exposures, and infectious agents.

#### a) Family history and genetics.

According to Schloss (2019) family history plays a role in the development of certain types of cancer, including breast, prostate, and colon cancer. There are specific genetic markers that medical professionals can test for to determine if a person has the genetic predisposition for the cancer, such as the BRCA gene in the case of breast cancer. Also, Jorde (2020) found that it is possible that a person may inherit a gene that increases their risk of developing cancer. This occurs when a cancer-causing mutation is passed down from one generation to the next. This is seen in families that have a high occurrence of a specific type of cancer, which is known as a "cancer family." Although this is rare, it shows that having a damaged gene can lead to cancer. If a person inherits the mutated gene, they have a high likelihood of developing a tumor because the altered gene is present in all of their cells. An example of this is retinoblastoma, a type of childhood eye cancer, where those with a mutated version of the retinoblastoma gene have about a 90% chance of developing the disease.

#### b) Lifestyle.

An unhealthy lifestyle, including smoking, drinking alcohol, and eating a poor diet, can also increase the risk of developing cancer. According to Hunter (2020), a diet high in energy-dense foods and sugary drinks, as well as consumption of red meat and processed meat, can increase the risk of cancer. On the other hand, he also mentions, eating mostly plant-based foods and limiting salt intake can reduce the risk. Obesity and being taller have also been linked to an increased risk of cancer, but the relationship can vary depending on the type of cancer. Alcohol consumption has been linked to several types of cancer, including mouth, throat, esophagus, liver, breast, and rectum. It is recommended to limit alcohol consumption to no more than one drink per day for women and two drinks per day for men. Additionally, Lewis (2021) states smoking is the main



reason for the increase in lung cancer and is also linked to cancers of the mouth, larynx, oesophagus, and bladder and in the United Kingdom, smoking is the leading cause of one-third of all cancer deaths. The direct relationship between smoking and lung cancer has been scientifically proven.

c) Environment exposures.

There are many environmental exposures that have been linked to an increased risk of cancer, including air pollution, UV radiation, exposure to ionizing radiation, chemicals and toxic substances. According to Hunter (2020), ionizing radiation is a known cause of leukaemia and thyroid cancer, although it accounts for only 2% of cancers in the UK. Hunter (2020) also found UV radiation is a major cause of skin cancers, including melanoma, and is estimated to account for 3-4% of cancers in the UK. Arsenic contamination of water supplies has been linked to lung and colon cancers in South-east Asia (Lewis et.al, 2021). Certain occupational exposures, such as exposure to industrial chemicals, are associated with higher risk of specific cancers and are estimated to be responsible for 4% of cancers in the UK. Risks may be higher in less developed economies due to less regulations and enforcement of employee protection (Lewis et.al,2021).

d) Infectious Agents.

According to Hunter (2020), about 15% of cancers globally are caused by infectious agents, with a higher percentage in less developed countries. Common infectious agents include *Helicobacter pylori* (stomach cancer), human papillomavirus (cervical cancer), hepatitis B and C viruses (liver cancers), and Epstein-Barr virus which is nasopharyngeal cancer (Lewis et.al, 2021). In the United Kingdom, only 3.6% of cancers are estimated to be due to infections. Chronic persistent infections and viruses containing transforming oncogenes can lead to cancer (Hunter,2020).

### 2.3.4 Cancer Type

#### a) Lung Cancer

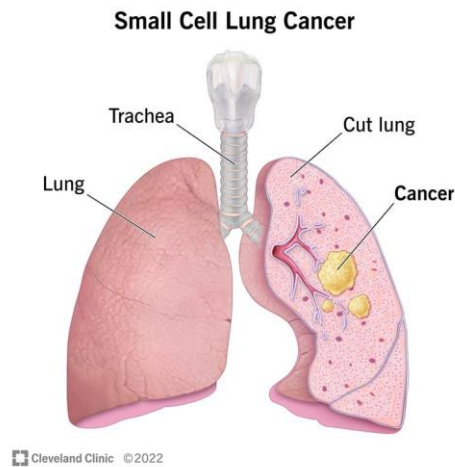


Figure 2.9: Lung cancer location

(Source Cleveland Clinic:<https://my.clevelandclinic.org/health/diseases/6202-small-cell-lung-cancer>)

According to Holt 2022, lung cancer is the most common cause of cancer death in both men and women. It is largely preventable, as 90% of lung cancer deaths are linked to smoking cigarettes. The 5-year survival rate depends heavily on when the cancer is diagnosed; 56% of patients with early-stage cancer will survive for five years or more, while only 5% of those with advanced cancer that has spread to distant parts of the body will still be alive after five years. Unfortunately, most people diagnosed with lung cancer die within one year due to being diagnosed late. According to Rivera et al. (2021), common symptoms include weight loss, cough, and dyspnea. Regional adenopathy and compression of nearby structures may result in superior vena cava syndrome, hoarseness, dysphagia, and obstruction of a central bronchus may result in post-obstructive pneumonia. Lung cancer treatments include surgery, radiation, and chemotherapy. The recommended therapy is determined by cell type, stage, and the patient's overall condition (Holt, 2022).

## b) Breast Cancer

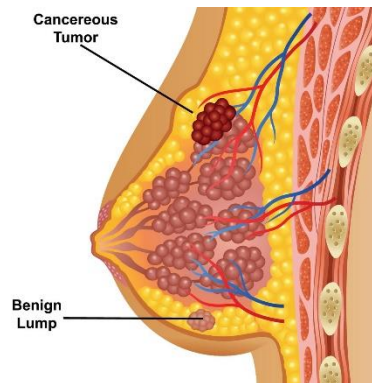


Figure 2.10: breast cancer

(Source: <https://www.vejthani.com/diseases-conditions/breast-cancer/>)

Breast cancer has a 12% lifetime risk among women in the United States. Breast cancer is responsible for around 3% of all deaths. Early menarche, late menopause, late first pregnancy, and nulliparity are all reproductive risk factors that raise the risk of breast cancer because they are thought to prolong estrogen exposure in the breasts (Holt, 2022). Smith (2018) has outlined the indications and manifestations of breast cancer, which include a new lump in the breast or armpit, thickening or swelling of a section of the breast, irritation or dimpling of the skin, redness or flakiness in the nipple area or breast, abnormal nipple discharge (including blood), alterations in the size or shape of the breast, and discomfort in any area of the breast. Moreover, breast conservation therapy consists of lumpectomy, radiation therapy, simple mastectomy, and radical mastectomy (Holt, 2022).

### c) Liver Cancer

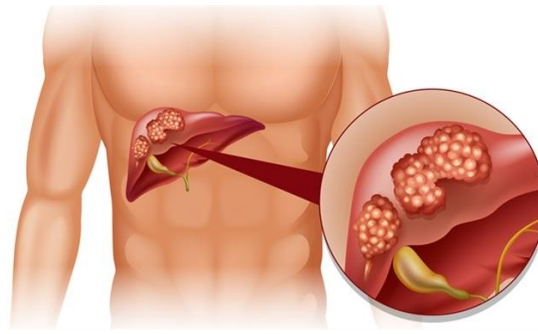


Figure 2.11: Liver cancer location

(Source: <https://www.news-medical.net/health/Secondary-Liver-Cancer.aspx>)

Hepatic neoplasm, also known as liver cancer, is a type of cancer that begins and spreads in the liver. According to Hou and Orhuru, liver cancer is the third leading cause of cancer-related deaths and the fifth most common cancer in the world (2021). Hepatoblastoma and hepatocellular carcinoma (HCC) are the two most common types of liver tumors. The most common type of liver cancer is HCC, which accounts for 80% of all malignant liver tumours (Chung, 2022). Newton et al. also state that the incidence of HCC has rapidly increased in many countries, including the United States and Europe, as a result of hepatitis C virus (HCV) infection and non-alcoholic steatohepatitis (NASH). HCC occurs at a rate of 4.9 cases per 100,000 people, with a 5-year survival rate of less than 12% (2017). Other risk factors, aside from HCV and NASH, include male gender, obesity, diabetes, alcohol and tobacco abuse, and ingestion of aflatoxin, a mycotoxin produced by moulds found in peanuts, soybeans, and corn. At the time of diagnosis, the majority of HCC patients have advanced disease and underlying liver dysfunction. Surgery is the only curative treatment for HCC, but only 15% of patients are candidates for surgery. Even for patients with small, early-stage tumours who undergo surgical resection, the 5-year survival rate for HCC patients is 47% to 53%. The 2-year recurrence rate after surgical resection is 55% (Newton et al., 2017).

For hepatoblastoma, the most common symptom of hepatoblastoma is a painless palpable abdominal mass. Anorexia, weight loss and failure to thrive, abdominal pain, anaemia, and abdominal distention are examples of nonspecific symptoms. Except in the most advanced tumours, jaundice is uncommon because liver function remains

relatively normal. (Chung, 2022). There are currently two main treatment options for hepatoblastoma, a type of liver cancer which are surgical removal of the tumour, and the second method is tumour biopsy first, followed by chemotherapy, and delaying surgical resection (Chung,2022).

#### d) Prostate Cancer

According to Chowdry (2021), in the United States, the prostate is the most common disease diagnosed in males and the second highest cause of cancer mortality in men. Prostate cancer is predicted to be interpreted in 1 in 8 men throughout their lives. In addition, he discovered that almost 191,930 cases of prostate cancer were detected in 2020, accounting for more than 21% of new cancer cases in men. Genetics plays a role in prostate cancer. The potential that vasectomy is linked to an increased risk of prostate cancer has not been proven. Adenocarcinoma of the prostate is almost usually present (Holt,2022). Other than genetics and vasectomy, the risk factor for prostate cancer are age, race, and diet (Chowdry,2021). Yeager (2019) states that the signs and symptoms of prostate cancer are related to its growth, which affects the urethra, causing weak or interrupted urine flow, difficulty starting or stopping urine flow, frequent urination, especially at night, and urinary problems similar to those of infection or benign prostatic hypertrophy. As cancer advances, it can lead to pain in the hips, spine, and ribs, impotence, weakness or numbness in the legs, and incontinence in bowel and bladder function. Besides, Holt (2020) found that screening can detect changes that may signal cancers. PSA-based screening has altered how prostate cancer is diagnosed. In asymptomatic males, elevated serum PSA levels may suggest the existence of prostate cancer. She added that the cancer stage and other indicators, such as the Gleason score and PSA values, would determine the treatment method. Surgery, radiation therapy, hormone therapy, and chemotherapy are all options for treatment.

#### e) Oesophagus cancer

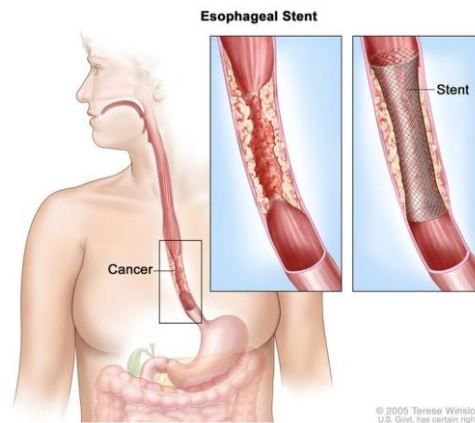


Figure 2.12: Oesophagus cancer location  
(Source: National Cancer Institute)

Cancer oesophagus is the sixth most common cancer worldwide, with approximately 412,000 new cases reported each year. More than 80% of these cases are found in developing countries (Spencer et al., 2019). According to Godfrey et al. (2021), oesophageal cancer is the sixth leading cause of cancer-related death in the United Kingdom. Oesophageal cancer, also known as esophagogastric junction cancer, develops in the upper gastrointestinal tract. It frequently manifests as advanced local disease with a bleak prognosis. Squamous cell carcinoma and adenocarcinoma are the two primary histological types (Godfrey et al., 2021). While squamous cell carcinomas are more common in developing countries, adenocarcinomas are more common in developed countries (Spencer et al., 2019). According to Holt (2022), excessive alcohol consumption and chronic cigarette smoking both increase the risk of oesophageal squamous cell carcinoma. Individuals with Barrett's oesophagus, a complication of gastroesophageal reflux disease, are at the highest risk for adenocarcinoma. Swallowing difficulty or pain is a common symptom. Dysphagia and weight loss are common early symptoms of oesophageal cancer, and they are frequently accompanied by malnutrition. This swallowing difficulty can result in regurgitation and an increased risk of aspiration. Clinical symptoms usually appear after the disease has spread due to the absence of a protective serosal layer around the oesophagus and its extensive lymphatic system (Holt, 2022). Endoscopy is commonly used to initially diagnose oesophageal cancer, allowing for histological confirmation via biopsies. Despite aggressive interventions, the 5-year survival rate for oesophageal squamous cell carcinoma remains low, at 15%

to 20%. The treatment that is suggested to treat oesophagus cancer is surgery, chemotherapy, radiation, and chemotherapy (Godfrey et al., 2021). Surgery to remove the oesophagus (esophagectomy) is common, but it carries significant risks. Chemotherapy and radiation may be used prior to surgery. Radiotherapy does not work well for adenocarcinomas, but combining chemotherapy and surgery can improve survival rates. Palliative measures may include the surgical placement of a feeding tube, bougienage to treat oesophageal strictures, or the placement of an endoscopic stent (Holt, 2022).

#### f) Bladder cancer



Figure 2.13: Liver cancer location

(Source: <https://www.narayanahealth.org/blog/bladder-cancer-causes-symptoms-and-treatment-options/> )

Bladder cancer is the second most prevalent type of urinary system cancer. Urothelial carcinoma is the most frequent type of bladder cancer, accounting for approximately 90% of all cases. Urine and serum indicators are used to diagnose bladder cancer early (Bai et al., 2021). Griebing (2021) found that bladder cancer is one of the most prevalent urologic malignancies observed in the elderly, and its incidence and prevalence rise dramatically with age. Age is one of the most important independent risk factors for the development of bladder cancer due to the lengthy latency of carcinogen exposure, with a median age at diagnosis of more than 70 years. According to Holt (2020), bladder cancer is more common in men. It is related to cigarette smoking and long-term exposure to chemicals used in the dye (aniline), leather, and rubber industries. Haematuria is the most common presenting characteristic. Treatment for bladder cancer is surgery, chemotherapy, immunotherapy, targeted therapy, and radiotherapy (Bai et al., 2021).

### 2.3.5 Cancer Staging

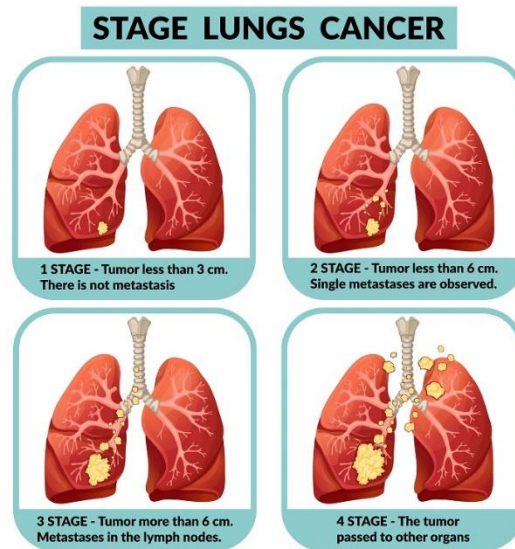


Figure 2.14: Cancer stage  
(Source: Narayana Health)

As Chu (2022) emphasises, accurate staging is critical in determining both prognosis and treatment strategies. Initially, cancer is usually diagnosed by analysing cellular materials such as blood samples, Pap smears, fine needle aspirations, or tumour tissue samples. Once a diagnosis has been confirmed, staging is used to assess disease extent, prognosis, and treatment options. As stated by Symonds et al. (2019), this staging process, which is applicable to various types of cancers, employs the tumour, node, and metastasis (TNM) classification system developed by the American Joint Committee on Cancer (AJCC) and the Union for International Cancer Control (UICC). The TNM system, which is widely used, considers factors such as cell type, primary tumour location, tumour size, grade, number, lymph node involvement, and metastasis presence (Jo, 2021). TNM staging differs depending on the type of cancer, but general principles apply. The primary lesion is classified as T1 to T4 based on its size and invasion of neighbouring structures. T0 indicates a lesion within the basement membrane. N0 (absence) to N3 (increasing involvement) are used to represent nodal involvement. M0 denotes the absence of distant metastases, whereas M1 or M2 denotes the presence and estimated number of metastases, as defined by Kumar et al. (2023). According to Symonds (2019), staging allows for comparisons of similar patient groups across different oncology centres around the world. The simplified stage classification



system consists of four stages: Stage 1: tumour contained within the originating organ; Stage 2: invasion of local lymph nodes; Stage 3: involvement of distant nodes or local spread beyond the origin; Stage 4: presence of blood-borne metastasis. The tumour is usually localised in the early stages (I or II), giving a better chance of cure through local or regional treatments. Higher tumour stages (III or IV) indicate non-localized cancer, where curative treatments are unlikely, according to Jo (2021). Figure 2.14 provides a visual representation of the cancer stage.

### **2.3.6 Cancer Prevention**

According to Schloss (2019), preventing cancer is a complex topic, and it's important to note that no matter how well someone takes care of their health, they may still be diagnosed with cancer. However, adopting a healthy lifestyle that includes a balanced diet with plenty of fruits and vegetables, regular exercise, maintaining a healthy weight, reducing stress, getting adequate sleep, engaging in enjoyable activities, limiting alcohol and substance use, practicing safe sex, having spiritual beliefs and moderation in all things, can help reduce the risk of cancer. Schloss (2019) also mentioned, estimated that around a third of all cancers can be linked to diet and lifestyle factors. On the other hand, Hunter (2020) states that secondary prevention aims to prevent cancer death in those diagnosed with cancer by treating it early through screening. The most successful screening program for cancer has been cervical cancer, which has significantly decreased deaths in countries like the US. Screening for colorectal and breast cancer is also beneficial (Hunter,2020).

### **2.3.7 Cancer Treatment**

According to Yeager (2019), surgery, radiation therapy, chemotherapy, targeted therapy, and immunotherapy are the five main types of cancer treatment. Yeager also mentioned, the treatments may be used individually or in combination and the selection of a treatment plan is based on factors such as the stage and type of cancer, the biology of the cancer cells, and the patient's overall health. The purpose of the treatment can also play a role in determining the type of therapy, with the aim being to cure or eliminate the cancer, control its growth, or reducing symptoms.

#### **a) Surgery**

According to Yeager (2019), Surgery, the most traditional method of cancer treatment, is recommended for the majority of solid tumours. Initially, surgery is used to diagnose the disease by determining tumour type and staging the disease by determining its extent, using sophisticated biopsy and exploratory techniques. Moreover, surgery can be used to prevent cancer by removing precancerous lesions or organs that have a high risk of developing cancer, such as undergoing bilateral mastectomy in people who are genetically predisposed to breast cancer (Sohal, 2022). Surgeries also play an important role in cancer treatment by removing the main source of the disease, reconstructing any affected body parts, and alleviating symptoms through palliative procedures (Yeager, 2019).

#### b) Chemotherapy.

According to Yeager (2019), Chemotherapy involves the administration of drugs with the aim of destroying cancer cells. The drugs work by either disrupting the DNA of cancer cells, hindering the synthesis of DNA, or preventing cell division. Sohal (2022) stated that chemotherapy can be further divided into two categories: cell cycle-nonspecific and cell cycle-specific agents. Cell cycle-nonspecific agents impact both cycling and non-cycling cells, whereas cell cycle-specific agents only affect cells that are cycling. Chemotherapy drugs are also classified based on their mode of action, including alkylating agents, antimetabolites, antitumor antibiotics, and mitotic spindle inhibitors.

#### c) Radiation Therapy

According to Yeager (2019), radiation therapy is a type of cancer treatment that uses high-energy radiation to destroy cancer cells. It can be used for different purposes such as to cure cancers, control the growth of cancer, reduce the size of a tumor before surgery, or provide relief from symptoms. Some of the cancers that can be treated with radiation therapy include skin, prostate, colorectal, lung, cervical, and Hodgkin's cancers. In addition, Yeager (2019) also mentioned radiation therapy can be used in combination with other treatments such as chemotherapy, surgery, or both. Palliative radiation therapy is used to relieve pain and other symptoms caused by cancer, such as pathologic fractures or central nervous system problems. Approximately half of all cancer patients receive radiation therapy as part of their treatment according to the National Cancer Institute (2009) as cited in Yeager (2019).

#### d) Targeted Therapy

Targeted cancer therapies are drugs that aim to stop the growth and spread of cancer by targeting specific molecules involved in cancer growth and progression (Yeager, 2019). Unlike traditional chemotherapy that affects both normal and cancerous cells, targeted therapies only target specific molecules associated with cancer (Sohal, 2022). According to Yeager (2019), targeted therapy is seen as a major development in the field of precision medicine and often result in fewer side effects compared to chemotherapy. The drugs are designed to target growth factors, signalling molecules, cell-cycle proteins, regulators of apoptosis, and molecules involved in host-tumour interactions such as angiogenesis and tumour immunity. Some examples of targeted therapies include monoclonal antibodies, receptor tyrosine kinase inhibitors, signal transduction inhibitors, and gene therapies (Sohal, 2022).

#### e) Immunotherapy

According to Sohal (2022), immunotherapy is a type of cancer treatment that boosts the immune system to fight cancer. It involves using substances made from living organisms to treat cancer, such as monoclonal antibodies that bind to specific targets in the body, causing an immune response that destroys cancer cells. Yeager (2019) also found another form of immunotherapy is adoptive cell transfer, which involves isolating T cells from the patient's body, growing them in the laboratory, and giving them back to the patient as an intravenous infusion. Cytokines, proteins made by the body's cells, can also be used to treat cancer. Treatment vaccines are also used to boost the immune system's response to cancer cells, while Bacillus Calmette-Guérin (BCG) is an immunotherapy that is used to treat bladder cancer by causing an immune response against cancer cells (Yeager, 2019).

More effective treatments have significantly increased cancer survival. However, the use of these stronger treatments is associated with toxicities and side effects that can affect nearly every organ system. Some of these effects are temporary, while others have long-term consequences. All of them have significant implications for cancer patients' perioperative care. (Holt, 2022). According to Lewis et.al (2021), frequently experienced side effects of cancer treatment include discomfort and exhaustion, a decrease in red blood cells causing anaemia, difficulties in the mouth, changes in appetite and weight, and issues with hair, skin, and nails.

## 1.4 Big Data

According to Jay et al. (2019), the Big data concept refers to data sets of a scale and complexity that requires the use of specialized analytical and statistical methods. While Abdallah (2019) refers to big data as a word used to describe a large quantity of data representing a daily business or system that can be structured, unstructured, or semi-structured. In addition, big data cannot be processed and managed by using typical data processing technologies because data is too complicated and large (Mehta & Pandit, 2018, as cited in Tan et al., 2014

### 2.4.1 Big Data Characteristics

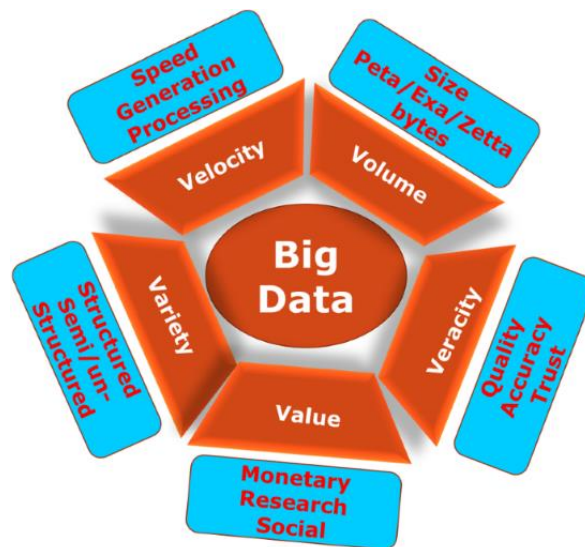


Figure 2.15 Big data characteristics

(Source: Younas, 2019)

Five main characteristics can be used to describe big data. Knowing the characteristics of big data is important for understanding how it works and how to use it. The 5V's characteristics are velocity, volume, veracity, value, and variety.

a) Velocity

Big data velocity is the rate at which data is generated, processed, and transferred across various systems and devices (Younas,2019). The speed of data processing is represented in terms of batch, near-real-time, real-time, and streaming, emphasising that the speed at which data is processed must match the speed at which data is produced (Bowers et al., 2019). Moreover, according to Hasani et al. (2019), a key indicator of big data's real-time character, velocity, is the rate at which the data are created or updated.

b) Volume

According to Ristevski & Chen (2018), volume refers to the amount of data. The large amount of data generated, gathered, and processed can be measured in petabytes, exabytes, and zettabytes. Twitter, for example, receives and processes millions of tweets on a daily basis. (Younas, 2019). Large amounts of big data require innovative methods for gathering, storing, and analysing (Hasani eet al. (2019).

c) Variety

According to Younas (2019), Variety refers to the various sorts of data that can be used together to achieve desired information or results. Structured, semi-structured, and unstructured data are examples of big data types and formats. Structured data is generally well-organized and easily sorted, but unstructured data is not. As an example, consider data stored in a relational database. Unstructured data, such as text and multimedia information, is random and difficult to evaluate. While semi-structured data has tags to separate data pieces, the database user is responsible for enforcing this structure. NoSQL databases are an example of semi-structured data (Hariri et al.,2019).

d) Veracity

"veracity" refers to the data's quality, relevance, uncertainty, dependability, and predictive value (Ristevski & Chen, 2018). Younas (2019) defines data quality

as accuracy, consistency, trust, security, and reliability. Therefore, data must be accurate and consistent, and it must be generated through a reliable system. Furthermore, huge data assets have complicated structures that make them ambiguous, imprecise, and inconsistent. For example, data relating to customer opinions expressed on social media can be skewed, misleading, and unclear (Hasani et al., 2019).

e) Value

The value of big data is defined by its coherent analysis (Ristevski & Chen, 2018). According to Hariri (2019), Value reflects the context and use of data for decision-making, whereas the previous V's are primarily concerned with representing big data difficulties. For example, Facebook, Google, and Amazon have all used analytics to maximize the value of big data in their products. According to Younas (2019), value refers to the various forms of benefits that can be derived through big data processing and analysis. Examples include financial value, societal value, research, or educational value.

#### **2.4.2 Big Data Challenges**

According to Ashabi et al. (2020), it is typical for opportunities to be followed by challenges. Every second, Abdullah et al. (2019) found that Big Data is acquired from many sources and in various forms, such as structured, semi-structured, or unstructured data, and the increasing volume, variety, and velocity of data brings opportunities as well as threats that demand further strategy. Three main challenges in Big Data will be discussed data process, data storage and management, and data privacy and security.

The first challenge in big data is the data process challenge. Many organizations need help dealing with the increasing volume of data. Organizations must determine whether to maintain all expanding data or only the valuable data to address this problem. Organizations must build methods to discover the highest possible business value extracted from Big Data (Abdullah et al. 2019, as cited in Gao et al. 2019).

The second challenge is the data storage and management challenge. The huge amount of data generated daily should be stored for analytical purposes and in compliance with laws and service level agreements to protect and save data (Ashabi et

al., 2020). Singh et al. (2022) found that Big Data typically contains unstructured or structured data in various formats, making it difficult for organizations and scientists to centralize this data into a linked database and then decentralize it based on information uniqueness. Moreover, organizations also encounter a variety of problems related to Big Data management, such as capturing, managing, and administering Big Data. The most difficult task in data management is capturing and managing vast amounts of data in real-time and multiple formats (Abdullah et al., 2019).

The next challenge is data privacy and security. According to Desai (2018), one of the primary concerns in big data is data privacy. Large amounts of data are generated on social media every day as users share private information, locations, images, and so on, revealing users' identities and potentially being used for criminal activities and fraud. Because Big Data contains massive amounts of individual, relational information, information security is the most difficult task. Data theft or hacking can lead to a variety of cybercrimes or phishing attacks. As a result, it must be handled with caution (Singh, 2022).

### **2.4.3 Big Data Tools and method**

#### **a) Big data Visualisation**

According to Liang et.al (2020), data visualization is a scientific approach to presenting data in a visual form and it focuses on showing only the relevant information that the users need and doesn't display all the data. It also can be considered as a method of extracting crucial information from data in the form of abstract that consists mainly of various variables and attributes. Mohammed et.al (2022) found the goal of visualisation is to make information easy to understand and by using shapes and graphics, visualisation allows for the demonstration or clarification of complex construction designs. There are a few visualisations technique include line chart, maps, bar graph, pie chart and scatter plot.

### i) Line chart

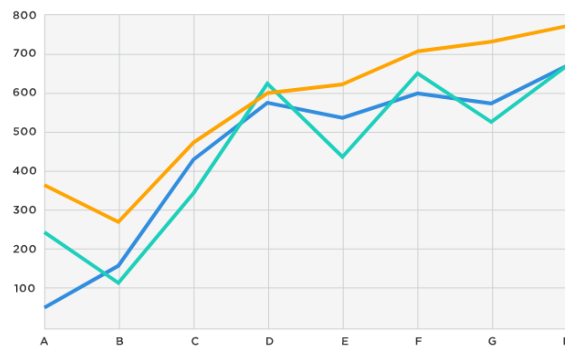


Figure 2.16 Line Chart

(Source: <https://www.tibco.com/reference-center/what-is-a-line-chart>)

Line charts are utilized to demonstrate patterns, changes in speed (increase or decrease), and fluctuations. Variations in data over a period of time can be depicted using these graphs, which display the relationships between the data points (Mohammed et.al, 2022).

### ii) Maps

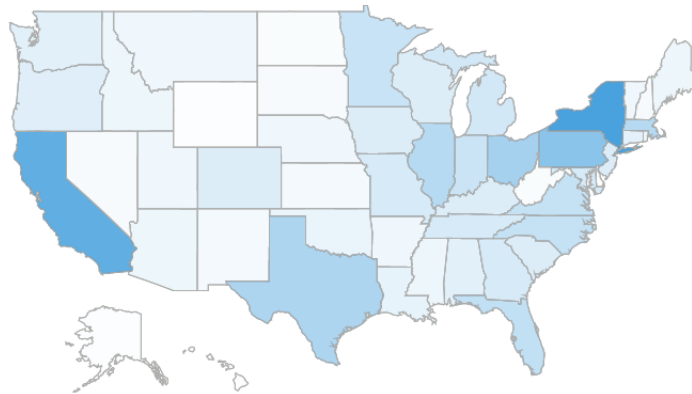


Figure 2.17 Map

(<https://www.inetsoft.com/blog/visualizing-data-map/>)

Maps are an effective tool for visualizing geographical data, which is why they are frequently used. They are typically presented as coloured area maps or bubble maps to display information about a specific location (Mohammed et.al, 2022).



### iii) Bar graph

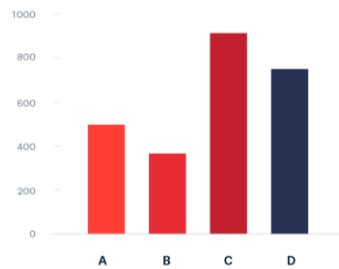


Figure2.18 Bar graph

(Source: <https://www.tibco.com/reference-center/what-is-a-line-chart>)

Bar charts are a useful tool for comparing data between different categories or groups. They are constructed with one axis showing numerical values and the other axis displaying the categories being compared. The height of the bars represents the magnitude of the data being displayed and the comparison of data between categories can be easily seen.

### iv) Pie chart

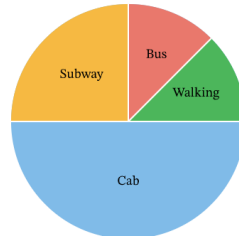


Figure 2.19 Pie chart

(Source: <https://www.nagwa.com/en/explainers/245194820905/>)

Pie charts are a simple and common way of visualizing data, showing the proportion of different parts of a whole. These charts are best for audiences that are not familiar with the data or just want the key highlights (Miller, 2019). According to Mohammed et.al (2022), the use of pie charts, despite its limitations in functionality, remains popular in data visualization due to its ability to effectively illustrate proportionality. These charts are used to show the composition of a variable over a set period of time. Additionally, pie charts are easily recognizable by a general audience and can evoke positive emotions, as the circular shape is pleasing to the eye. Overall, pie charts are a useful tool in communicating data stories to a diverse audience.

v) Scatter plot

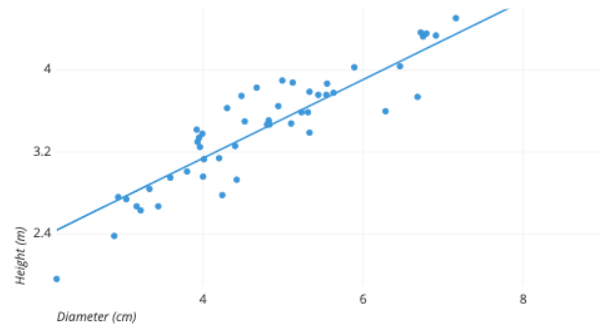


Figure 2.20 Scatter plot

(Source: <https://chartio.com/learn/charts/what-is-a-scatter-plot/>)

According to Miller (2019) scatter plot is a commonly used data visualization technique that shows the relationship between two variables represented by points plotted on a horizontal and vertical axis. This type of visualization is useful in identifying correlations or trends in data, and it works best with a relatively large data set. The closer the data points are grouped together, the stronger the correlation or trend is likely to be. Mohammed et.al state the data in scatter plot is plotted on two axes, with the dependent variable on the y-axis and the independent variable on the x-axis. The resulting scatter plot can reveal the nature of the relationship between the variables, whether it is positive, negative, or nonexistent. Adding a trend line to the scatter plot can help to better illustrate the relationship and its level of statistical significance.

b) Data Analytics

According to Singh et.al (2022), Big Data Analytics (BDA) involves the analysis of large data sets to uncover new patterns and insights. It combines technologies and approaches that require new methods of integration to extract valuable information from large, complex data sets. Descriptive Analytics is the first phase of data analysis that helps identify patterns and provide understanding through identifying patterns and probabilities. Predictive Analytics uses existing data to predict future events and patterns and identify trends. Predictive analytics predicts future outcomes with a high degree of accuracy. Nti et.al (2022) state the aims of big data analytics to identify patterns and find knowledge and intelligence within big data. Big data analytics is a current trend in Business Intelligence (BI) as it combines big data and analytics to provide valuable insights.

### c) Recommendation System

According to Almatrooshi (2020), recommendation systems are algorithms designed to suggest items to a user based on their preferences as determined by certain characteristics or features, within a search-based application. Besides, artificial intelligence (AI) methods are used by recommender systems (RSs) to provide users with item recommendations (Portugal, 2018). According to Natrajan (2020), many websites, including Flipkart, Amazon, and YouTube, utilize recommendation systems to provide tailored products and services to their customers. Such personalized recommendations can increase user satisfaction and loyalty. Recommender systems are categorized into three types: collaborative filtering, content-based, and hybrid.

According to Portugal (2018), collaborative filtering RSs take into account user data when making recommendations. For instance, an online music store's RS can analyze a user's profile to determine the user's preferences and suggest songs that have been bought by similar users. Additionally, Natrajan (2020) found collaborative filtering relies on other users' opinions to make recommendations and is widely used in many applications. However, it has limitations such as data sparsity and the cold start issue. Data sparsity arises when users interact with only a small portion of items, and it makes it difficult for the collaborative filtering model to make accurate recommendations and the cold start issue occurs when there is a lack of data about new entities like a new item or user, making it difficult for the system to compute similarity (Natrajan, 2020).

Content-based filtering RSs rely on item data when generating recommendations. For example, an online store's RS can gather information about a particular computer a user is interested in, and then search for computers with similar attributes (price, CPU speed, etc.) to suggest to the user (Portugal, 2018).

Hybrid filtering RSs combine the collaborative and content-based filtering approaches by considering both user and item data. For instance, on a social network, an RS may recommend profiles similar to a user's profile by comparing interests, and then consider the recommended profiles as items to access their data to search for more similar profiles (Portugal, 2018).

#### d) Machine Learning

According to Ngiam and Khor (2019), machine learning is a type of AI that uses algorithms to allow computers to solve problems without being explicitly programmed. Furthermore, machine learning is a subset of artificial intelligence that focuses on the development of algorithms that allow computers to learn and predict based on previous experiences and patterns (Li et al, 2021). Also, the accuracy of the machine learning model depends on the reliability of the data used to train it (Ngiam and Khor, 2019). Li et. al (2021) found machine learning draws upon various disciplines such as statistics, approximation theory, and probability to develop algorithms that allow computers to make informed decisions. The increasing interest in machine learning has led to its rapid growth and widespread applications across various industries.

#### **2.4.4 Big Data application in healthcare**

According to Kumar and Singh (2019), data in the healthcare system is defined by its heterogeneity and variety as a result of the integration of a diverse set of biological data sources, including sensor data, images, gene arrays, laboratory tests, free text, and demographics. The majority of data in the healthcare system, such as doctor's notes, lab test results, and clinical data, is unstructured and not maintained electronically, so it exists only in hard copies, and its volume is rapidly increasing. Desai (2018) discovered that in today's world, it is important to digitized data in health care organisations. This large amount of information might include a patient's past medical history, X-rays, drug specifics for study, audio surgery footage, health regulations, and output data from medical equipment, among other things. With structured and unstructured data are involved, and using big data analytics, doctors can discover hidden knowledge, analyse trends to advance treatment, and make more accurate diagnoses at a lower cost and in less time.

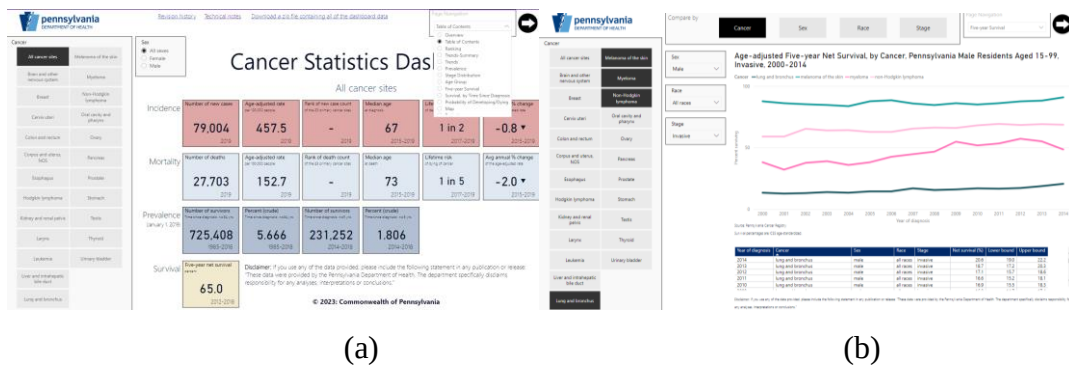
#### a) Big Data Analytics in healthcare

Kumar and Singh (2019) found, the execution of computer programming differentiates traditional health analysis from big-data health analytics. Big data analytics is already being used to predict the consequences of doctor decisions, such as the outcome of a heart operation for a problem based on the patient's age, current state,

and health status. Big data also improve the healthcare sector quality. Big data's responsibility in the health sector is important to handle data sets connected to healthcare that are complicated and difficult to manage with current technology, software, and management methods. According to Zeitschrift and Chen (2019), various benefits can be derived from the application of data analytics in healthcare. These benefits include early detection of disease outbreaks, better understanding of disease causes, monitoring the quality of healthcare institutions, and developing more effective treatment strategies, all of which contribute to improving patient-centered care. With improve healthcare quality by focusing on patient-centric services, healthcare team can detect diseases sooner, improving treatment procedures, and evaluating hospital quality with derived information (Desai,2018). According to the Kumar & Singh (2019), predictive analytics in healthcare, machine learning in healthcare, and electronic health records (EHR) are the frameworks available for analysing healthcare data. One of the most important areas is predictive analytics, which uses predictive method to identify if a patient is at danger of readmission. Using this information, clinicians can make important decisions patient treatment. Predictive analysis requires an understanding of and application of machine learning, which is commonly used in this technique. Machine learning, on the other hand, uses that data to improve the program's understanding. Machine learning involves identifying data patterns and modifying program functions accordingly. In healthcare, the most frequent application of big data is in electronic health records (EHRs). EHRs contain individual patients' medical information, including their medical history, allergies, diagnosis, symptoms, and laboratory test results.

## 2.5 Review on Existing System

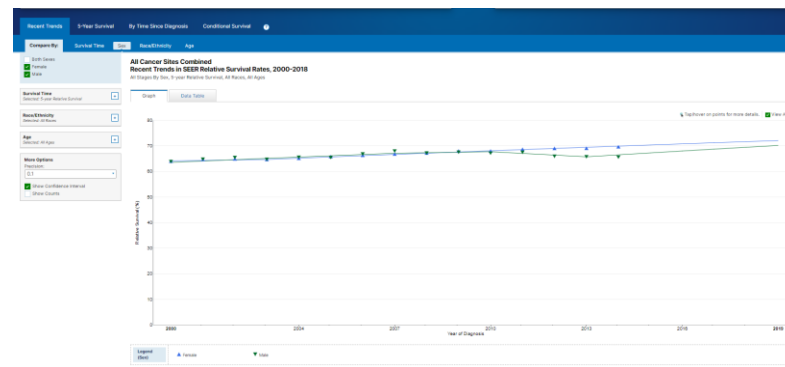
### a) Cancer Statistics Dashboard



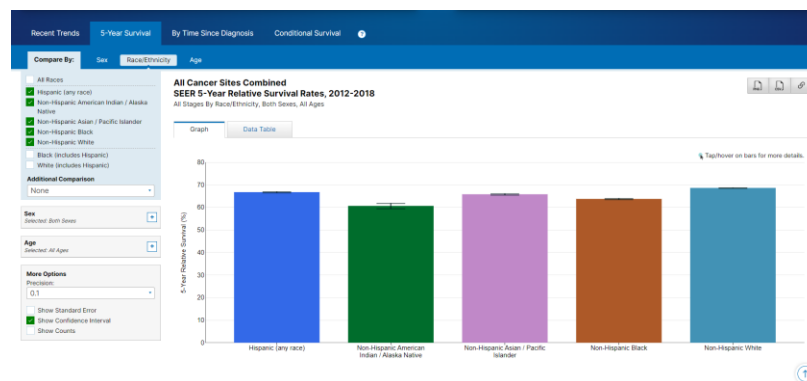
(a) (b)  
Figure 2.21 Cancer Statistics Dashboard  
(Source: Pennsylvania department of health)

Figure 2.21 shows the Pennsylvania cancer dashboard that visualizes and provides information on the cancer statistics for all cancer sites. This dashboard provides information about cancer incidence, mortality, survival, and prevalence. Figure 2.21(a) shows the text box with a summary of data, and the user also can choose the information they want to see in the visualization in the table of content. Figure 2.21(b) shows the graph of survival visualization; the user can interact with the dashboard with filter the information based on their preference such as type of cancer, gender and age. The limitation is that the interface is a bit confusing since the user needs to drop down to see the table of content (navigation) and click “next “to go to the information they want, such as cancer survival or age group visualization information, so the new user needs to study first before using this dashboard (Figure 2.21(a)). Next limitation is that the cancer survival case data is not the latest since the data is from the year 2000 -2014.

## b) SEER\*Explorer Website



(a)



(b)

Figure 2.22 SEER\*Explorer Website  
(Source: National Cancer Institute)

Figure 2.22 shows the cancer survival visualization from the National cancer institute website. The website allows users to visualize data on cancer survival in the United States, including cancer incidence, mortality, survival, and prevalence. Figure 2.22 shows a visualization of cancer survival data based on survival time, and at the top of the page is the navigation of the dashboard visualization. Users can choose to see the survival trend over time, five-year survival (figure 2.22(b)), analysis visualization by time since diagnosis, and conditional survival. The dashboard allowed users to explore and compare the visualization by gender, survival time, and age. Limitations are that the data is based on cancer in the United States and is not up to date since the latest year of data is 2018. Another limitation is that the visualization analysis is based on general cancer cases, and users cannot compare by cancer type.

### c) United States Cancer Statistics Dashboard

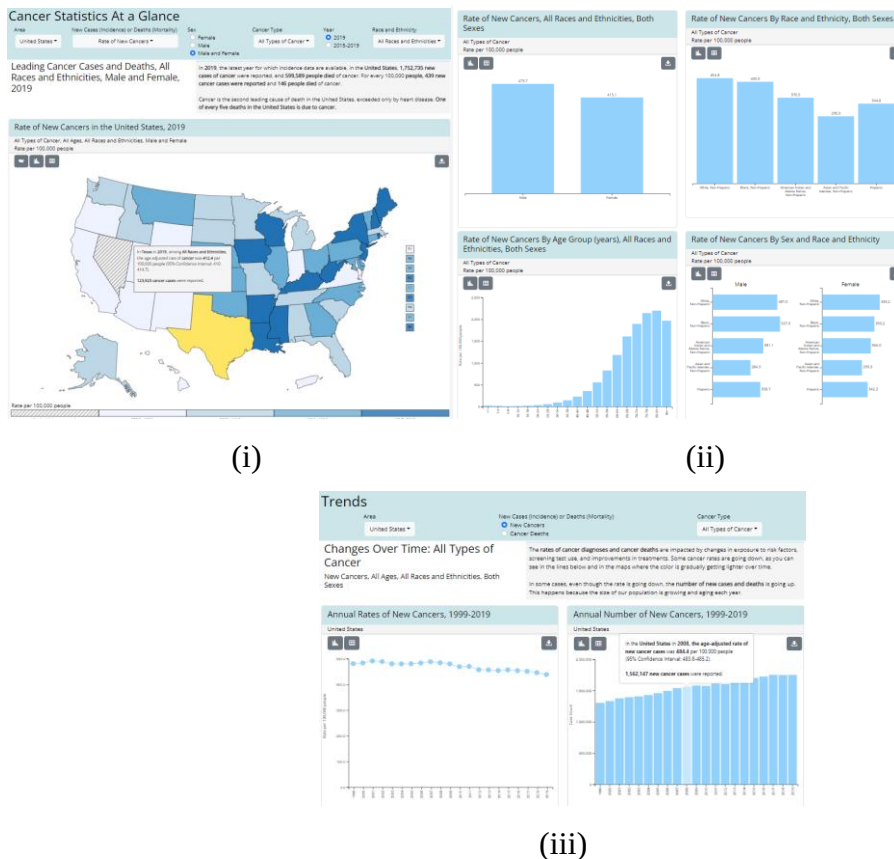


Figure 2.23 United States Cancer Statistics Dashboard  
Source: United States Cancer Statistics Website

The United States Cancer Statistics (USCS) website provides an interactive dashboard that allows users to explore and visualize cancer incidence and mortality rates for different cancer types, demographic groups, and geographic regions in the United States. The dashboard includes an "At A Glance" section for a brief overview of key statistics, an "Explore the Data" section for generating tables, charts, and maps of the data and a "Trends and Maps" section for viewing trends in incidence and mortality rates over time. dashboard offers a user-friendly interface for exploring and visualizing cancer statistics in the United States. Its features include the ability to generate custom tables, charts, and maps of the data quickly, a variety of filters and customization options for exploring the data from multiple perspectives, and trend charts and maps for analyzing changes in cancer incidence and mortality rates over time. The limitation is the cancer data is only in 2019 and the data is based on cancer cases reported to cancer registries in the United States, which may not capture all cases of cancer.



## d) Virginia Department of Health Dashboard

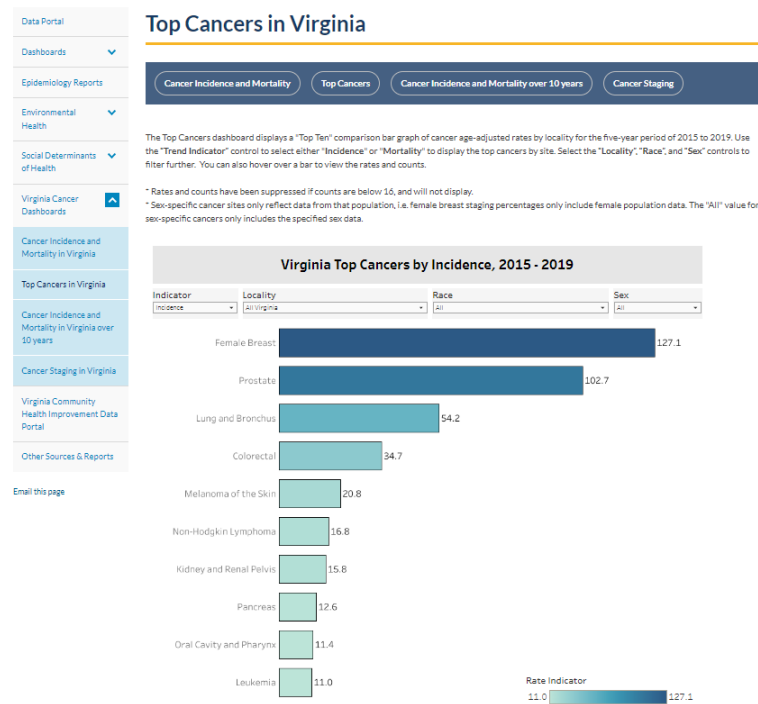
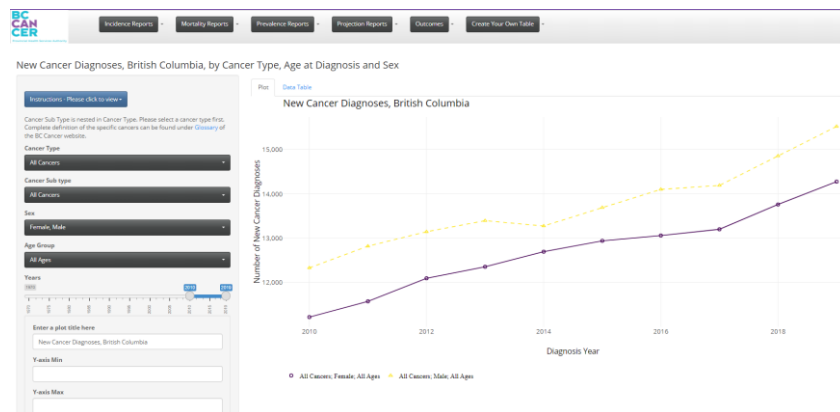


Figure 2.24 Virginia Department of Health Dashboard  
Source: Virginia Department of Health

The Virginia Cancer Dashboards is a website that provides access to a variety of interactive dashboards and visualizations that present cancer-related data for the state of Virginia. The website includes dashboards for cancer incidence, mortality, survival, screening, and risk factors, as well as detailed data on cancer types, demographic groups, and geographic regions within Virginia. Users can interact with the dashboards by selecting different filters and variables to generate custom charts, tables, and maps of the data. The limitation is the data presented on the website is limited to the state of Virginia and the data year is from 2015-2019.

## e) BC Cancer Dashboard



(i)

Show 20 entries

| Cancer Type | Cancer Sub Type | Sex    | Age Group | Year | Count  |
|-------------|-----------------|--------|-----------|------|--------|
| All Cancers | All Cancers     | Female | All Ages  | 2010 | 11,220 |
| All Cancers | All Cancers     | Female | All Ages  | 2011 | 11,575 |
| All Cancers | All Cancers     | Female | All Ages  | 2012 | 12,095 |
| All Cancers | All Cancers     | Female | All Ages  | 2013 | 12,355 |
| All Cancers | All Cancers     | Female | All Ages  | 2014 | 12,695 |
| All Cancers | All Cancers     | Female | All Ages  | 2015 | 12,940 |
| All Cancers | All Cancers     | Female | All Ages  | 2016 | 13,060 |
| All Cancers | All Cancers     | Female | All Ages  | 2017 | 13,200 |
| All Cancers | All Cancers     | Female | All Ages  | 2018 | 13,760 |
| All Cancers | All Cancers     | Female | All Ages  | 2019 | 14,275 |
| All Cancers | All Cancers     | Male   | All Ages  | 2010 | 12,330 |
| All Cancers | All Cancers     | Male   | All Ages  | 2011 | 12,820 |
| All Cancers | All Cancers     | Male   | All Ages  | 2012 | 13,145 |
| All Cancers | All Cancers     | Male   | All Ages  | 2013 | 13,395 |
| All Cancers | All Cancers     | Male   | All Ages  | 2014 | 13,275 |
| All Cancers | All Cancers     | Male   | All Ages  | 2015 | 13,690 |
| All Cancers | All Cancers     | Male   | All Ages  | 2016 | 14,100 |
| All Cancers | All Cancers     | Male   | All Ages  | 2017 | 14,190 |
| All Cancers | All Cancers     | Male   | All Ages  | 2018 | 14,855 |
| All Cancers | All Cancers     | Male   | All Ages  | 2019 | 15,520 |

(ii)

Figure 2.25 BC cancer Dashboard  
Source: BC Cancer website

The BCCA (British Columbia Cancer Agency) Incidence Counts dashboard is an interactive website that provides users with information on cancer incidence rates in the province of British Columbia, Canada. The dashboard allows users to visualize cancer incidence counts and rates by cancer type, sex, and age. It also provides a feature that allows users to download the data for further analysis (Figure 2.25(i)). The dashboard also provide data table (Figure 2.25(ii)) that show the information of cancer data from the visualisation graph. The limitation of the dashboard is that the data is only available for the province of British Columbia and is not latest since the data is until 2018.

f) Global Cancer Observatory (GCO) SurvivalMark dashboard

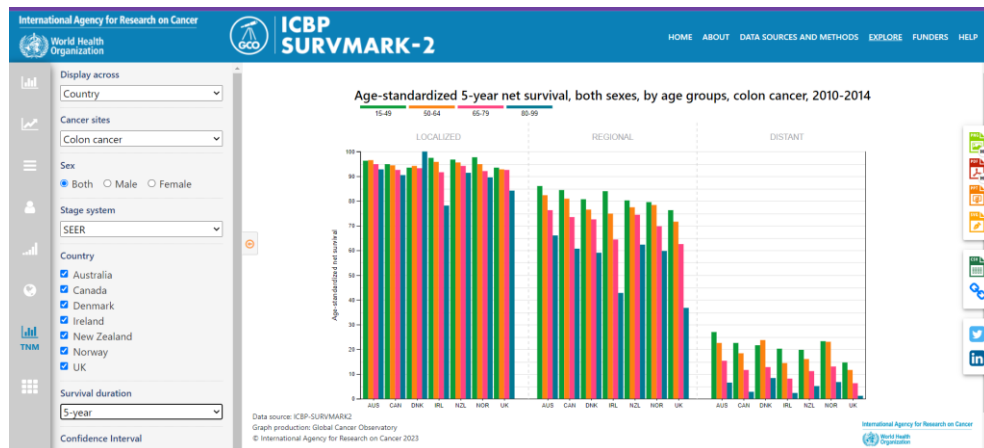


Figure 2.26 Global Cancer Observatory (GCO) SurvivalMark dashboard  
Source: International Agency for Research on Cancer (IARC)

The Global Cancer Observatory (GCO) SurvivalMark dashboard is an online tool developed by World Health Organization (WHO) that provides information on cancer survival rates by country, cancer type, and other demographic characteristics. One key feature of the SurvivalMark dashboard is its interactive data visualization tools, which allow users to explore and compare survival rates across different countries, cancer types, and periods of diagnosis. Users can also filter the data by demographic variables such as age, sex, and cancer stage. It features interactive data visualization tools, allowing users to compare and explore survival rates across different countries, cancer types, and diagnosis periods. Users can filter the data by demographic variables such as age, sex, and cancer stage, and view both observed and relative survival rates. The limitation is the recent data is until 2014 and the data is from Australia, Canada, Denmark, Ireland, New Zealand, Norway, and United Kingdom.

## g) IBM Watson for oncology

**Figure 2.27(i): IBM Watson for Oncology (WFO) - Patient Cases and Clinical Information**

The screenshot shows the 'Patient Cases for Beta' interface. On the left, a list of cases is displayed, including 'Stage IIB resective with Case Notes' and 'Stage IIB resective with Case Notes'. On the right, the 'Breast Cancer - Clinical Information' panel is visible, showing patient characteristics such as Gender (Female), Age (62), and Menopausal status (Postmenopausal). The 'Staging characteristics' section shows 'M Stage' as 'M0 - No clinical or radiologic evidence of distant disease' and 'cT Stage' as 'T3 - Tumor greater than 2 cm but not more than 5 cm'. The 'Treatment Plan Options' panel on the far right provides a summary of recommended treatment options, including 'Chemotherapy followed by surgery followed by endocrine therapy and radiation therapy'.

(i)

(ii)

**Figure 2.27(iii): IBM Watson for Oncology (WFO) - Get Clinical Trials**

The screenshot shows the 'Get Clinical Trials' dialog box. It prompts the user to provide patient information to query clinicaltrials.gov for matching potential clinical trials. The fields include: Age (62), Terms (lobular II), Gender (Female), Cancer type (Breast Cancer), Recruitment (Recruiting), Study type (Interventional Studies), and Country and state (Optional). The 'Submit' button is highlighted.

(iii)

(iv)

Figure 2.27 IBM Watson for Oncology (WFO) system

Source: IBM Watson

IBM Watson for Oncology (WFO) is a clinical decision support system based on artificial intelligence (AI) developed by IBM. The system helps healthcare professionals, especially oncologists, make treatment decisions for cancer patients. To provide personalized treatment recommendations and evidence-based insights, IBM Watson for Oncology analyses and interprets vast amounts of medical literature, patient records, and other data sources. Based on the data entered into the system, it generates individualised treatment plans and recommendations, it improve treatment quality, and potentially prevent doctor-patient contention. One of the limitations of WFO is the lack of consistency between WFO and multidisciplinary team (MDT) decisions, particularly in patients with advanced cancer. Differences in treatment schemes, drug choices, complications, and economic factors can all have an impact on consistency. A multidisciplinary team (MDT) is a group of health and care workers from various organisations and professions.

## h) JAMA Network – DeepSurv



Figure 2.28 DeepSurv system

Source: JAMA Network

DeepSurv is a system by JAMA Network, the system can predict cancer survival and recommend treatment for lung cancer. DeepSurv is a deep learning network system designed to predict survival outcomes in non-small cell lung cancer (NSCLC) patients. It applies deep learning principles, a branch of machine learning that focuses on training artificial neural networks to learn intricate patterns and associations from large datasets. DeepSurv can detect complex associations that linear models may miss by analysing a wide range of factors such as age, gender, histology, and treatment options. DeepSurv is notable for its ability to generate treatment recommendations based on calculated risk. DeepSurv can provide individualised treatment suggestions that aim to improve survival outcomes by incorporating Cox proportional hazards. As a result, it has the potential to be a useful tool for clinical decision-making in lung cancer treatment. DeepSurv's limitation is that the decision-making process within deep learning networks can be difficult to interpret due to their complexity. Furthermore, the model's performance should be externally validated to ensure its efficacy across various patient populations and clinical settings.

## i) ASCO CancerLinQ

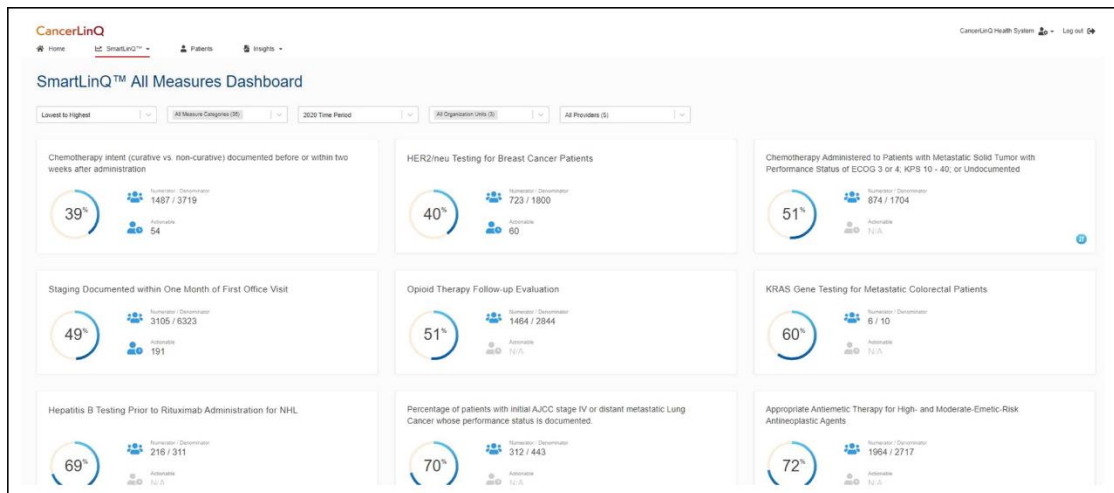


Figure 2.29 ASCO CancerLinQ system

Source: The American Society of Clinical Oncology (ASCO)

CancerLinQ is a system project initiated by the American Society of Clinical Oncology (ASCO). The system recommends cancer treatment. It is a non-profit, physician-led big-data platform dedicated to oncology. It accomplishes this by extracting and aggregating data from participating medical oncology organizations' electronic health records (EHRs) to generate insights and recommendations for clinicians. It can provide oncologists with real-time feedback on treatment options and outcomes, allowing them to make informed decisions based on the most recent evidence. CancerLinQ's limitation is that data sources are limited. CancerLinQ is based on data extracted from participating medical oncology organizations' electronic health records (EHRs).

## j) Nomogram by Memorial Sloan Kettering Cancer Center (MSKCC)

Researchers at Memorial Sloan Kettering have pioneered the use of prediction tools known as nomograms to help patients and physicians make important treatment decisions.

These tools are based on information from hundreds or even thousands of people with cancer. The tools can be used to predict cancer outcomes or assess risk based on specific characteristics of a patient and of his or her disease.

To use our prediction tools online, select from the list below.

- Bladder Cancer
- Breast Cancer
- Clinical Trials: Serious Drug-Related Toxicity in Phase I Clinical Trials
- Colorectal Cancer
- Endometrial Cancer
- Gastric Carcinoma
- Gastrointestinal Stromal Tumor (GIST)
- Liver Cancer
- Lung Cancer Screening Decision Tool
- Melanoma
- Ovarian Cancer
- Prostate Cancer
- Renal Cell Carcinoma
- Sarcoma
- Uterine Leiomyosarcoma

(a)

**Enter Your Information**  
All fields are required unless noted optional

What is the patient's diagnosis?  
Select diagnosis

Is there a need for an extrahepatic organ resection during the hepatectomy?  
☐ Yes ☒ No

What is the planned number of liver segments to be resected during the hepatectomy?  
3 (0 to 6)

What is the patient's hemoglobin level before the hepatectomy?  
20 g/L (3 to 30)

What is the patient's platelet count before the hepatectomy?  
25  $\times 10^3/\text{mm}^3$  (20 to 1000)

**Calculate** **Clear**

**RISK OF PERIOPERATIVE TRANSFUSION IN HEPATECTOMY**

**10%**

This number shows, as a percentage, the probability that the patient will need a transfusion of red blood cells before, during, or shortly after undergoing a hepatectomy for liver cancer. This probability means that for every 100 patients like this, we expect that 10 will need a transfusion of red blood cells before, during, or shortly after undergoing surgery and 90 will not need a transfusion.

Results produced by this tool are based on data from patients treated at MSK, a large research institution with surgeons who perform a high volume of liver cancer procedures. All results must be understood in the context of each patient's specific treatment plan. Patients and caregivers using this tool should discuss the result with the patient's physician.

**Supporting Publications**

Sima CS, Jemagni WR, Fong Y, Elkin E, Fischer M, Wuert D, D'Angelica M, DeMatteo RP, Blumgart LH, Gonen M. Predicting the risk of perioperative transfusion for patients

(b)

(c)

Figure 2.30 Nomogram system  
Source: Memorial Sloan Kettering Cancer Center (MSKCC)

Memorial Sloan Kettering researchers pioneered the use of prediction tools known as nomograms to assist patients and physicians in making critical treatment decisions. These tools are based on data from hundreds or even thousands of cancer patients. The tools can be used to predict cancer outcomes or assess risk based on specific patient and disease characteristics. The websites can predict the survival of many cancer types, including Bladder Cancer, Breast Cancer, Colorectal Cancer, Endometrial Cancer, Gastric Carcinoma, Gastrointestinal Stromal Tumour (GIST), Liver Cancer, Lung Cancer Screening Decision Tool, Melanoma, Ovarian Cancer, Prostate Cancer, Renal Cell Carcinoma, Sarcoma, and Uterine Leiomyosarcoma, as well as provide suggestions for treatment. One limitation is that the prediction survival rate and treatment recommendations are not consolidated into a single system or website. Instead, multiple separate websites (as shown in Figure 2.30 (a)) are provided for different functions, which can lead to a lack of integration and difficulties for users in understanding and navigating the various tools.

## CHAPTER THREE

### METHODOLOGY

#### 3.1 Introduction

This chapter discusses the methodology used in the research project to develop the CANCERVIVE: Data visualization of cancer survival and treatment recommendation. The methodology is a systematic method for gathering information and data to guide the development, structuring, planning, and control processes of the framework. The agile methodology is the primary methodology for the CANCERVIVE system project because it allows for iterative and collaborative development. This project also includes the OSEMN framework for data management, which involves various data pipeline activities. Figure 3.1 shows methodology that used in CANCERVIVE project.

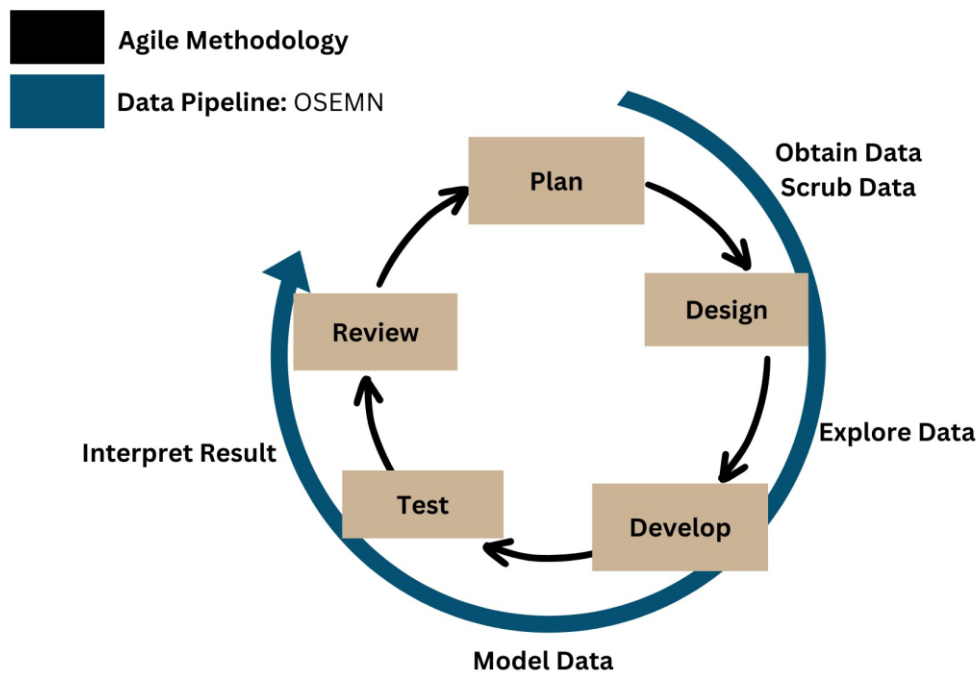


Figure 3.1 Methodology in CANCERVIVE



### 3.2 Methodology

The methodology use in this project is agile methodology. Agile methodology is a method of project management that divides a project into phases. The phases of this project will be plan, design, develop, test, and review. Agile methodology was chosen because it is flexible and efficient method for managing data science projects. It also allows for changes to be made throughout the project. Figure 3.2 is agile methodology diagram with phases that will use in this project.

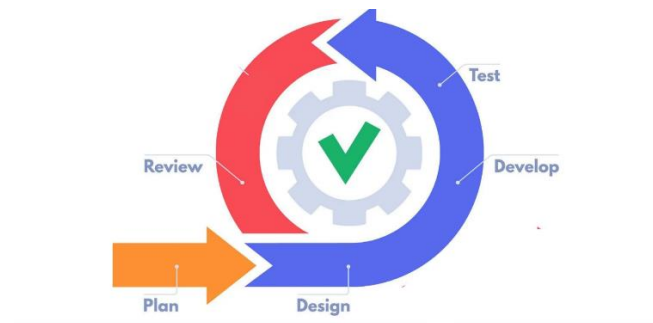


Figure 3.2 : Agile methodology.

(Note : Adapted From Mobilejon. (2021, April 5). Applying the agile methodology to the modern workplace. Mobile Jon's [Image]. <https://mobile-jon.com/2021/04/05/applying-the-agile-methodology-to-the-modern-workplace/>)

OSEMN framework is used in the project for data management and analysis process, which also involves in data pipeline activities. OSEMN is acronym that defines each step of the data analysis process, the steps are Obtain Data, Scrub Data, Explore Data, Model Data, and Interpret Results. Data pipeline is a method for analysing raw data from various data sources and then transferring it to a data store, such as a data lake or data warehouse. Data pipelines ensure consistent data quality, which is crucial for reliable business insights. Figure 3.3 show OSEMN framework.

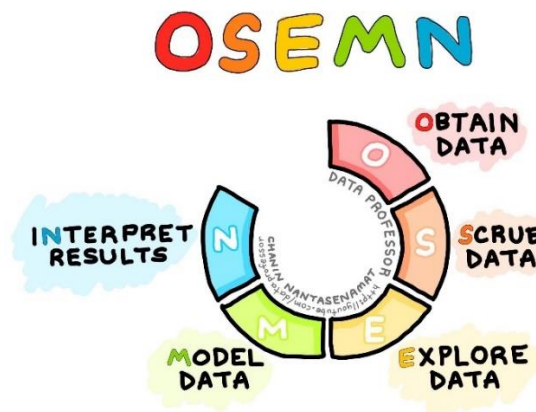


Figure 3.3 : OSEMN framework.

(Note: Adopted From Chanin Nantasenamat. (2020, August 26). The data science process [Image]. The Data Science Process. <https://resources.experfy.com/bigdata-cloud/the-data-science-process/>)

### 3.3 Plan

The first phase in the methodology is plan. The objective of the plan phase is to identify system requirements for Cancervive: Data visualization of cancer survival and treatment recommendation. This includes determining the problem statement, project objective, scope, system features, hardware and software list, and datasets we want to use in the project. Activities in the plan phase are defining problems, ideate solutions, and data pipeline which is data extraction. Table 3.1 shows the summary of the plan phase, including the objective, activities, tools, and techniques to be used in the project, as well as the plan phase deliverable.

Table 3.1: plan phase

| Phase | Objective                           | Activities     | Tools / techniques                                                                            | Deliverables       |
|-------|-------------------------------------|----------------|-----------------------------------------------------------------------------------------------|--------------------|
| Plan  | To identify system requirements for | Define Problem | Techniques:<br>Literature Review.<br>Document Analysis<br><br>Tools: Corpus,<br>Clinical Key, | Problem Statements |

|  |                                                                                 |                                   |                                                                                     |                                                                                         |
|--|---------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
|  | Cancervive: Data visualization of cancer survival and treatment recommendation. |                                   | Science Direct, ACS Journal and IEEE Xplore                                         |                                                                                         |
|  |                                                                                 | Ideate Solution                   | Techniques:<br>System Review, Literature Review                                     | Project Objectives<br>Project Scope<br>System Features<br>List of hardware and software |
|  |                                                                                 | Data Pipeline:<br>Data Extraction | Techniques:<br>Obtain Data, Scrub Data<br>Tools: google datasets, Jupyter Notebook. | Obtain raw datasets for system development                                              |

### 3.4 Design phase.

The next phase is the design phase. This phase is to design the overall system for this project before the development phases. Data transformation, design dashboard database, Outline system features, design interaction flow, and design database. In this phase, the deliverables will be the formatted datasets for system development and data mart architecture. Next, for design dashboard activities, the deliverables are interface design and context use of the project's dashboard. In this phase, outline the system features that will show the outline list of functionality and non-functionality. This phase will also design interaction flow such as a use case diagram, flowchart, and sequence diagram that will show the user and system interaction overview. The last activity in the design phase is the design database that will show the system architecture for big data.

Table 3.2: Design phase

| Phase  | Objective                                                                                    | Activities                         | Tools/techniques                                                | Deliverables                                                                               |
|--------|----------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Design | To Design the Cancervive: Data visualization of cancer survival and treatment recommendation | Data Pipeline: Data Transformation | Techniques: Explore Data, Model Data<br>Tools: Jupyter Notebook | Formatted data sets for system development (The V's of Big Data)<br>Data Mart Architecture |
|        |                                                                                              | Design dashboard database          | Technique: Wireframe<br>Tools: Hand sketch                      | Dashboard user interface design and context of use<br>Illustration                         |
|        |                                                                                              |                                    | Technique: Storyboard<br>Tools: Canva                           |                                                                                            |
|        |                                                                                              | Outline system features            | Technique: Requirement analysis                                 | List of functionality and non-functionality requirement.                                   |
|        |                                                                                              |                                    | Technique: Flowchart diagramming<br>Tools: Draw.io              | Flowchart of the system (business logic + business                                         |

|  |  |                            |                                                                           |                                     |
|--|--|----------------------------|---------------------------------------------------------------------------|-------------------------------------|
|  |  |                            |                                                                           | rule)                               |
|  |  | Design<br>interaction flow | Technique: Use Case<br>Diagramming<br>Tools: Draw.io                      | Use Case Diagram                    |
|  |  |                            | Technique: Sequence<br>Diagramming<br>Tools: Draw.io                      | Sequence Diagram                    |
|  |  | Design database            | Technique:<br>Conceptual Database<br>Design<br>Tools: Database<br>Modeler | System Architecture<br>for Big Data |

### 3.5 Develop

After the design phases, which design the outline of the system develop phase that will develop the system, the data pipeline activity is data load. Also in this phase, the dashboard system will be developed.

Table 3.3: Develop phases.

| Phase   | Objective                                                                                     | Activities                  | Tools/<br>techniques                                                                                                      | Deliverables                                                                           |
|---------|-----------------------------------------------------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| Develop | To develop the Cancervive: Data visualization of cancer survival and treatment recommendation | Data Pipeline:<br>Data load | Techniques:<br>Data Interpretation<br>Tools: Jupyter Notebook, Apache Hadoop                                              | Fact Table<br>Data store (data warehouse)<br>Database model<br>Data mart model /Schema |
|         |                                                                                               |                             | Data Programming:<br>Predictive Analysis<br>System programming:<br>Frontend and backend coding, system integration coding | Coding Algorithm                                                                       |

|  |  |                      |                                                                                                 |                  |
|--|--|----------------------|-------------------------------------------------------------------------------------------------|------------------|
|  |  | Develop<br>Dashboard | Technique:<br>Process and<br>Visualize<br>Tools: Jupyter<br>Notebook,<br>VSCode and<br>Tableau. | Dashboard System |
|--|--|----------------------|-------------------------------------------------------------------------------------------------|------------------|

### 3.6 Test

The next phase is the system test, where the activities system testing will check the dashboard system and the usability test the real person user will test and give feedback about the developed dashboard.

Table 3.4: Test phases

| Phase | Objective                                                                                        | Activities     | Tools / techniques                                                                                   | Deliverables                   |
|-------|--------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------|--------------------------------|
| Test  | To develop the Cancervive:<br>Data visualization of cancer survival and treatment recommendation | System Testing | Techniques:<br>Integration<br>Checking,<br>Debugging<br>Tool: Automated<br>Testing Tool              | Checked<br>Dashboard<br>System |
|       |                                                                                                  | Usability Test | Technique: System<br>Usability Scale<br>Testing<br><br>Tool: System<br>Usability Scale<br>(SUS) Tool | User Feedback                  |

### 3.7 Review

After the test phase, the review phase takes place to assess whether the product or feature developed meets the necessary requirements and quality standards. During this phase, revisions and improvements to the design and development of the dashboard are made based on a list of system errors and user feedback, resulting in an improved dashboard system.

Table 3.5: Test phase

| Phase  | Objective                                                                                     | Activities                       | Tools/techniques                                                                                    | Deliverables              |
|--------|-----------------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------|
| Review | To develop the Cancervive: Data visualization of cancer survival and treatment recommendation | Design and Development Iteration | Revision or improvement on design and development based on list of system errors and user feedback. | Improved Dashboard System |

### 3.8 Summary

The project uses agile methodology to implement the project output. The phases involved in this project are plan, design, develop, test, and review. OSEMN framework is involve data pipeline activities in every phase. This chapter includes the summary table objective, activities, tools or techniques, and deliverables for every phase.



## **CHAPTER FOUR**

### **RESULT AND DISCUSSION**

#### **4.1 Introduction**

On this chapter, the discussion would be focusing on the expected results, current results, and the studies on the development of the dashboard system on CANCEVIVE.

#### **4.2 Identification of Data and System Requirements**

In this development of CANCEVIVE visualisation dashboard, It is critical to identify the data and system requirements to ensure that the project can be implemented with a suitable approach. To meet the project's objectives, a large amount of data is required. The system requirements should cover the system requirements for the datasets required to develop the system. Additionally, system requirements define the function or performance that must be provided in order for a project to run smoothly. As a result, both data and system requirements are essential to the project.

##### **4.2.1 Data requirements**

Data The 5 V's of big data technology are volume, velocity, value, veracity, and variety. The desired output of the CANCEVIVE dashboard system is to be able to provide users, who are doctors or oncologists, with a dashboard that can help users predict cancer survival rates and plan treatment for their patients, as well as analyse previous cases involving cancer survival rates. With this information, the user's patients and the user can make decisions about the next steps to cure the cancer. It is important to lay out what data is required for system development before obtaining the desired output. In this project there are two data involve which is cancer survival rate data and cancer treatment data. In this section I will explain the data process of cancer survival rate data and cancer treatment data.

## a) Cancer Survival Rate data

### i) Data Model

|    | A              | B           | C       | D                   | E                | F                        | G                     | H        | I            | J            | K            |
|----|----------------|-------------|---------|---------------------|------------------|--------------------------|-----------------------|----------|--------------|--------------|--------------|
|    | Diagnosis Year | Cancer Site | Gender  | Stage               | Age At Diagnosis | Standardisation Type     | Years Since Diagnosis | Patients | Survival (%) | Lower 95% CI | Upper 95% CI |
| 1  | 2015 to 2019   | Anus        | Persons | All stages combined | 15 to 44         | Non - standardised       | 1                     | 122      | 87.8         | 82           | 93.7         |
| 2  | 2015 to 2019   | Anus        | Persons | All stages combined | 45 to 54         | Non - standardised       | 1                     | 268      | 92.2         | 88.9         | 95.5         |
| 3  | 2015 to 2019   | Anus        | Persons | All stages combined | 55 to 64         | Non - standardised       | 1                     | 353      | 86.3         | 82.6         | 90           |
| 4  | 2015 to 2019   | Anus        | Persons | All stages combined | 65 to 74         | Non - standardised       | 1                     | 336      | 84           | 79.8         | 88.1         |
| 5  | 2015 to 2019   | Anus        | Persons | All stages combined | 75 to 99         | Non - standardised       | 1                     | 281      | 73.4         | 67.5         | 79.3         |
| 6  | 2015 to 2019   | Anus        | Persons | All stages combined | All ages         | Non - standardised       | 1                     | 1360     | 84.4         | 82.3         | 86.4         |
| 7  | 2015 to 2019   | Anus        | Persons | All stages combined | All ages         | Age-standardised (5 gps) | 1                     | 1360     | 82.4         | 80           | 84.9         |
| 8  | 2015 to 2019   | Anus        | Persons | All stages combined | 15 to 44         | Non - standardised       | 2                     | 122      | 78.2         | 70.7         | 85.7         |
| 9  | 2015 to 2019   | Anus        | Persons | All stages combined | 45 to 54         | Non - standardised       | 2                     | 268      | 82.3         | 77.5         | 87.1         |
| 10 | 2015 to 2019   | Anus        | Persons | All stages combined | 55 to 64         | Non - standardised       | 2                     | 353      | 77.8         | 73.2         | 82.5         |
| 11 | 2015 to 2019   | Anus        | Persons | All stages combined | 65 to 74         | Non - standardised       | 2                     | 336      | 74.6         | 69.4         | 79.8         |
| 12 | 2015 to 2019   | Anus        | Persons | All stages combined | 75 to 99         | Non - standardised       | 2                     | 281      | 57.2         | 50.2         | 64.2         |
| 13 | 2015 to 2019   | Anus        | Persons | All stages combined | All ages         | Non - standardised       | 2                     | 1360     | 73.7         | 71.1         | 76.3         |
| 14 | 2015 to 2019   | Anus        | Persons | All stages combined | All ages         | Age-standardised (5 gps) | 2                     | 1360     | 70.8         | 67.8         | 74           |
| 15 | 2015 to 2019   | Anus        | Persons | All stages combined | 15 to 44         | Non - standardised       | 3                     | 122      | 73.3         | 65.1         | 81.6         |
| 16 | 2015 to 2019   | Anus        | Persons | All stages combined | 45 to 54         | Non - standardised       | 3                     | 268      | 74.7         | 68.9         | 80.4         |
| 17 | 2015 to 2019   | Anus        | Persons | All stages combined | 55 to 64         | Non - standardised       | 3                     | 353      | 74.9         | 69.8         | 80           |
| 18 | 2015 to 2019   | Anus        | Persons | All stages combined | 65 to 74         | Non - standardised       | 3                     | 336      | 69.6         | 63.7         | 75.6         |
| 19 | 2015 to 2019   | Anus        | Persons | All stages combined | 75 to 99         | Non - standardised       | 3                     | 281      | 50.9         | 43.1         | 58.8         |
| 20 | 2015 to 2019   | Anus        | Persons | All stages combined | All ages         | Non - standardised       | 3                     | 1360     | 68.4         | 65.4         | 71.4         |
| 21 | 2015 to 2019   | Anus        | Persons | All stages combined | All ages         | Age-standardised (5 gps) | 3                     | 1360     | 65.4         | 62           | 69.1         |
| 22 | 2015 to 2019   | Anus        | Persons | All stages combined | 15 to 44         | Non - standardised       | 4                     | 122      | 69.5         | 60.6         | 78.5         |
| 23 | 2015 to 2019   | Anus        | Persons | All stages combined | 45 to 54         | Non - standardised       | 4                     | 268      | 70           | 63.7         | 76.4         |
| 24 | 2015 to 2019   | Anus        | Persons | All stages combined | 55 to 64         | Non - standardised       | 4                     | 353      | 70.3         | 64.3         | 76.2         |
| 25 | 2015 to 2019   | Anus        | Persons | All stages combined | 65 to 74         | Non - standardised       | 4                     | 336      | 65.9         | 59.1         | 72.7         |
| 26 | 2015 to 2019   | Anus        | Persons | All stages combined | 75 to 99         | Non - standardised       | 4                     | 281      | 48.2         | 39.1         | 57.2         |
| 27 | 2015 to 2019   | Anus        | Persons | All stages combined | All ages         | Non - standardised       | 4                     | 1360     | 64.4         | 61.4         | 67.4         |

Figure 4.1: Cancer survival rate datasets

Figure 4.1 shows screenshot of the part of Cancer survival datasets. Before proceeding with further data processing, the data type and null values must be identified. The following figures shows the initial process of cancer survival data management.

```
#Load the data file
cancerS_df = pd.read_csv('C:/Users/Admin/anaconda3/cancer_survivals.csv', low_memory=False)

print(cancerS_df)
```

|        | Diagnosis Year | Cancer Site | Gender  | Stage               | \   |
|--------|----------------|-------------|---------|---------------------|-----|
| 0      | 2015 to 2019   | Anus        | Persons | All stages combined |     |
| 1      | 2015 to 2019   | Anus        | Persons | All stages combined |     |
| 2      | 2015 to 2019   | Anus        | Persons | All stages combined |     |
| 3      | 2015 to 2019   | Anus        | Persons | All stages combined |     |
| 4      | 2015 to 2019   | Anus        | Persons | All stages combined |     |
| ...    | ...            | ...         | ...     | ...                 | ... |
| 170755 | 2015 to 2019   | Vulva       | Female  | NaN                 |     |
| 170756 | 2015 to 2019   | Vulva       | Female  | NaN                 |     |
| 170757 | 2015 to 2019   | Vulva       | Female  | NaN                 |     |
| 170758 | 2015 to 2019   | Vulva       | Female  | NaN                 |     |
| 170759 | 2015 to 2019   | Vulva       | Female  | NaN                 |     |

|        | Age At Diagnosis | Standardisation Type     | Years Since Diagnosis | \   |
|--------|------------------|--------------------------|-----------------------|-----|
| 0      | 15 to 44         | Non - standardised       | 1                     |     |
| 1      | 45 to 54         | Non - standardised       | 1                     |     |
| 2      | 55 to 64         | Non - standardised       | 1                     |     |
| 3      | 65 to 74         | Non - standardised       | 1                     |     |
| 4      | 75 to 99         | Non - standardised       | 1                     |     |
| ...    | ...              | ...                      | ...                   | ... |
| 170755 | 55 to 64         | Non - standardised       | 1                     |     |
| 170756 | All ages         | Non - standardised       | 2                     |     |
| 170757 | All ages         | Age-standardised (4 gps) | 3                     |     |
| 170758 | 65 to 74         | Non - standardised       | 3                     |     |
| 170759 | All ages         | Non - standardised       | 3                     |     |

|        | Patients | Survival (%) | Lower 95% CI | Upper 95% CI |
|--------|----------|--------------|--------------|--------------|
| 0      | 122.0    | 87.8         | 82.0         | 93.7         |
| 1      | 268.0    | 92.2         | 88.9         | 95.5         |
| 2      | 353.0    | 86.3         | 82.6         | 90.0         |
| 3      | 336.0    | 84.0         | 79.8         | 88.1         |
| 4      | 281.0    | 73.4         | 67.5         | 79.3         |
| ...    | ...      | ...          | ...          | ...          |
| 170755 | 125.0    | 93.3         | 88.8         | 97.9         |
| 170756 | 811.0    | 79.5         | 76.1         | 82.8         |
| 170757 | 811.0    | 77.6         | 74.0         | 81.4         |
| 170758 | 201.0    | 82.1         | 75.8         | 88.5         |
| 170759 | 811.0    | 76.8         | 72.9         | 80.6         |

[170760 rows x 11 columns]

Figure 4.2: cancer survival data Load

Figure 4.2 shows the cancer survival data load process from the Excel file into the Jupyter Notebook for the data cleaning process. The datasets are in CSV format.

```
#check for information concerning the dataset
cancerS_df.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 4668 entries, 1216 to 19409
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Diagnosis Year         4668 non-null   object
1   Cancer type            4668 non-null   object
2   Gender                 4668 non-null   object
3   Stage                  4668 non-null   object
4   Age At Diagnosis       4668 non-null   object
5   Years Since Diagnosis  4668 non-null   int64
6   Patients               4668 non-null   float64
7   Survival (%)           4668 non-null   float64
8   Lower 95% CI          4668 non-null   float64
9   Upper 95% CI          4668 non-null   float64
dtypes: float64(4), int64(1), object(5)
memory usage: 401.2+ KB
```

Figure 4.3: cancerS\_df data Types

The details of the cancer survival datasets are shown in Figure 4.3, including the range index, id, column name, non-null value counts, and datatype of each column.

```
cancerS_df.describe()
```

|              | Years Since Diagnosis | Patients      | Survival (%)  | Lower 95% CI  | Upper 95% CI  |
|--------------|-----------------------|---------------|---------------|---------------|---------------|
| <b>count</b> | 170760.000000         | 170603.000000 | 121609.000000 | 121609.000000 | 121609.000000 |
| <b>mean</b>  | 3.005470              | 1357.911350   | 62.623951     | 57.037437     | 68.229601     |
| <b>std</b>   | 1.416752              | 5473.317266   | 23.846716     | 24.262336     | 24.095763     |
| <b>min</b>   | 1.000000              | 1.000000      | 1.300000      | 0.800000      | 1.800000      |
| <b>25%</b>   | 2.000000              | 85.000000     | 46.900000     | 39.000000     | 53.600000     |
| <b>50%</b>   | 3.000000              | 275.000000    | 66.700000     | 60.300000     | 73.100000     |
| <b>75%</b>   | 4.000000              | 903.000000    | 81.200000     | 75.500000     | 87.600000     |
| <b>max</b>   | 5.000000              | 221216.000000 | 111.100000    | 105.100000    | 131.900000    |

Figure 4.4: Generate descriptive statistics cancerS\_df

Figure 4.4 shows summary statistics for the DataFrame's numerical columns, including count, mean, standard deviation, minimum value, 25th percentile, median (50th percentile), 75th percentile, and maximum value.

```
print(cancerS_df.columns.values)

['Diagnosis Year' 'Cancer Site' 'Gender' 'Stage' 'Age At Diagnosis'
 'Years Since Diagnosis' 'Patients' 'Survival (%)' 'Lower 95% CI'
 'Upper 95% CI']
```

Figure 4.5: cancerS\_df column names

Figure 4.5 shows all the column names in CancerS\_df dataframe.

```
cancerS_df.rename(columns = {'Cancer Site':'Cancer type'}, inplace = True)

print(cancerS_df.columns.values)

['Diagnosis Year' 'Cancer type' 'Gender' 'Stage' 'Age At Diagnosis'
 'Years Since Diagnosis' 'Patients' 'Survival (%)' 'Lower 95% CI'
 'Upper 95% CI']
```

Figure 4.6: cancerS\_df change column name

Figure 4.6 shows we change column name “Cancer site” to “Cancer type”. The column was renamed "Cancer type" to provide a clearer description of the data contained within the column and to ensure consistency with the treatment dataset, which also uses the same data name for cancer types.

```
print("Number of NaN values for each columns: ")
cancerS_df.isna().sum()

Number of NaN values for each columns:
Diagnosis Year          0
Cancer type            0
Gender                 0
Stage                 30505
Age At Diagnosis       0
Years Since Diagnosis  0
Patients               55
Survival (%)          10727
Lower 95% CI          10727
Upper 95% CI          10727
dtype: int64
```

Figure 4.7: Check Null values in cancerS\_df

In figure 4.7 shows the process of analysing the missing values in the cancerS\_df dataframe. It will then return the number of null values in each of the column in the dataset.

## II) Data Cleaning

```
#show all type of cancer and count
cancerS_df['Cancer type'].value_counts()

Colorectal      8429
Colon            8383
Liver            8375
Stomach          8344
Rectal           8333
Melanoma         8234
Kidney           8215
Oesophagus       8190
Lung             8186
Bladder          8157
Myeloma          8113
Myeloid leukaemia 7940
Pancreas         7833
Leukaemia        7797
Small intestine  7762
Large cell (diffuse) lymphoma 7748
Brain            7730
Non-Hodgkin lymphoma 7710
Breast           2935
Prostate         2804
Ovary            2790
Cervix           2787
Vulva            2778
Larynx           2718
Uterus           2712
Mesothelioma     1217
Anus             1216
Thyroid          1203
Hodgkin lymphoma 1124
Follicular (nodular) NHL 585
Testis           412
Name: Cancer type, dtype: int64
```

Figure 4.8: Count cancer types

In figure 4.8, we count the number of cancer types in cancerS\_df to see the number of cancer data based on cancer types.

```
#drop all rows except 5 type of cancer
cancerS_df = cancerS_df.query("(('Cancer type' == 'Liver') | ('Cancer type' == 'Lung') | ('Cancer type' == 'Bladder') | ('Cancer type' == 'Breast') | ('Cancer type' == 'Oesophagus'))")

#show all type of cancer and count
cancerS_df['Cancer type'].value_counts()

Liver      8375
Oesophagus 8190
Lung       8186
Bladder    8157
Breast     2935
Name: Cancer type, dtype: int64
```

Figure 4.9: drop unrelated cancer type.

In Cancer survival datasets have 30 type of cancer but in this project I only focusing in 5 type of cancer which are:-

- Liver Cancer
- Lung Cancer
- Bladder Cancer
- Breast Cancer
- Prostate Cancer

Figure 4.9 shows the removal of cancer types that are unrelated to this project,

except for the five cancer types listed above. The top cancer type chosen for this project is the top cancer type in Malaysia.

```
print("Number of NaN values for each columns: ")
cancerS_df.isna().sum()
```

```
Number of NaN values for each columns:
Diagnosis Year      0
Cancer type        0
Gender             0
Stage              30505
Age At Diagnosis   0
Years Since Diagnosis 0
Patients           55
Survival (%)       10727
Lower 95% CI       10727
Upper 95% CI       10727
dtype: int64
```

Figure 4.10: count number of null values in dataframe

The figure 4.10 shows count of null values in each column. From the results shows there are null vales in stage, survival (%) , Lower 95% CI and Upper 95% CI column.

```
#drop all the null values
cancerS_df.dropna(inplace=True)
cancerS_df.isnull().sum()
```

Figure 4.11: drop null value

Next, in Figure 4.11 we drop all null values in dataframe.

```
Diagnosis Year      0
Cancer type        0
Gender             0
Stage              0
Age At Diagnosis   0
Years Since Diagnosis 0
Patients           0
Survival (%)       0
Lower 95% CI       0
Upper 95% CI       0
dtype: int64
```

Figure 4.12 recheck null value.

After dropping null values in the dataframe in figure 4.11, we recheck whether the null values were successfully removed from the dataframe in figure 4.12. The figure above shows that there are no null values in the dataframe.

```
# count records from the dataframe
cancerS_df.value_counts()
```

| Diagnosis Year | Cancer type | Gender  | Stage               | Age At Diagnosis | Years Since Diagnosis | Patients | Survival (%) | Low |
|----------------|-------------|---------|---------------------|------------------|-----------------------|----------|--------------|-----|
| 2015-2019      | Bladder     | Persons | All stages combined | 75 to 99         | 2                     | 5432.0   | 53.0         | 51. |
| 4              | 54.6        | 2       |                     |                  |                       |          |              |     |
|                |             | Female  | 1                   | 15 to 44         | 1                     | 23.0     | 95.7         | 87. |
| 5              | 103.8       | 1       |                     |                  |                       |          |              |     |
|                | Lung        | Male    | All stages combined | 65 to 74         | 4                     | 6112.0   | 20.4         | 19. |
| 2              | 21.6        | 1       |                     |                  |                       |          |              |     |
|                |             |         |                     |                  | 5                     | 6112.0   | 17.6         | 16. |
| 2              | 19.0        | 1       |                     |                  |                       |          |              |     |
|                |             |         |                     |                  |                       | 4828.0   | 18.9         | 17. |
| 3              | 20.5        | 1       |                     |                  |                       |          |              |     |
| ..             |             |         |                     |                  |                       |          |              |     |
|                | Liver       | Female  | 6 - Unstageable     | All ages         | 2                     | 119.0    | 40.9         | 31. |
| 8              | 50.0        | 1       |                     |                  |                       |          |              |     |
|                |             |         |                     |                  | 1                     | 119.0    | 48.5         | 39. |
| 5              | 57.5        | 1       |                     |                  |                       |          |              |     |
|                |             |         |                     | 65 to 74         | 2                     | 36.0     | 45.3         | 29. |
| 1              | 61.5        | 1       |                     |                  |                       |          |              |     |
|                |             |         |                     |                  | 1                     | 36.0     | 50.6         | 34. |
| 4              | 66.8        | 1       |                     |                  |                       |          |              |     |
|                | Oesophagus  | Persons | X - Unknown/missing | All ages         | 5                     | 6885.0   | 15.6         | 14. |
| 0              | 17.3        | 1       |                     |                  |                       |          |              |     |

Length: 4667, dtype: int64

Figure 4.13 Detect duplicate data.

Duplicate data can interrupt the flow of the data because if we did not perform the cleaning data properly it can affect the decision making from the higher organization. So, we need to search for the duplicate data from the dataframe. It can help us to detect the repeating data from one data in each cell.

```
# check for duplicates
print(cancerS_df.duplicated())
```

```
1216    False
1217    False
1218    False
1219    False
1220    False
...
19405    False
19406    False
19407    False
19408    False
19409    False
Length: 4136, dtype: bool
```

```
duplicates = cancerS_df.duplicated()
print(duplicates.sum())
```

```
1
```

Figure 4.14 Find the duplicate data.

In figure 4.14 shows, we find the duplicate data in the dataframe, and the results show there are 1 duplicate data in the dataframe.

```
# drop duplicates
cancerS_df.drop_duplicates(inplace=True)
```

```
duplicates = cancerS_df.duplicated()
print(duplicates.sum())
```

0

Figure 4.15: remove duplicate data.

In figure 4.15, we remove duplicate data and show there is 0 duplicate data in the dataframe.

```
cancerS_df['Stage'].unique()
array(['All stages combined', '1', '2', '3', '4', '6 - Unstageable',
      'X - Unknown/missing'], dtype=object)
```

Figure 4.16: all value in stage column

Figure 4.16 shows the value in the Stage column and count. The results shows there stage 1-4 of cancer, 6- unstageable, All stage combines, X-unknown/ missing.

```
# remove the row where Stage equals X - Unknown/missing because is consider as NaN and 6 unstageable because the stage is not qual
cancerS_df = cancerS_df.drop(cancerS_df[(cancerS_df['Stage'] == 'X - Unknown/missing') |
                                         |(cancerS_df['Stage'] == '6 - Unstageable')].index)
```

Figure 4.17: remove unknown stage in dataframe.

In Figure 4.17, we remove the stages "X-unknown/missing" and "6-unstageable" because the values in the datasets are considered missing and unconfirmed data.

```
cancerS_df['Stage'].unique()
array(['All stages combined', '1', '2', '3', '4'], dtype=object)
```

Figure 4.18: check updated value in column stage.

In figure 4.18, we then recheck the remaining value in the stage column to see if it is dropped or not.



```
#show all type of cancer and count
cancerS_df['Age At Diagnosis'].value_counts()
```

```
All ages      1238
65 to 74      647
55 to 64      635
75 to 99      624
45 to 54      410
15 to 44      406
15 to 54      175
Name: Age At Diagnosis, dtype: int64
```

```
# Replace "to" with "-"
cancerS_df['Diagnosis Year'] = cancerS_df['Diagnosis Year'].str.replace(' to ', '-')
```

Figure 4.19: replace sign.

```
#show all type of cancer and count
cancerS_df['Age At Diagnosis'].value_counts()
```

```
All ages      1238
65-74         647
55-64         635
75-99         624
45-54         410
15-44         406
15-54         175
Name: Age At Diagnosis, dtype: int64
```

Figure 4.20: age range category.

In Figure 4.19, we replace the word "to" with a "-" sign in the Age at diagnosis year column. Figure 4.20 shows how to check if the data in the dataframe has been updated. Following that, we can see the age range category in the cancerS\_df dataframe in figure 4.20. The age range is not consistent in this dataframe.

```
age_ranges_to_drop = ['15-54', '75-84', '85-99']
cancerS_df = cancerS_df[~cancerS_df['Age At Diagnosis'].isin(age_ranges_to_drop)]

#show all type of cancer and count
cancerS_df['Age At Diagnosis'].unique()

array(['15-44', '45-54', '55-64', '65-74', '75-99', 'All ages'],
      dtype=object)
```

Figure 4.21: remove inconsistence age range.

In figure 4.21, the inconsistent age ranges such as '15-54,' '75-84,' and '85-99' are removed from the dataframe. Because of the repeated occurrence of age 15 and the irregularity of age ranges following it, the age range '15-54' is identified as inconsistent. Furthermore, the age ranges '75-84' and '85-99' are considered inconsistent. Following that, we display the 'age at diagnosis' data in the dataframe.

```
cancerS_df['Gender'].unique()
array(['Persons', 'Female', 'Male'], dtype=object)

cancerS_df = cancerS_df[cancerS_df['Gender'] != 'Persons']

cancerS_df['Gender'].unique()
array(['Female', 'Male'], dtype=object)

#show all type of cancer and count
cancerS_df['Stage'].unique()
array(['All stages combined', '1', '2', '3', '4'], dtype=object)

#show all type of cancer and count
cancerS_df['Diagnosis Year'].value_counts()

2015-2019    2665
Name: Diagnosis Year, dtype: int64
```

Figure 4.22: display column unique values

In figure 4.22 shows the column values in dataframe. The consistency of the data in column gender, stage and diagnosis year were checked. From the output, all the value is consistent.

```
print(cancerS_df.columns.values)
['Diagnosis Year' 'Cancer type' 'Gender' 'Stage' 'Age At Diagnosis'
'Years Since Diagnosis' 'Patients' 'Survival (%)' 'Lower 95% CI'
'Upper 95% CI']

#drop unuse column
columns_to_drop = ['Lower 95% CI', 'Upper 95% CI']
cancerS_df = cancerS_df.drop(columns_to_drop, axis=1)

print(cancerS_df.columns.values)
['Diagnosis Year' 'Cancer type' 'Gender' 'Stage' 'Age At Diagnosis'
'Years Since Diagnosis' 'Patients' 'Survival (%)']
```

Figure 4.23: drop unuse column from dataframe

```
print(cancerS_df)
```

```

      Diagnosis Year Cancer type Gender      Stage \
1615      2015-2019      Bladder  Female All stages combined
1616      2015-2019      Bladder  Female All stages combined
1617      2015-2019      Bladder  Female All stages combined
1618      2015-2019      Bladder  Female All stages combined
1619      2015-2019      Bladder  Female All stages combined
...
19405      2015-2019  Oesophagus   Male All stages combined
19406      2015-2019  Oesophagus   Male All stages combined
19407      2015-2019  Oesophagus   Male All stages combined
19408      2015-2019  Oesophagus   Male All stages combined
19409      2015-2019  Oesophagus   Male All stages combined

      Age At Diagnosis  Years Since Diagnosis  Patients  Survival (%)
1615              15-44                      1      37.0      40.6
1616              45-54                      1     109.0      62.3
1617              55-64                      1     298.0      70.0
1618              65-74                      1     586.0      61.8
1619              75-99                      1    1229.0      50.9
...
19405              55-64                      5   5362.0      21.2
19406              65-74                      5   9405.0      20.2
19407              75-99                      5   9618.0       9.9
19408              All ages                    5  26592.0      16.9
19409              All ages                    5  26592.0      16.9

```

[2665 rows x 8 columns]

Figure 4.24: print cancerS\_df dataframe

In figure 4.23, the unused columns 'Lower 95% CI' and 'Upper 95% CI' are removed from the dataframe. Lastly, in figure 4.24, shows data in cancerS\_df dataframe.

## b) Cancer Treatment Data.

### i) Data Model

| A                | B                 | C                       | D                | E                  | F                  | G                      | H                         |
|------------------|-------------------|-------------------------|------------------|--------------------|--------------------|------------------------|---------------------------|
| Cancer type      | Year of Diagnosis | Factor of interest      | Levels of factor | Chemotherapy (Y/N) | Radiotherapy (Y/N) | Tumour Resection (Y/N) | Number of tumours treated |
| Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | No                 | No                     | 52                        |
| Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | No                 | Yes                    | 107                       |
| Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | Yes                | No                     | 3                         |
| Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | Yes                | Yes                    | 14                        |
| Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | Yes                | No                 | No                     | 1                         |
| Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | Yes                | No                 | Yes                    | 1                         |
| Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | No                 | No                     | 73                        |
| Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | No                 | Yes                    | 122                       |
| Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | Yes                | No                     | 3                         |
| Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | Yes                | Yes                    | 13                        |
| Benign Endocrine | 2015              | Age at cancer diagnosis | 50-59            | No                 | No                 | No                     | 55                        |
| Benign Endocrine | 2015              | Age at cancer diagnosis | 50-59            | No                 | No                 | Yes                    | 103                       |
| Benign Endocrine | 2015              | Age at cancer diagnosis | 50-59            | No                 | Yes                | No                     | 2                         |
| Benign Endocrine | 2015              | Age at cancer diagnosis | 50-59            | No                 | Yes                | Yes                    | 8                         |
| Benign Endocrine | 2015              | Age at cancer diagnosis | 50-59            | Yes                | No                 | Yes                    | 1                         |
| Benign Endocrine | 2015              | Age at cancer diagnosis | 50-59            | Yes                | Yes                | Yes                    | 1                         |
| Benign Endocrine | 2016              | Age at cancer diagnosis | 50-59            | No                 | No                 | No                     | 56                        |
| Benign Endocrine | 2016              | Age at cancer diagnosis | 50-59            | No                 | No                 | Yes                    | 108                       |
| Benign Endocrine | 2016              | Age at cancer diagnosis | 50-59            | No                 | Yes                | No                     | 6                         |
| Benign Endocrine | 2016              | Age at cancer diagnosis | 50-59            | No                 | Yes                | Yes                    | 11                        |
| Benign Endocrine | 2016              | Age at cancer diagnosis | 50-59            | Yes                | No                 | No                     | 1                         |
| Benign Endocrine | 2016              | Age at cancer diagnosis | 50-59            | Yes                | No                 | Yes                    | 1                         |
| Benign Endocrine | 2017              | Age at cancer diagnosis | 50-59            | No                 | No                 | No                     | 69                        |
| Benign Endocrine | 2017              | Age at cancer diagnosis | 50-59            | No                 | No                 | Yes                    | 122                       |
| Benign Endocrine | 2017              | Age at cancer diagnosis | 50-59            | No                 | Yes                | No                     | 2                         |
| Benign Endocrine | 2017              | Age at cancer diagnosis | 50-59            | No                 | Yes                | Yes                    | 11                        |

Figure 4.25: Cancer Treatment data.

Figure 4.25 shows screenshot of the part of Cancer treatment datasets. Before proceeding with further data processing, the data type and null values must be identified. The following figures shows the initial process of data management of cancer treatment data.

```
cancerT_df = pd.read_csv('Treatment_raw_data.csv')
```

Figure 4.26: Data Load

Figure 4.26 shows the data load process from the Excel file into the Jupyter Notebook for the data cleaning process. The datasets are in CSV format.

```
cancerT_df.head(10)
```

|   | Cancer type      | Year of Diagnosis | Factor of interest      | Levels of factor | Chemotherapy (Y/N) | Radiotherapy (Y/N) | Tumour Resection (Y/N) | Number of tumours treated |
|---|------------------|-------------------|-------------------------|------------------|--------------------|--------------------|------------------------|---------------------------|
| 0 | Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | No                 | No                     | 52                        |
| 1 | Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | No                 | Yes                    | 107                       |
| 2 | Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | Yes                | No                     | 3                         |
| 3 | Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | No                 | Yes                | Yes                    | 14                        |
| 4 | Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | Yes                | No                 | No                     | 1                         |
| 5 | Benign Endocrine | 2013              | Age at cancer diagnosis | 50-59            | Yes                | No                 | Yes                    | 1                         |
| 6 | Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | No                 | No                     | 73                        |
| 7 | Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | No                 | Yes                    | 122                       |
| 8 | Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | Yes                | No                     | 3                         |
| 9 | Benign Endocrine | 2014              | Age at cancer diagnosis | 50-59            | No                 | Yes                | Yes                    | 13                        |

Figure 4.27: data in cancerT\_df dataframe

Figure 4.27 show first ten data of cancer treatment in cancerT\_df dataframe after the data uploaded in jupyter notebook.

```
cancerT_df.dtypes
```

```
Cancer type           object
Year of Diagnosis     object
Factor of interest     object
Levels of factor       object
Chemotherapy (Y/N)    object
Radiotherapy (Y/N)    object
Tumour Resection (Y/N) object
Number of tumours treated int64
dtype: object
```

Figure 4.28: cancerT\_df Data Types

The details of the cancer survival datasets are shown in Figure 4.28, including the range index, id, column name, non-null value counts, and datatype of each column.

```
#describe the dataframe with showing the count, mean, standard deviation, min, percentiles, and max.
cancerT_df.describe()
```

| Number of tumours treated |               |
|---------------------------|---------------|
| count                     | 51361.000000  |
| mean                      | 2131.308035   |
| std                       | 13846.243463  |
| min                       | 1.000000      |
| 25%                       | 14.000000     |
| 50%                       | 79.000000     |
| 75%                       | 509.000000    |
| max                       | 881750.000000 |

Figure 4.29: Generate descriptive statistics cancer\_df.

Figure 4.29 shows summary statistics for the cancerT\_df dataframe's numerical columns, including count, mean, standard deviation, minimum value, 25th percentile, median (50th percentile), 75th percentile, and maximum value.

```
print(cancerT_df.columns.values)

['Cancer type' 'Year of Diagnosis' 'Factor of interest' 'Levels of factor'
 'Chemotherapy (Y/N)' 'Radiotherapy (Y/N)' 'Tumour Resection (Y/N)'
 'Number of tumours treated']
```

Figure 4.30: Column names

Figure 4.30 shows all the column names in CancerT\_df dataframe. It shows that the dataframe have 8 column which are 'Cancer type', 'Year of Diagnosis', 'Factor of interest', 'Levels of factor', 'Chemotherapy (Y/N)', 'Radiotherapy (Y/N)', 'Tumour Resection (Y/N)', and 'Number of tumours treated'.

## ii) Clean Data

```
#show all type of cancer and count
cancerT_df['Cancer type'].value_counts()
```

|                                                                     |      |
|---------------------------------------------------------------------|------|
| All defined malignant neoplasms, excluding non-melanoma skin cancer | 1648 |
| Breast                                                              | 1648 |
| All defined malignant neoplasms                                     | 1648 |
| Rectum                                                              | 1647 |
| Lung: Non-small cell lung cancer                                    | 1646 |
| Bladder                                                             | 1645 |
| Kidney                                                              | 1643 |
| Colon and rectosigmoid junction                                     | 1642 |
| Skin: cSCC (experimental)                                           | 1623 |
| Oesophagus                                                          | 1621 |
| Skin: Melanoma                                                      | 1619 |
| Head and neck: Oral cavity excluding lip and palate                 | 1616 |
| Liver                                                               | 1605 |
| Head and neck: Oropharynx, base of tongue, and tonsil               | 1596 |
| Head and neck: Larynx                                               | 1587 |
| Uterine                                                             | 1581 |
| Stomach                                                             | 1579 |
| Pancreatic                                                          | 1570 |
| Head and neck: Other head and neck excluding lip                    | 1567 |
| Prostate                                                            | 1529 |
| Skin: Rare                                                          | 1529 |
| Cervical                                                            | 1528 |
| Head and neck: Hypopharynx including pyriform sinus                 | 1474 |
| Skin: BCC (experimental)                                            | 1472 |
| Ovary                                                               | 1471 |
| Lung: Small cell lung cancer                                        | 1445 |
| Head and neck: Salivary glands                                      | 1430 |
| Brain: Malignant                                                    | 1388 |
| Testes                                                              | 1319 |
| All defined non-malignant neoplasms                                 | 1261 |
| Brain: Non-malignant                                                | 1237 |
| Non-Benign Endocrine                                                | 1003 |
| Benign Endocrine                                                    | 945  |
| Other: Malignant neoplasms                                          | 824  |
| Other: Non-malignant neoplasms                                      | 775  |
| Name: Cancer type, dtype: int64                                     |      |

Figure 4.31: list of cancer type in cancer\_df

Figure 4.31 shows all the cancer type contain in cancer\_df dataframe. From the figure above, they are 35 types of cancer contain in the dataframe.

```
cancerT_df['Cancer type'] = cancerT_df['Cancer type'].replace('Lung: Small cell lung cancer', 'Lung')
```

Figure 4.32: rename cancer type in cancerT\_df

```
#show all type of cancer and count
cancerT_df['Cancer type'].value_counts()
```

|                                                                     |      |
|---------------------------------------------------------------------|------|
| All defined malignant neoplasms, excluding non-melanoma skin cancer | 1648 |
| Breast                                                              | 1648 |
| All defined malignant neoplasms                                     | 1648 |
| Rectum                                                              | 1647 |
| Lung: Non-small cell lung cancer                                    | 1646 |
| Bladder                                                             | 1645 |
| Kidney                                                              | 1643 |
| Colon                                                               | 1642 |
| Skin: cSCC (experimental)                                           | 1623 |
| Oesophagus                                                          | 1621 |
| Skin: Melanoma                                                      | 1619 |
| Head and neck: Oral cavity excluding lip and palate                 | 1616 |
| Liver                                                               | 1605 |
| Head and neck: Oropharynx, base of tongue, and tonsil               | 1596 |
| Head and neck: Larynx                                               | 1587 |
| Uterine                                                             | 1581 |
| Stomach                                                             | 1579 |
| Pancreatic                                                          | 1570 |
| Head and neck: Other head and neck excluding lip                    | 1567 |
| Prostate                                                            | 1529 |
| Skin: Rare                                                          | 1529 |
| Cervical                                                            | 1528 |
| Head and neck: Hypopharynx including pyriform sinus                 | 1474 |
| Skin: BCC (experimental)                                            | 1472 |
| Ovary                                                               | 1471 |
| Lung                                                                | 1445 |
| Head and neck: Salivary glands                                      | 1430 |
| Brain: Malignant                                                    | 1388 |
| Testes                                                              | 1319 |
| All defined non-malignant neoplasms                                 | 1261 |
| Brain: Non-malignant                                                | 1237 |
| Non-Benign Endocrine                                                | 1003 |
| Benign Endocrine                                                    | 945  |
| Other: Malignant neoplasms                                          | 824  |
| Other: Non-malignant neoplasms                                      | 775  |
| Name: Cancer type, dtype: int64                                     |      |

Figure 4.33: recheck list of cancer type.

```
#drop all rows except 5 type of cancer
cancerT_df = cancerT_df.query("(('Cancer type' == 'Liver') | ('Cancer type' == 'Lung') |  
('Cancer type' == 'Bladder') | ('Cancer type' == 'Breast') | ('Cancer type' == 'Prostate'))")
```

Figure 4.34: drop unrelated cancer type.

As mention before, in this project we only focus in five types of cancer which are:-

- Liver Cancer
- Lung Cancer
- Bladder Cancer
- Breast Cancer
- Prostate Cancer

In figure 4.32 we identify the name of lung cancer is different than in the cancer treatment dataframe. To ensure consistency and integration with the cancer treatment

dataset, "Lung: small cell lung cancer" is renamed "Lung" in figure 4.33, as it is in the cancer treatment dataset. In Figure 4.34 we recheck the list of cancer types to ensure the name of lung cancer is changed. In figure 4.34 shows the removal of cancer types that are unrelated to this project, except for the five cancer types listed above.

```
#show all type of cancer and count
cancerT_df['Cancer type'].value_counts()

Breast      1648
Bladder     1645
Liver       1605
Prostate    1529
Lung        1445
Name: Cancer type, dtype: int64
```

Figure 4.35: cancer type remains in cancer\_df dataframe

Figure 4.35 shows the list of cancer types and the corresponding number of data counts that remain in the cancer\_df dataframe after removing cancer types that are not used in this project. This step was taken to focus the predictive model only on the relevant cancer types and ensure that the dataset contains only the information required for accurate analysis and personalized patient care.

```
print(cancerT_df["Factor of interest"].unique())

['Age at cancer diagnosis' 'Broad ethnic group'
 'Charlson comorbidity index'
 'Deprivation (Index of Multiple Deprivation)' 'Gender'
 'Stage at cancer diagnosis' 'Year of cancer diagnosis']
```

Figure 4.36: list of value in factor of interest column.

Figure 4.36 list of value in ccolumn in cancerT\_df. The “Factor of interest” column is the factor that influences the result of treatment is received by the patient in the cancer treatment datasets.

```
#drop all rows of Factor of interest that is not relates in the projects
cancerT_df = cancerT_df.query("('Factor of interest' == 'Age at cancer diagnosis') | ('Factor of interest' == 'Gender') | ('Factor of interest' == 'Stage at cancer diagnosis')")
```

Figure 4.37(a): drop unrelated factor

```
print(cancerT_df["Factor of interest"].unique())

['Age at cancer diagnosis' 'Gender' 'Stage at cancer diagnosis']
```

Figure 4.37(b): filtered factor of interest

Figure 4.37(a) shows that all unrelated factors have been filtered out except for



three: age at diagnosis, gender, and stage. The other factor is removed because there is no data in the cancer survival datasets; this is done to ensure data integration and consistency across both dataframes.

#### Handling null values of the dataframe

```
print("Number of NaN values for each columns: ")
cancerT_df.isna().sum()
```

Number of NaN values for each columns:

```
Cancer type          0
Year of Diagnosis    0
Factor of interest    0
Levels of factor      0
Chemotherapy (Y/N)   0
Radiotherapy (Y/N)   0
Tumour Resection (Y/N) 0
Number of tumours treated 0
dtype: int64
```

Figure 4.38: count number of null values in cancerT\_df dataframe

Figure 4.38 shows the number of null values in each column. The results show no null values in any of the columns in the cancerT\_df dataframe.

```
# count records from the dataframe
cancerT_df.value_counts()
```

| Cancer type | Year of Diagnosis | Factor of interest        | Levels of factor | Chemotherapy | Radiotherapy | Tumour Resection | Number of tumours treated |
|-------------|-------------------|---------------------------|------------------|--------------|--------------|------------------|---------------------------|
| Bladder     | 2013              | Age at cancer diagnosis   | 50-59            | No           | No           | No               | 90                        |
| Lung        | 2014              | Stage at cancer diagnosis | Stage 4          | Yes          | Yes          | No               | 835                       |
|             |                   |                           | Stage 3          | No           | Yes          | No               | 47                        |
|             |                   |                           |                  | Yes          | No           | No               | 242                       |
|             |                   |                           |                  |              |              | Yes              | 14                        |
| ...         |                   |                           |                  |              |              |                  |                           |
| Breast      | 2017              | Age at cancer diagnosis   | 80+              | Yes          | No           | No               | 133                       |
|             |                   |                           |                  |              |              | Yes              | 53                        |
|             |                   |                           |                  |              | Yes          | No               | 60                        |
|             |                   |                           |                  |              |              | Yes              | 93                        |
| Oesophagus  | 2019              | Stage at cancer diagnosis | Unknown Stage    | Yes          | Yes          | Yes              | 17                        |

Length: 3234, dtype: int64

```
# check for duplicates
print(cancerT_df.duplicated())
```

```
759    False
760    False
761    False
762    False
763    False
...
30303   False
30304   False
30305   False
30306   False
30307   False
Length: 3234, dtype: bool
```

```
duplicates = cancerT_df.duplicated()
print(duplicates.sum())
```

```
0
```

Figure 4.39 Detect duplicate data.

Duplicate data can interrupt the flow of the data because if we did not perform the cleaning data properly it can affect the decision making from the higher organization. Figure 4.39 shows how to find duplicate data and the results show that there is no duplicate data in the dataframe.

## Resolve inconsistencies

Next, we want to remove the inconsistencies occurred in this dataset. First, we need to display all of the column in the dataset.

```
: print(cancerT_df.columns.values)

['Cancer type' 'Year of Diagnosis' 'Factor of interest' 'Levels of factor'
 'Chemotherapy (Y/N)' 'Radiotherapy (Y/N)' 'Tumour Resection (Y/N)'
 'Number of tumours treated']

cancerT_df = cancerT_df.rename(columns={'Chemotherapy (Y/N)': 'Chemotherapy', 'Radiotherapy (Y/N)': 'Radiotherapy',
                                       'Tumour Resection (Y/N)': 'Tumour Resection'})

print(cancerT_df.columns.values)

['Cancer type' 'Year of Diagnosis' 'Factor of interest' 'Levels of factor'
 'Chemotherapy' 'Radiotherapy' 'Tumour Resection'
 'Number of tumours treated']
```

Figure 4.40: rename column name

Figure 4.40 shows the treatment column renamed to remove the "Y/N" words from the column name. Following the renaming, the three treatment column names are 'Chemotherapy,' 'Radiotherapy,' and 'Tumour Resection.'. After the process we recheck the column name again, the final all column name in treatment cancerT\_df dataframe are: -

- Cancer type
- Year of Diagnosis
- Factor of interest
- Chemotherapy
- Radiotherapy
- Tumour Resection
- Number of tumours treated.

```
cancerT_df['Year of Diagnosis'].unique()
array(['2013', '2014', '2015', '2016', '2017', '2018', '2019',
       '2013-2019'], dtype=object)
```

Figure 4.41: year of diagnosis column unique value

Figure 4.41 shows the unique value in the Year of Diagnosis column. The values in the results above range from year 2013 to 2019. However, inconsistency and duplicate years were discovered in the year of diagnosis column.

```
year_to_drop = ['2013-2019']
cancerT_df = cancerT_df[~cancerT_df['Year of Diagnosis'].isin(year_to_drop)]

cancerT_df['Year of Diagnosis'].unique()
array(['2013', '2014', '2015', '2016', '2017', '2018', '2019'],
      dtype=object)
```

Figure 4.42: remove Year of Diagnosis value duplicate.

The year "2018\_2019" is removed to resolve data inconsistency and duplicates in the cancerT\_df dataframe. Figure above shows the removal of duplicate years "2013-2019". After removing the duplicate year, the results show the unique value in the "Year of Diagnosis" column.

```
cancerT_df['Cancer type'].unique()
array(['Bladder', 'Breast', 'Liver', 'Oesophagus', 'Lung'], dtype=object)

cancerT_df['Year of Diagnosis'].unique()
array(['2013', '2014', '2015', '2016', '2017', '2018', '2019'],
      dtype=object)

cancerT_df['Factor of interest'].unique()
array(['Age at cancer diagnosis', 'Gender', 'Stage at cancer diagnosis'],
      dtype=object)

cancerT_df['Levels of factor'].unique()
array(['50-59', '60-69', '70-79', '80+', 'U50', 'female', 'male',
       'Stage 1', 'Stage 2', 'Stage 3', 'Stage 4', 'Unknown Stage'],
      dtype=object)
```

Figure 4.43: check cancerT\_df column unique value

The diagram above shows that all of the columns in the cancerT\_df column were checked. Except for the "levels of factor column" value, which has a null value and an unclear value, all of the values in cancer type, year of diagnosis, and factor of interest are consistent and have no duplicates. This must be addressed in order to ensure the dataset's quality and accuracy.

```
# Replace 'U50' with '0-49'
cancerT_df.loc[cancerT_df['Levels of factor'] == 'U50', 'Levels of factor'] = '0-49'
```

Figure 4.44: range age U50

```
cancerT_df = cancerT_df[cancerT_df['Levels of factor'] != 'Unknown Stage']
print(cancerT_df['Levels of factor'].unique())

['50-59' '60-69' '70-79' '80+' '0-49' 'female' 'male' 'Stage 1' 'Stage 2'
 'Stage 3' 'Stage 4']
```

Figure 4.45: remove unknown stage.

Figure 4.44 shows the age of patients labelled "U50," or those under the age of 50. The age range "0-50" was chosen to improve clarity and consistency in the age data. The unknown stage values were removed from the dataframe's "Levels of Factor" column in Figure 4.45. This step ensures that the dataset contains only clear and relevant stage information, eliminating any ambiguity or inconsistency. Finally, after removing the inconsistent data, Figure 4.45 displays the unique values in the "levels of factor" column.

#### 4.2.2 System Requirement

In this section, we will be discussing on two types of system requirements which are functional and non-functional requirements. The functional requirement is the features that will have in the dashboard. While the non-functional requirement is a set of specifications that describe the dashboard's operation capabilities and constraints and attempt to improve its functionality.

##### A. Functional and non-functional requirements.

Table 4.1: Functional requirements

| Functional Requirement | Description                                                                                                                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Filtering         | The filter allows the user to drill down into specific data visualizations of cancer survival rates. The filter function is also used to personalize the visualization of the cancer survival data dashboard based on user input about the cancer characteristics of the patient.      |
| Interactive Dashboard  | The data displayed on the dashboard can be interacted with by the user. By interacting with the data in the dashboard, the dashboard will drill down into the detailed data information, filter the data, and make the visualisation dashboard custom based on the user's preferences. |
| Prediction             | Based on input from the user, this dashboard's system will predict the cancer survival rate. The dashboard will display the prediction result.                                                                                                                                         |

Table 4.2: Non-functional requirements

| Non-Functional Requirement | Description                                     |
|----------------------------|-------------------------------------------------|
| Usability                  | The dashboard is easy to use and user friendly. |

|                        |                                                                                                                                                                              |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Performance            | The dashboard loads and displays data quickly.                                                                                                                               |
| Responsive             | When the user clicks the button, the dashboard will take them to the right page. Aside from that, the system provides specific output when the user enters some information. |
| Platform Compatibility | Any browser can access this web-based visualisation dashboard.                                                                                                               |

#### B. IT Infra Component of CANCEVIVE Dashboard.

Table 4.3: Hardware Component.

| Hardware    | Specification    |                                                      |
|-------------|------------------|------------------------------------------------------|
| Laptop / PC | Processor        | Intel(R) Core(TM) i5-8300H<br>CPU @ 2.30GHz 2.30 GHz |
|             | RAM              | 8.00 GB                                              |
|             | Storage          | 500GB                                                |
|             | Operating System | Windows 11 Pro                                       |
|             | Graphic card     | NVIDIA® GeForce® GTX<br>1050, 4GB GDDR5              |

Table 4.4: software Component.

| No. | Software                     | Description                                                |
|-----|------------------------------|------------------------------------------------------------|
| 1.  | Tableau                      | To develop an interactive dashboard and data visualisation |
| 2.  | Anaconda Jupyter Notebook    | To clean and filter datasets and modelling prediction.     |
| 3.  | Microsoft Excel              | All the data format in xlsx and CSV                        |
| 4.  | Draw.io                      | To create flowchart and use case diagram                   |
| 5.  | Canva                        | To create storyboard and wireframe                         |
| 6.  | Visual Studio Code (VS Code) | To make a prediction website.                              |

## 4.3 Designing CANCERVIVE: Data Visualisation of Cancer Survival and Treatment Recommendation.

### 4.3.1 Storyboard

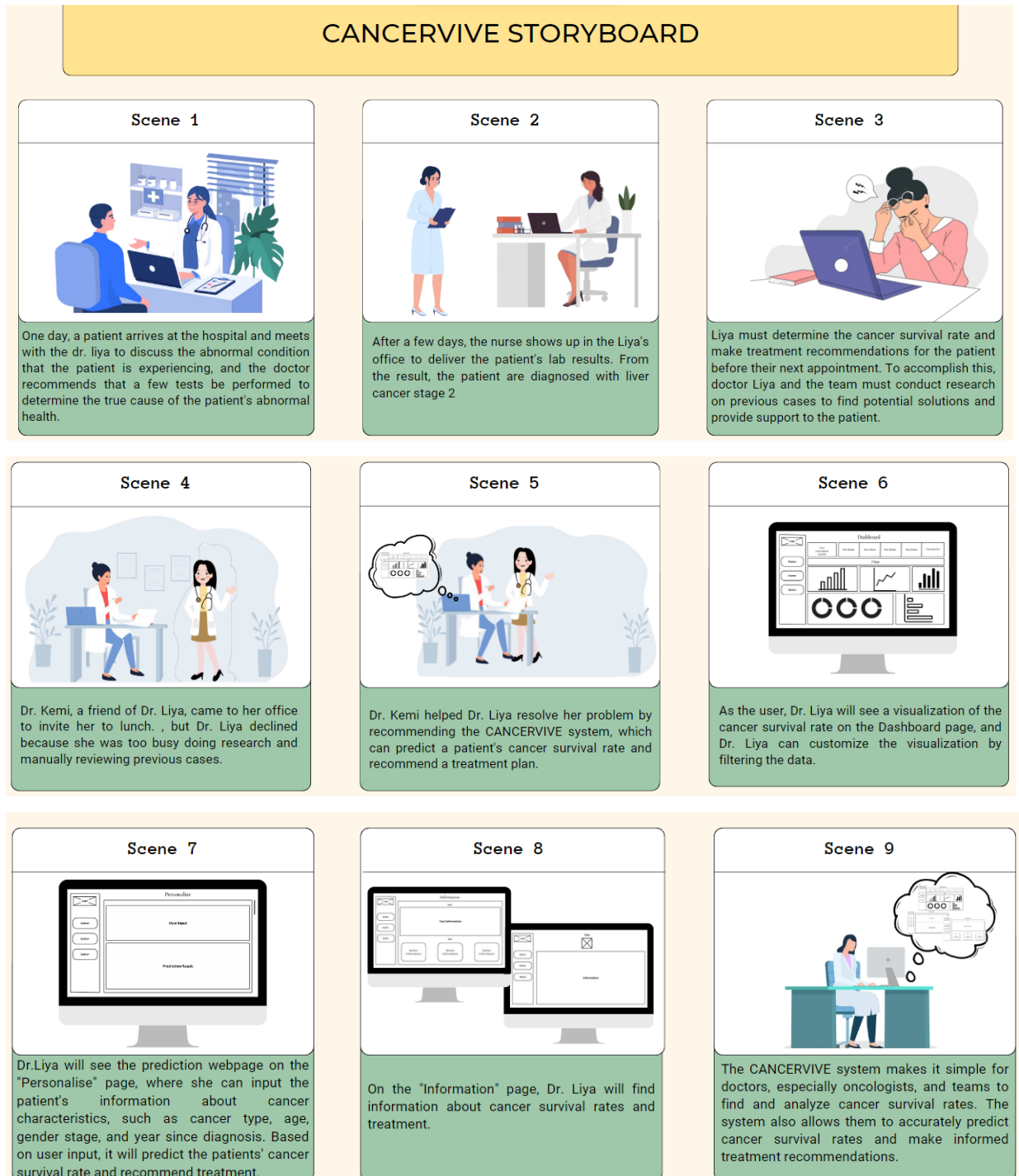


Figure 4.46: Storyboard

### 4.3.2 CANCERVIVE Flowchart

The Figure below shows the flowchart illustration of CANCERVIVE system dashboard. The flowchart gives insight the flow of the system uses and condition when using the system features.

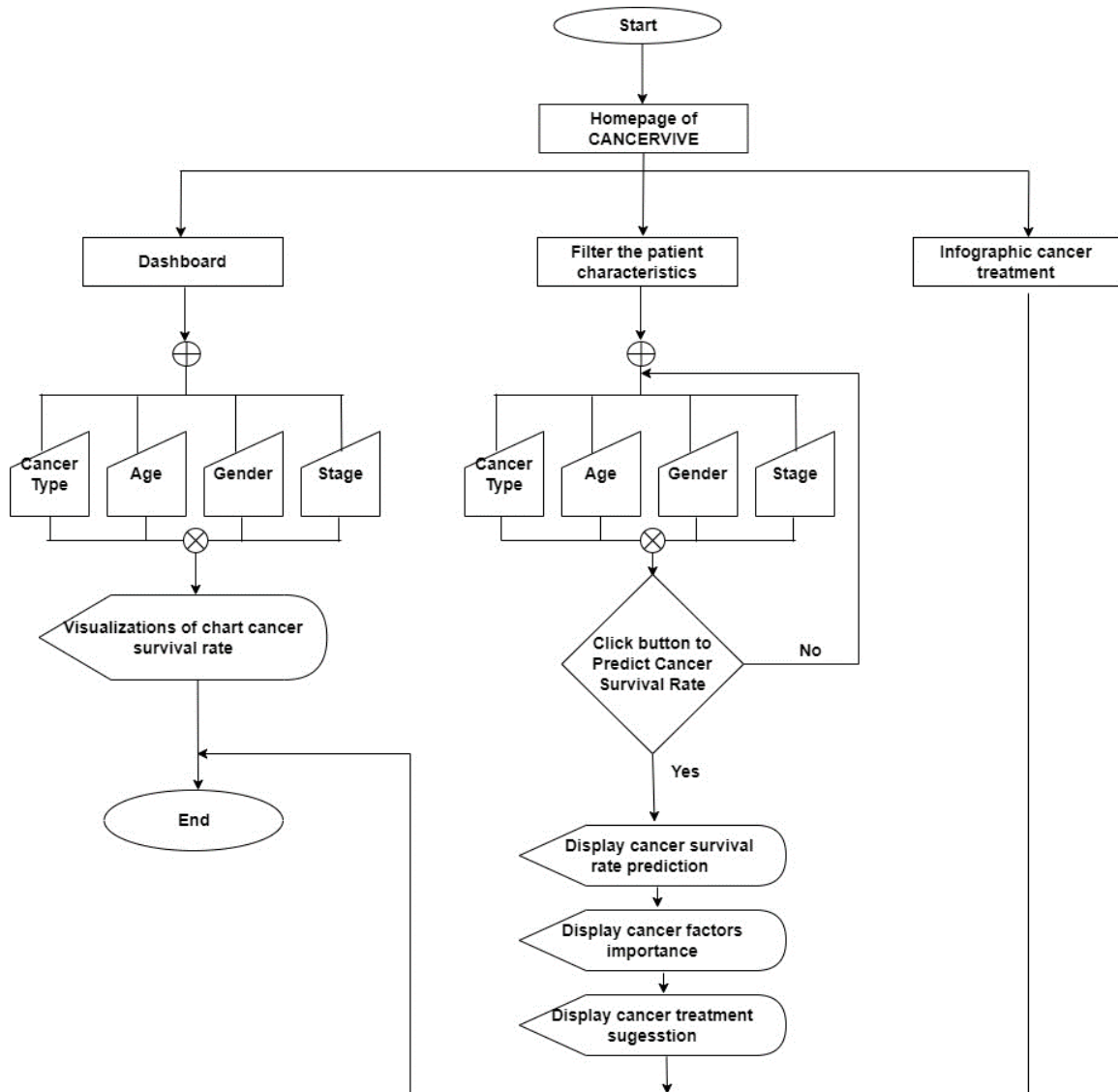
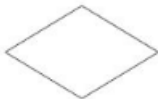
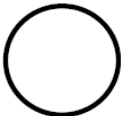


Figure 4.47: Flowchart Diagram



Table 4.5: symbol in flowchart.

| Symbol                                                                              | Description                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | These symbols represent the system's starting point.                                                                                                                         |
|    | This symbol represents the user's navigation process or action within the system.                                                                                            |
|    | This symbol represents the input information the user has entered into the system, where data manipulation such as filtering, can be performed.                              |
|    | The flowchart symbol represents a stage in which manual input has been provided or is required by the user.                                                                  |
|   | In a flowchart, a rhombus represents a decision (a this-or-that or an if-then condition). It has the ability to divide the flow of your diagram into two or more directions. |
|  | This symbol represents the connection between multiple pages within the system.                                                                                              |
|  | This symbol shows a "or" where the user can select how to manipulate the data using the tools provided, such as a data slicer or a click.                                    |
|  | This symbol represents the summoning junction symbol, which is used to generate a single process from the system's multiple inputs.                                          |
|  | This symbol represents the display symbol, which is used to display the visualisation within the system.                                                                     |

### 4.3.3 CANCERVIVE Use Case Diagram

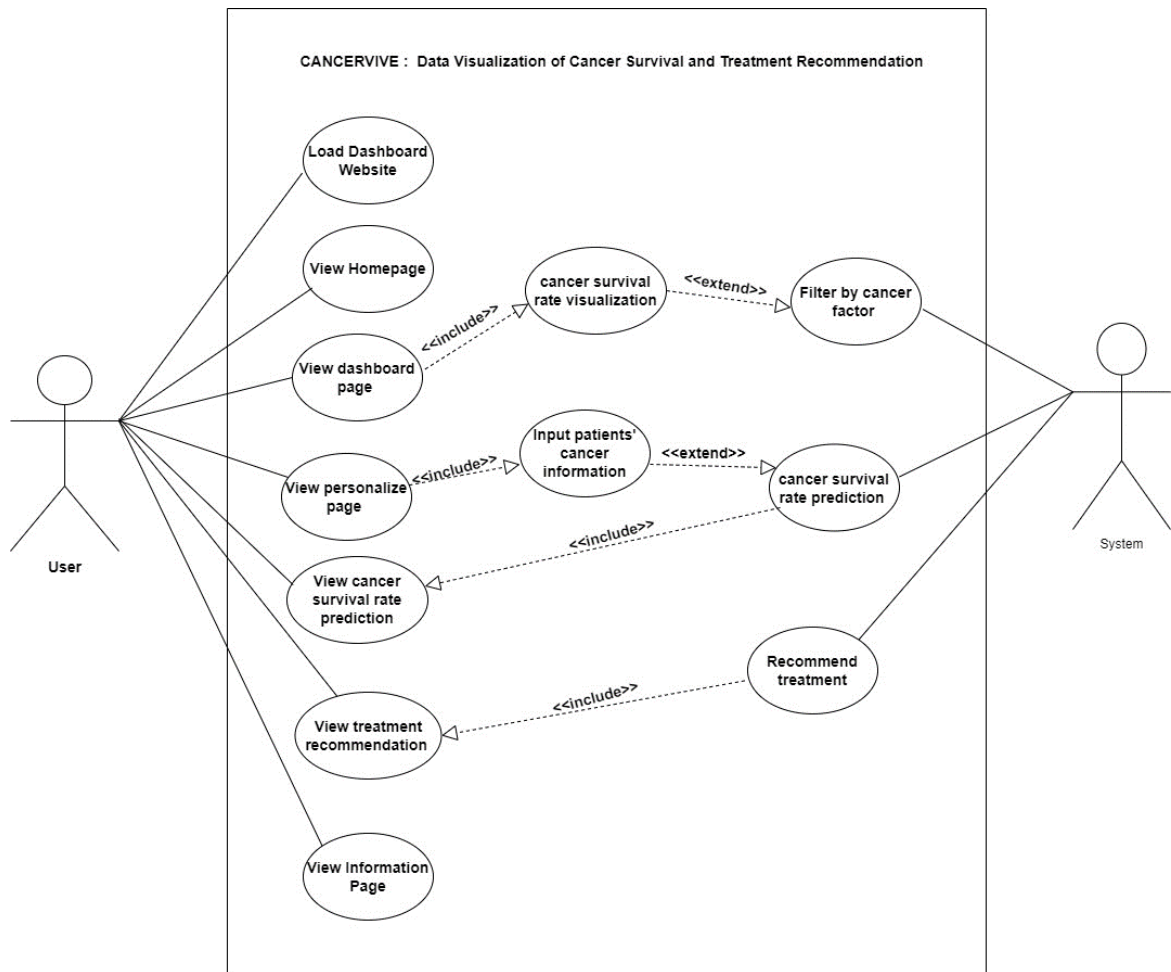


Figure 4.48: Use case Diagram

Table 4.6: use case description.

| Actor | Action                 | Description                                                                                                                                                                                                                                                                                  |
|-------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User  | Load Dashboard Website | The dashboard will load the website by obtaining the datasets from Excel before showing it to the user and enabling user interaction.                                                                                                                                                        |
|       | View Homepage          | At the start of the dashboard, users can view the homepage. On the homepage, users will see the welcome message and the CANCERVIVE logo, as well as three buttons labelled "Dashboard," "Personalise," and "Information," which, when clicked, will navigate them to their respective pages. |
|       | View                   | The user will be directed to the dashboard page if they                                                                                                                                                                                                                                      |

|        |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | Dashboard Page                  | <p>click the "Dashboard" button on the main page or in the navigation bar. This page displays a visualization of the cancer survival rate based on factors such as cancer type, stage, age, gender, and year since diagnosis.</p> <ul style="list-style-type: none"> <li>The user can use the dashboard's filter feature to narrow down the information displayed for the cancer survival rate.</li> </ul>                                                                                                                       |
|        | View personalizes page          | <p>The dashboard will direct to the personalize page when a user clicks the "Personalize" button on the home page or in the navigation bar. Users can predict the cancer survival rate based on user input on this page and recommend a treatment plan tailored to each individual based on that information.</p> <p>Extend:</p> <ul style="list-style-type: none"> <li>The user can enter patient information such as cancer type, stage, age, gender, and year since diagnosis to predict the cancer survival rate.</li> </ul> |
|        | View treatment recommendation   | The user can view the personalized treatment recommendation based on their input.                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|        | View Information Page           | The user will find information on the cancer survival rate and three types of cancer treatment, which are "Radiotherapy," "Chemotherapy," and "Tumour Resection Surgery" on this information page.                                                                                                                                                                                                                                                                                                                               |
| System | Filter by cancer factor         | The system will filter the cancer survival rate visualization based on the type of cancer, cancer stage, age, gender, and year since diagnosis.                                                                                                                                                                                                                                                                                                                                                                                  |
|        | Cancer survival rate prediction | The system will predict the patients' cancer survival rate based on user input. The system will also display the weight of each factor in the prediction.                                                                                                                                                                                                                                                                                                                                                                        |
|        | Recommend treatment plan        | The system will display treatment recommendations based on the user input that are suitable for their patients.                                                                                                                                                                                                                                                                                                                                                                                                                  |

Table 4.7: Load Dashboard Website use case

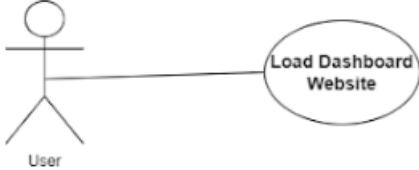
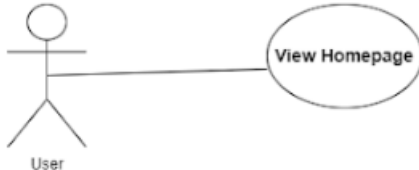
|                   |                                                                                                                                                                                                                     |                                                                                                                                                                                         |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name:    | Load Dashboard Website                                                                                                                                                                                              |                                                                                                                                                                                         |
| Actors:           | Users                                                                                                                                                                                                               |                                                                                                                                                                                         |
| Use Case Diagram: |                                                                                                                                   |                                                                                                                                                                                         |
| Preconditions:    | User open the dashboard                                                                                                                                                                                             |                                                                                                                                                                                         |
| Postconditions:   | Users can view the dashboard's homepage                                                                                                                                                                             |                                                                                                                                                                                         |
| Normal Flow:      | Actor                                                                                                                                                                                                               | System                                                                                                                                                                                  |
|                   | <ol style="list-style-type: none"> <li>1. The user opens the dashboard</li> <li>2. The user views the prompt and waits.</li> <li>3. Users can navigate the from dashboard</li> <li>4. This use case ends</li> </ol> | <ol style="list-style-type: none"> <li>1. The system will display a prompt indicating that the Excel file is being loaded to the dashboard</li> <li>2. The prompt disappears</li> </ol> |
| Exceptions:       | The error message will appear                                                                                                                                                                                       |                                                                                                                                                                                         |

Table 4.8: View homepage use case.

|                   |                                                                                                                                                                                      |                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name:    | View homepage                                                                                                                                                                        |                                                                                                                                                                                                                |
| Actors:           | User                                                                                                                                                                                 |                                                                                                                                                                                                                |
| Use Case Diagram: |                                                                                                  |                                                                                                                                                                                                                |
| Preconditions:    | The dashboard system successfully loaded the Excel file to the dashboard                                                                                                             |                                                                                                                                                                                                                |
| Postconditions:   | To proceed to the next page, the user needs to click one from three the navigation button at the homepage.                                                                           |                                                                                                                                                                                                                |
| Normal Flow:      | Actor                                                                                                                                                                                | System                                                                                                                                                                                                         |
|                   | <ol style="list-style-type: none"> <li>1. The user views the homepage.</li> <li>2. Users view the three buttons, which are “Dashboard”, “Personalize”, and “Information”.</li> </ol> | <ol style="list-style-type: none"> <li>1. System displays the homepage.</li> <li>2. The system displays the three buttons on the homepage, which are “Dashboard”, “Personalize”, and “Information”.</li> </ol> |

|             |      |
|-------------|------|
| Exceptions: | None |
|-------------|------|

Table 4.9: View dashboard use case.

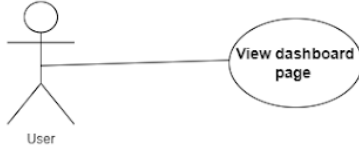
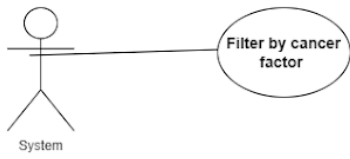
|                   |                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                            |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name:    | View dashboard Page                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                            |
| Actors:           | User                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                            |
| Use Case Diagram: |  <pre> graph LR     User((User)) --- UC((View dashboard page)) </pre>                                                                                                                                                                                                 |                                                                                                                                                                                            |
| Preconditions:    | The user clicks the “Dashboard” button on the homepage or navigation bar.                                                                                                                                                                                                                                                                               |                                                                                                                                                                                            |
| Postconditions:   | <ol style="list-style-type: none"> <li>1. The user can view the visualization of cancer survival rate.</li> <li>2. The user can filter the visualization of cancer survival rate based on user needs.</li> </ol>                                                                                                                                        |                                                                                                                                                                                            |
| Normal Flow:      | Actor                                                                                                                                                                                                                                                                                                                                                   | System                                                                                                                                                                                     |
|                   | <ol style="list-style-type: none"> <li>1. The user clicks the “Dashboard” button on the homepage.</li> <li>2. The user views the dashboard page.</li> <li>3. The use case will proceed to visualize the cancer survival rate and filter.</li> <li>4. To navigate to another page, the user clicks the "Personalise" or "Information" button.</li> </ol> | <ol style="list-style-type: none"> <li>1. The system displays the dashboard page.</li> <li>2. The system will display the visualization of the cancer survival rate and filter.</li> </ol> |
| Exceptions:       | None                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                            |

Table 4.10: View cancer survival rate visualization use case.

|                   |                                                                                                                                                                          |  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Use Case Name:    | View cancer survival rate visualization on the dashboard page                                                                                                            |  |
| Actors:           | User                                                                                                                                                                     |  |
| Use Case Diagram: |  <pre> graph LR     User((User)) --- UC((cancer survival rate visualization)) </pre> |  |
| Preconditions:    | The user clicks the “Dashboard” button on the homepage                                                                                                                   |  |
| Postconditions:   | <ol style="list-style-type: none"> <li>1. The user can filter the visualization of cancer survival rate based on user needs.</li> </ol>                                  |  |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                      |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | 2. The user can navigate to another page by clicking the "Personalise" or "Information" buttons.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                      |
| Normal Flow: | Actor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | System                                                                                                                                                                                                                                               |
|              | <ol style="list-style-type: none"> <li>1. The user clicks the "Dashboard" button on the homepage</li> <li>2. The user views the dashboard page.</li> <li>3. The user will view the visualization of the cancer survival rate.</li> <li>4. The user can filter the visualization by using the filter function.</li> <li>5. To visualize a personalized dashboard based on filter input, the user selects a single value of cancer factors (Cancer type, patient age, gender, stage, and years since diagnosis) from the filter function.</li> <li>6. To navigate to another page, the user clicks the "Personalise" or "Information" button.</li> <li>7. This use case ends</li> </ol> | <ol style="list-style-type: none"> <li>1. The system displays the dashboard page.</li> <li>2. The system will display the visualization of the cancer survival rate.</li> <li>3. The system visualizes the dashboard based on the filter.</li> </ol> |
| Exceptions:  | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                      |

Table 4.11: Filter Visualisation use case.

|                   |                                                                                                                                                                                                                                                                          |                                                                                                                                                                                     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name:    | Filter Visualisation                                                                                                                                                                                                                                                     |                                                                                                                                                                                     |
| Actors:           | System                                                                                                                                                                                                                                                                   |                                                                                                                                                                                     |
| Use Case Diagram: |  <pre> graph LR     Actor[System] --- UseCase((Filter by cancer factor)) </pre>                                                                                                      |                                                                                                                                                                                     |
| Preconditions:    | The user use the filter function on the dashboard page.                                                                                                                                                                                                                  |                                                                                                                                                                                     |
| Postconditions:   | User view filtered visualizations                                                                                                                                                                                                                                        |                                                                                                                                                                                     |
| Normal Flow:      | Actor                                                                                                                                                                                                                                                                    | System                                                                                                                                                                              |
|                   | <ol style="list-style-type: none"> <li>1. Select one value of the cancer factor (Cancer type, patient age, gender, stage, and years since diagnosis )in the dropdown dashboard filter</li> <li>2. View filtered visualizations</li> <li>3. This use case ends</li> </ol> | <ol style="list-style-type: none"> <li>1. Received input from the user</li> <li>2. Filtered data based on input</li> <li>3. Send the filtered data to the visualizations</li> </ol> |

|            |      |
|------------|------|
| Exception: | None |
|------------|------|

Table 4.12: View personalizes Page use case.

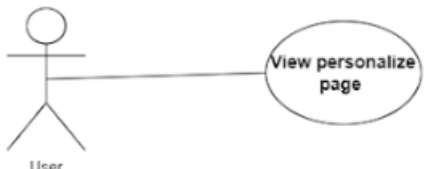
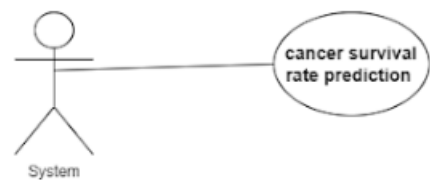
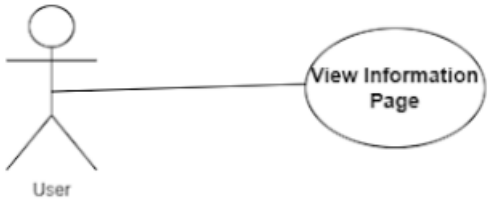
|                   |                                                                                                                                                                                                                                                                                                               |                                                                                                                                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Use Case Name:    | View personalizes Page                                                                                                                                                                                                                                                                                        |                                                                                                                                             |
| Actors:           | User                                                                                                                                                                                                                                                                                                          |                                                                                                                                             |
| Use Case Diagram: |  <pre> graph LR     User((User)) --- UC((View personalize page)) </pre>                                                                                                                                                     |                                                                                                                                             |
| Preconditions:    | The user clicks the “Personalize” button on the homepage or navigation bar.                                                                                                                                                                                                                                   |                                                                                                                                             |
| Postconditions:   | <ol style="list-style-type: none"> <li>1. The user can view the prediction webpage.</li> <li>2. The user can enter patient information to predict the cancer survival rate.</li> </ol>                                                                                                                        |                                                                                                                                             |
| Normal Flow:      | Actor                                                                                                                                                                                                                                                                                                         | System                                                                                                                                      |
|                   | <ol style="list-style-type: none"> <li>1. The user clicks the “Personalize” button on the homepage or navigation sidebar.</li> <li>2. The user views the prediction web page.</li> <li>3. The users input the patients’ information.</li> <li>4. The user clicks “Predict cancer survival” button.</li> </ol> | <ol style="list-style-type: none"> <li>1. The system displays the dashboard page.</li> <li>2. The system load prediction webpage</li> </ol> |
| Exceptions:       | <ol style="list-style-type: none"> <li>1. No internet connection</li> </ol>                                                                                                                                                                                                                                   |                                                                                                                                             |

Table 4.13: Cancer survival rate prediction use case.

|                   |                                                                                                                                                                           |                                                                                                |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Use Case Name:    | Cancer Survival Rate Prediction                                                                                                                                           |                                                                                                |
| Actors:           | System                                                                                                                                                                    |                                                                                                |
| Use Case Diagram: |  <pre> graph LR     System((System)) --- UC((cancer survival rate prediction)) </pre> |                                                                                                |
| Preconditions:    | The user clicks the “Personalize” button on the homepage or navigation bar and the prediction webpage load successfully.                                                  |                                                                                                |
| Postconditions:   | <ol style="list-style-type: none"> <li>1. User views the cancer survival rate of the patients and treatment recommendations.</li> </ol>                                   |                                                                                                |
| Normal Flow:      | Actor                                                                                                                                                                     | System                                                                                         |
|                   | <ol style="list-style-type: none"> <li>1. The user clicks the “Personalize” button on</li> </ol>                                                                          | <ol style="list-style-type: none"> <li>1. The system displays the personalize page.</li> </ol> |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <p>the homepage or navigation sidebar.</p> <ol style="list-style-type: none"> <li>The user views the prediction web page.</li> <li>The users input the patients' information.</li> <li>The user clicks the "Predict cancer survival" button.</li> <li>The user can see the cancer survival rate of the patients as well as the treatment plan for the cancer patient.</li> <li>To navigate to another page, the user clicks the "Dashboard" or "Information" button.</li> <li>The use case end</li> </ol> | <ol style="list-style-type: none"> <li>The system load prediction webpage</li> <li>The system received input from the user.</li> <li>The system predict the users' cancer survival rate and recommends the treatment.</li> <li>Shows the prediction result</li> </ol> |
| Exceptions: | 1) No internet connection                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                       |

Table 4.14: View Information Page use case.

|                   |                                                                                                                                                                       |                                           |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Use Case Name:    | View Information Page                                                                                                                                                 |                                           |
| Actors:           | User                                                                                                                                                                  |                                           |
| Use Case Diagram: |  <pre> graph LR     User((User)) --- ViewInfoPage([View Information Page]) </pre> |                                           |
| Preconditions:    | The user clicks the "Information" button on the homepage or navigation bar                                                                                            |                                           |
| Postconditions:   | The user views cancer survival rate and treatment recommendation.                                                                                                     |                                           |
| Normal Flow:      | Actor                                                                                                                                                                 | System                                    |
|                   | The user clicks the "Personalize" button on the homepage or navigation sidebar. The user views cancer survival rate and treatment information.                        | The system displays the information page. |



|             |                                                                                                                |  |
|-------------|----------------------------------------------------------------------------------------------------------------|--|
|             | To navigate to another page, the user clicks the "Dashboard" or "Personalize" button.<br><br>The use case end. |  |
| Exceptions: | none                                                                                                           |  |

#### 4.3.5 Data Design for CANCERVIVE: Data Visualisation of Cancer Survival and Treatment Recommendation.

##### A. Data Model.

This section describes the procedure following data scrubbing. Machine learning is used in my project to predict patients' cancer survival rates and recommend treatment based on patient information as a doctor's guideline. As mentioned in section 4.3.1, this project involves two datasets: cancer survival rate datasets and cancer treatment datasets. Both datasets contain previous cancer case records from the United Kingdom. Machine learning is used to predict cancer survival rates and to provide personalized patient care. This section will also explain the steps after scrubbing in the OSEMN framework, which include data exploration before to the actual model data.

##### a) Cancer Survival rate

##### I) Data Preparation.

| A              | B           | C      | D                   | E                | F                     | G        | H            |
|----------------|-------------|--------|---------------------|------------------|-----------------------|----------|--------------|
| Diagnosis Year | Cancer type | Gender | Stage               | Age At Diagnosis | Years Since Diagnosis | Patients | Survival (%) |
| 2015-2019      | Bladder     | Female | All stages combined | 15-44            |                       | 1        | 37           |
| 2015-2019      | Bladder     | Female | All stages combined | 45-54            |                       | 1        | 109          |
| 2015-2019      | Bladder     | Female | All stages combined | 55-64            |                       | 1        | 298          |
| 2015-2019      | Bladder     | Female | All stages combined | 65-74            |                       | 1        | 586          |
| 2015-2019      | Bladder     | Female | All stages combined | 75-99            |                       | 1        | 1229         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 1        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 1        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | 55-64            |                       | 2        | 298          |
| 2015-2019      | Bladder     | Female | All stages combined | 65-74            |                       | 2        | 586          |
| 2015-2019      | Bladder     | Female | All stages combined | 75-99            |                       | 2        | 1229         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 2        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 2        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | 55-64            |                       | 3        | 298          |
| 2015-2019      | Bladder     | Female | All stages combined | 65-74            |                       | 3        | 586          |
| 2015-2019      | Bladder     | Female | All stages combined | 75-99            |                       | 3        | 1229         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 3        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 3        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | 55-64            |                       | 4        | 298          |
| 2015-2019      | Bladder     | Female | All stages combined | 65-74            |                       | 4        | 586          |
| 2015-2019      | Bladder     | Female | All stages combined | 75-99            |                       | 4        | 1229         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 4        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 4        | 2259         |
| 2015-2019      | Bladder     | Female | All stages combined | 55-64            |                       | 5        | 298          |
| 2015-2019      | Bladder     | Female | All stages combined | 65-74            |                       | 5        | 586          |
| 2015-2019      | Bladder     | Female | All stages combined | 75-99            |                       | 5        | 1229         |
| 2015-2019      | Bladder     | Female | All stages combined | All ages         |                       | 5        | 2259         |

Figure 4.49: Cancer Survival Clean Data

Figure 4.49 shows the Cancer Survival clean data that has been scrubbed before. In

this section, the dataset is explored.

```
#Load the data file
cancerS_df = pd.read_csv('C:/Users/Admin/anaconda3/cancer_survivals.csv', low_memory=False)
```

Figure 4.50: Load Cancer Survival Clean Data

```
cancerS_df.head()
```

|      | Diagnosis Year | Cancer type | Gender | Stage               | Age At Diagnosis | Years Since Diagnosis | Patients | Survival (%) |
|------|----------------|-------------|--------|---------------------|------------------|-----------------------|----------|--------------|
| 1615 | 2015-2019      | Bladder     | Female | All stages combined | 15-44            | 1                     | 37.0     | 40.6         |
| 1616 | 2015-2019      | Bladder     | Female | All stages combined | 45-54            | 1                     | 109.0    | 62.3         |
| 1617 | 2015-2019      | Bladder     | Female | All stages combined | 55-64            | 1                     | 298.0    | 70.0         |
| 1618 | 2015-2019      | Bladder     | Female | All stages combined | 65-74            | 1                     | 586.0    | 61.8         |
| 1619 | 2015-2019      | Bladder     | Female | All stages combined | 75-99            | 1                     | 1229.0   | 50.9         |

```
cancerS_df.tail()
```

|       | Diagnosis Year | Cancer type | Gender | Stage               | Age At Diagnosis | Years Since Diagnosis | Patients | Survival (%) |
|-------|----------------|-------------|--------|---------------------|------------------|-----------------------|----------|--------------|
| 19405 | 2015-2019      | Oesophagus  | Male   | All stages combined | 55-64            | 5                     | 5362.0   | 21.2         |
| 19406 | 2015-2019      | Oesophagus  | Male   | All stages combined | 65-74            | 5                     | 9405.0   | 20.2         |
| 19407 | 2015-2019      | Oesophagus  | Male   | All stages combined | 75-99            | 5                     | 9618.0   | 9.9          |
| 19408 | 2015-2019      | Oesophagus  | Male   | All stages combined | All ages         | 5                     | 26592.0  | 16.9         |
| 19409 | 2015-2019      | Oesophagus  | Male   | All stages combined | All ages         | 5                     | 26592.0  | 16.9         |

Figure 4.51: cancer survival rate data

In figure 4.50 shows the cancer survival data is loaded in the jupyter notebook. Figure 4.51 shows the first five and last five data in cancerS\_df dataframe.

```
cancerS_df.shape
```

```
(2665, 8)
```

Figure 4.52: cancer survival rate data shape

```
cancerS_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2665 entries, 1615 to 19409
Data columns (total 8 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Diagnosis Year       2665 non-null   object
1   Cancer type          2665 non-null   object
2   Gender               2665 non-null   object
3   Stage                2665 non-null   object
4   Age At Diagnosis     2665 non-null   object
5   Years Since Diagnosis 2665 non-null   int64
6   Patients             2665 non-null   float64
7   Survival (%)         2665 non-null   float64
dtypes: float64(2), int64(1), object(5)
memory usage: 187.4+ KB
```

Figure 4.53:check cancerS\_df dataframe info

```
#describe the dataframe with showing the count, mean, standard deviation, min, percentiles, and max.
cancerS_df.describe()
```

|       | Years Since Diagnosis | Patients      | Survival (%) |
|-------|-----------------------|---------------|--------------|
| count | 2665.000000           | 2665.000000   | 2665.000000  |
| mean  | 2.798874              | 8189.351595   | 46.502251    |
| std   | 1.390146              | 18874.882560  | 27.118002    |
| min   | 1.000000              | 12.000000     | 1.700000     |
| 25%   | 2.000000              | 630.000000    | 23.600000    |
| 50%   | 3.000000              | 2259.000000   | 41.600000    |
| 75%   | 4.000000              | 7515.000000   | 66.600000    |
| max   | 5.000000              | 214123.000000 | 100.100000   |

Figure 4.54: describe cancerS\_df dataframe

Figure 4.52, 4.53 and Figure 4.54 describes the content of the cancerS\_df dataframe before cleaning process.

## II) Clean Data

```
cancerS_df.isnull().sum()
```

```
Diagnosis Year      0
Cancer type        0
Gender              0
Stage              0
Age At Diagnosis    0
Years Since Diagnosis 0
Patients            0
Survival (%)        0
dtype: int64
```

Figure 4.55: check Null values.

```
duplicates = cancerS_df.duplicated()
print(duplicates.sum())
```

```
10
```

Figure 4.56: check duplicate value in cancerS\_df dataframe

```
# drop duplicates
cancerS_df.drop_duplicates(inplace=True)
```

```
duplicates = cancerS_df.duplicated()
print(duplicates.sum())
```

```
0
```

Figure 4.57: remove duplicate value

In figure 4.55, null values are checked in the dataframe; the results show that there are no null values in the dataframe. The duplicate values are then checked, as shown in figure 4.56, and the results show that there are 10 duplicate data in the dataframe. The duplicate data is removed in figure 4.57.

### III) Explore Data

```
cancerS_df['Diagnosis Year'].unique()
array(['2015-2019'], dtype=object)

cancerS_df['Cancer type'].unique()
array(['Bladder', 'Breast', 'Liver', 'Lung', 'Oesophagus'], dtype=object)

cancerS_df['Gender'].unique()
array(['Female', 'Male'], dtype=object)

cancerS_df['Stage'].unique()
array(['All stages combined', '1', '2', '3', '4'], dtype=object)

cancerS_df['Age At Diagnosis'].unique()
array(['15-44', '45-54', '55-64', '65-74', '75-99', 'All ages'],
      dtype=object)

cancerS_df['Years Since Diagnosis'].unique()
array([1, 2, 3, 4, 5], dtype=int64)

cancerS_df['Survival (%)'].unique()
array([ 40.6,  62.3,  70. ,  61.8,  50.9,  56.6,  58.4,  56.9,  51.1,
        37. ,  43.8,  46.6,  52.8,  46.9,  32.6,  39.5,  42.3,  50.1,
        27.3,  35.2,  38.1,  47.3,  41.6,  25.5,  33.2,  71.9,  59.2,
        70.3,  66.9,  47.2,  55.8,  60.6,  58. ,  52.2,  60.4,  37.1,
        45.8,  50.4,  57.6,  52.3,  32.1,  41.5,  46.5,  55.9,  30.2,
        44.6,  53.8,  49.2,  27.1,  65.3,  70.2,  68.1,  71.2,  52.7,
        59.6,  64.1,  61. ,  53.7,  57.1,  61.4,  49.1,  53.3,  51.5,
        57.7,  37.2,  44.2,  48.1,  41.8,  53. ,  33. ,  40.3,  68.5,
        71.6,  73.6,  73.9,  52.6,  66.3,  65.9,  62. ,  42. ,  49.8,
        56.2,  62.4,  37.9,  45.4,  52.1,  54.4,  35.4,  43.2,  54.3,
        33.9,  81.3,  83. ,  81.5,  74.9,  64.3,  71.8,  65.2,  44.3,
        52.5,  69.4,  38.3,  68.8,  36. ,  44.1,  50.8,  66.1,  34.3,
        42.1,  95.7,  97.4,  95.5,  94. ,  87. ,  90.3,  92.8,  92.1,
        88.9,  78.7,  83.7,  87.5,  89.8,  84.7,  73.1,  79.1,  82.7,
        68.7,  75.7,  86.8,  80.5,  66. ,  73.5,  60.9,  81.7,  78. ,
        70.4,  56. ,  62.7,  68. ,  51.9,  66.7,  55.1,  39.1,  46.3,
        52. ,  56.4,  48.8,  32.3,  40.1,  47. ,  54.1,  45.2,  29.1,
        36.8,  43.4,  50.3,  44.5,  27. ,  34.8,  64.9,  41. ,  56.5,
        54. ,  46.8,  28.2,  41.7,  45.6,  42.8,  24.1,  34.5,  36.9,
        40. ,  21.3,  32.4,  37.3,  20.2,  30.7,  35.6,  30.3,  34.7,
        18.5,  24.7,  27.4,  23.2,  17.8,   8.6,  13.6,  15.2,  20.8,
        12.2,   6.1,  10. ,  19.7,   9.8,   5.6,   8.9,  16.4,   7.6,
```

Figure 4.58: check unique value in cancer survival dataframe

The unique value of each column in the cancer survival dataframe is then displayed in figure 4.58. The results show that the value in the dataframe is consistent and that no duplicate data exists. Next, in the figure below, the outliers of the dataframe is check.

We will use box plot to determine if there is an outlier in the dataset.

```
|: import plotly.express as px
#create box plot for 'Survival (%)' column
fig = px.box(cancerS_df, y="Survival (%)")
fig.show()
```

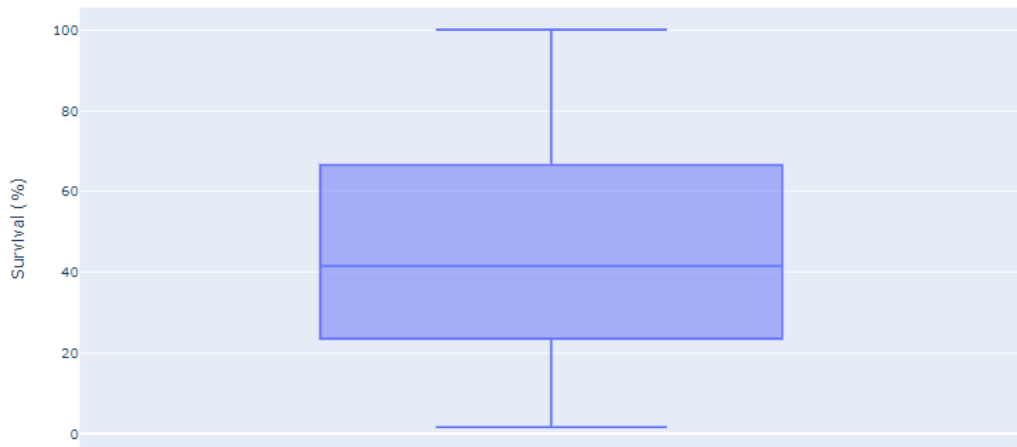


Figure 4.59: outlier box plot.

There seems to be no outliers in the dataset. However, we will try to find the outliers using statistical method. Statistical method can give a clear output of the outlier rather than the coding from the above. The figure below shows the steps to find outliers using statistical method.

```
#finding outliers using statistical method and define Q1 and Q3
def find_outliers_IQR(cancerS_df):
    q1=cancerS_df.quantile(0.25)
    q3=cancerS_df.quantile(0.75)
    IQR=q3-q1
    outliers=cancerS_df[((cancerS_df<(q1-1.5*IQR))|(cancerS_df>(q3+1.5*IQR)))]
    return outliers
```

```
#display the number of outliers for 'product_id' column
outliers=find_outliers_IQR(cancerS_df["Survival (%)"])
print("number of outliers: " + str(len(outliers)))
print("max outlier value: " + str(outliers.max()))
print("min outlier value: " + str(outliers.min()))
outliers
```

```
number of outliers: 0
max outlier value: nan
min outlier value: nan
```

Figure 4.60: finding outliers using statistical method.

statistical methods called interquartile range (IQR) is used to determine the outliers.

Using IQR, we can determine the outlier if the data falls below  $Q1-1.5$  or  $Q3+1.5$ .

- $Q1$ : 25th percentile

- $Q3$ : 75th percentile

The formula to calculate the IQR is:  $Q3-Q1$

As seen in figure 4.60, by using statistical method, we have identified that there is no outlier in the dataset.

From the output, the max outlier value and min outlier value both return NaN, NaN here indicates that the value is not an outlier.

#### IV) Model Data (Machine Learning)

```
Cancer_type = {'Bladder': 0, 'Breast': 1, 'Liver': 2, 'Lung':3, 'Oesophagus':4}
cancerS_df['Cancer type'] = cancerS_df['Cancer type'].map(Cancer_type)

Gender = {'Persons': 0, 'Female': 1, 'Male':2}
cancerS_df['Gender'] = cancerS_df['Gender'].map(Gender)

Stage = {'All stages combined': 0, '1':1, '2':2, '3':3, '4':4}
cancerS_df['Stage'] = cancerS_df['Stage'].map(Stage)

#Age
Age = {'All ages': 0, '15-44':1, '45-54':2, '55-64':3, '65-74':4, '75-99':5}
cancerS_df['Age At Diagnosis'] = cancerS_df['Age At Diagnosis'].map(Age)
```

Figure 4.61: encode the data

The data in Figure 4.61 is encoded into numerical form because machine learning algorithms can only read and process numerical data as input.

```
features = ['Cancer type', 'Gender', 'Stage', 'Age At Diagnosis', 'Years Since Diagnosis' ]
X = cancerS_df[features]
y = cancerS_df['Survival (%)']
```

Figure 4.62: Data preparation for machine learning

```
print(X)
```

|       | Cancer type | Gender | Stage | Age At Diagnosis | Years Since Diagnosis |
|-------|-------------|--------|-------|------------------|-----------------------|
| 1615  | 0           | 1      | 0     | 1                | 1                     |
| 1616  | 0           | 1      | 0     | 2                | 1                     |
| 1617  | 0           | 1      | 0     | 3                | 1                     |
| 1618  | 0           | 1      | 0     | 4                | 1                     |
| 1619  | 0           | 1      | 0     | 5                | 1                     |
| ...   | ...         | ...    | ...   | ...              | ...                   |
| 19405 | 4           | 2      | 0     | 3                | 5                     |
| 19406 | 4           | 2      | 0     | 4                | 5                     |
| 19407 | 4           | 2      | 0     | 5                | 5                     |
| 19408 | 4           | 2      | 0     | 0                | 5                     |
| 19409 | 4           | 2      | 0     | 0                | 5                     |

```
[2665 rows x 5 columns]
```

```
print(y)
```

|       |      |
|-------|------|
| 1615  | 40.6 |
| 1616  | 62.3 |
| 1617  | 70.0 |
| 1618  | 61.8 |
| 1619  | 50.9 |
| ...   | ...  |
| 19405 | 21.2 |
| 19406 | 20.2 |
| 19407 | 9.9  |
| 19408 | 16.9 |
| 19409 | 16.9 |

```
Name: Survival (%), Length: 2665, dtype: float64
```

Figure 4.63: print X and Y data

The data preparation process for machine learning is shown in Figure 4.62. The dataset has been split down into two data structures, X and Y. The input features are contained in DataFrame X, while the target variable will be predicted in DataFrame Y.

```
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print( X.shape, X_train.shape, X_test.shape)

(2665, 5) (2132, 5) (533, 5)
```

Figure 4.64: train data

The cancerS\_df datarame is divided into training and testing sets in the figure above. This data separation into training and testing sets enables us to effectively train and evaluate our machine learning model, ensuring its predictability on new, unseen data.

```
from sklearn.metrics import classification_report, confusion_matrix

def evaluate_model(model, train_features, y_train, test_features, y_test):
    """
    Function: train and evaluate a machine learning classifier
    Args:
        model: machine learning classifier
        train_features: training data extracted features
        y_train: training data labels
        test_features: test data extracted features
        y_test: test data labels
    Returns:
        results (dictionary): a dictionary of classification report and confusion matrix
    """
    results = {}

    # Train the model
    model.fit(train_features, y_train)

    # Test the model
    train_predicted = model.predict(train_features)
    test_predicted = model.predict(test_features)

    # Classification report and Confusion Matrix
    results['classification_report'] = classification_report(y_test, test_predicted)
    results['confusion_matrix'] = confusion_matrix(y_test, test_predicted)

    return results
```

```
from sklearn.metrics import mean_squared_error

# Initialize the models
lr = LinearRegression()
rf = RandomForestRegressor(random_state=0)
dtree = DecisionTreeRegressor(random_state=1)

# Fit and evaluate models
results = {}
for model in [lr, rf, dtree]:
    model_name = model.__class__.__name__
    model.fit(X_train, y_train)
    train_predictions = model.predict(X_train)
    test_predictions = model.predict(X_test)

    train_mse = mean_squared_error(y_train, train_predictions)
    test_mse = mean_squared_error(y_test, test_predictions)
    overall_error_test = np.sqrt(mean_squared_error(y_test, test_predictions))
    overall_error_train = np.sqrt(mean_squared_error(y_train, train_predictions))

    results[model_name] = {'Train MSE': train_mse, 'Test MSE': test_mse, 'Overall Error (Train)':
                           overall_error_train, 'Overall Error (Test)': overall_error_test}
```

```

# Print regression metrics and Coefficient of Variation (CV)
for result in results:
    print(result)
    print()
    for metric in results[result]:
        print(metric + ":")
        print(results[result][metric])
        print()

# Calculate and print the Coefficient of Variation (CV)
y_test_mean = np.mean(y_test)
y_test_std = np.std(y_test)
test_predictions = model.predict(X_test)
test_error = np.mean(np.abs(y_test - test_predictions))
cv = (y_test_std / y_test_mean) * 100
print(f"CV: {cv:.2f}%")

print("-" * 20)
print()

```

Figure 4.65: Find the best machine learning model.

In figure 4.65, shown the process to find the best model for predict the cancer survival rate. It shows that three model were initialize which are Linear Regressor, random forest regressor and decision tree regressor. The result are shown below.

```

LinearRegression

Train MSE:
488.6401449638814

Test MSE:
481.0420621207201

Overall Error (Train):
22.105206286390576

Overall Error (Test):
21.932671112309148

CV: 57.94%
-----

RandomForestRegressor

Train MSE:
7.901119550631744

Test MSE:
25.024118201861643

Overall Error (Train):
2.810893016575292

Overall Error (Test):
5.00241123877892

CV: 57.94%
-----

DecisionTreeRegressor

Train MSE:
6.500893626539813

Test MSE:
32.972980599789935

Overall Error (Train):
2.5496850053565074

Overall Error (Test):
5.742210428031172

CV: 57.94%
-----

```

Figure 4.66: Report Results



The value of the evaluation metrics shown above includes Mean Squared Error (MSE) on training and testing data, the square root of MSE (Overall Error) on both training and testing data, and R-squared values for training and testing data, which are used to find the best model for predicting cancer survival rate. The correlation variation of the three models is 57.94% based on the results, but in this project, we chose the Random forest regressor because it has the lowest overall error (both train and test) and the highest R-squared value for the test set, indicating better performance and a stronger ability to generalise to new data compared to the other models. The random forest regressor is the most accurate model for predicting cancer survival rate.

## b) Cancer Treatment Dataset

### I) Data Preparation.

| Cancer type | Year of Diagnosis | Factor of interest      | Levels of factor | Chemotherapy | Radiotherapy | Tumour Resection | Number of tumours treated |
|-------------|-------------------|-------------------------|------------------|--------------|--------------|------------------|---------------------------|
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | No           | No               | 90                        |
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | No           | Yes              | 179                       |
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | Yes          | No               | 14                        |
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | Yes          | Yes              | 4                         |
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | Yes          | No           | No               | 48                        |
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | Yes          | No           | Yes              | 188                       |
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | Yes          | Yes          | No               | 56                        |
| Bladder     | 2013              | Age at cancer diagnosis | 50-59            | Yes          | Yes          | Yes              | 19                        |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | No           | No           | No               | 72                        |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | No           | No           | Yes              | 185                       |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | No           | Yes          | No               | 17                        |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | No           | Yes          | Yes              | 4                         |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | Yes          | No           | No               | 25                        |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | Yes          | No           | Yes              | 192                       |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | Yes          | Yes          | No               | 49                        |
| Bladder     | 2014              | Age at cancer diagnosis | 50-59            | Yes          | Yes          | Yes              | 18                        |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | No           | No           | No               | 62                        |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | No           | No           | Yes              | 168                       |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | No           | Yes          | No               | 20                        |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | No           | Yes          | Yes              | 8                         |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | Yes          | No           | No               | 38                        |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | Yes          | No           | Yes              | 180                       |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | Yes          | Yes          | No               | 49                        |
| Bladder     | 2015              | Age at cancer diagnosis | 50-59            | Yes          | Yes          | Yes              | 10                        |
| Bladder     | 2016              | Age at cancer diagnosis | 50-59            | No           | No           | No               | 72                        |
| Bladder     | 2016              | Age at cancer diagnosis | 50-59            | No           | No           | Yes              | 165                       |

Figure 4.67: Cancer Treatment clean dataset

Figure 4.67 shows the Cancer treatment clean data that has been scrubbed before. In this section, the dataset is explored.

```
#read file without outliers
cancerT_df = pd.read_csv('clean_treatment.csv')
```

Figure 4.68: Load Treatment dataset in jupyter notebook

```
cancerT_df.head()
```

|     | Cancer type | Year of Diagnosis | Factor of interest      | Levels of factor | Chemotherapy | Radiotherapy | Tumour Resection | Number of tumours treated |
|-----|-------------|-------------------|-------------------------|------------------|--------------|--------------|------------------|---------------------------|
| 759 | Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | No           | No               | 90                        |
| 760 | Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | No           | Yes              | 179                       |
| 761 | Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | Yes          | No               | 14                        |
| 762 | Bladder     | 2013              | Age at cancer diagnosis | 50-59            | No           | Yes          | Yes              | 4                         |
| 763 | Bladder     | 2013              | Age at cancer diagnosis | 50-59            | Yes          | No           | No               | 48                        |

Figure 4.69: cancerT\_df dataframe

In Figure 4.68, the cancer treatment data is loaded into the Jupyter Notebook and stored in the 'cancerT\_df' DataFrame. Figure 4.69 displays the first five rows of the 'cancerT\_df' DataFrame.

```
cancerT_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 2974 entries, 759 to 30268
Data columns (total 8 columns):
#   Column                                Non-Null Count  Dtype
---  ---                                ---
0   Cancer type                          2974 non-null   object
1   Year of Diagnosis                    2974 non-null   object
2   Factor of interest                   2974 non-null   object
3   Levels of factor                     2974 non-null   object
4   Chemotherapy                        2974 non-null   object
5   Radiotherapy                        2974 non-null   object
6   Tumour Resection                    2974 non-null   object
7   Number of tumours treated            2974 non-null   int64
dtypes: int64(1), object(7)
memory usage: 209.1+ KB
```

Figure 4.70: cancerT\_df info

```
cancerT_df.describe()
```

| Number of tumours treated |              |
|---------------------------|--------------|
| count                     | 2974.000000  |
| mean                      | 482.225286   |
| std                       | 1329.090752  |
| min                       | 1.000000     |
| 25%                       | 20.000000    |
| 50%                       | 99.000000    |
| 75%                       | 379.750000   |
| max                       | 16002.000000 |

Figure 4.71: describe statistics value of cancerT\_df

Figure 4.70 and Figure 4.71 describes the content of the cancerT\_df dataframe before cleaning process.

## II) Data Cleaning

```
cancerT_df.isnull().sum()
```

```
Cancer type          0
Year of Diagnosis     0
Factor of interest    0
Levels of factor      0
Chemotherapy          0
Radiotherapy          0
Tumour Resection      0
Number of tumours treated  0
dtype: int64
```

Figure 4.72: check cancerT\_df null value

```
# check for duplicates
print(cancerT_df.duplicated())
```

```
759    False
760    False
761    False
762    False
763    False
...
30264   False
30265   False
30266   False
30267   False
30268   False
Length: 2974, dtype: bool
```

```
duplicates = cancerT_df.duplicated()
print(duplicates.sum())
```

```
0
```

Figure 4.73: check cancerT\_df duplicate value

In figure 4.73, null values are checked in the dataframe; the results show that there are no null values in the dataframe. The duplicate values are then checked, as shown in figure 4.56, and the results show there is no duplicate data in cancerT\_df dataframe.

## III) Data Explore

```
print(cancerT_df.columns.values)

['Cancer type' 'Year of Diagnosis' 'Factor of interest' 'Levels of factor'
 'Chemotherapy' 'Radiotherapy' 'Tumour Resection'
 'Number of tumours treated']

#get all unique values in 'Cancer type' column
cancerT_df['Cancer type'].unique()

array(['Bladder', 'Breast', 'Liver', 'Oesophagus', 'Lung'], dtype=object)

cancerT_df['Year of Diagnosis'].unique()

array(['2013', '2014', '2015', '2016', '2017', '2018', '2019'],
      dtype=object)

cancerT_df['Factor of interest'].unique()

array(['Age at cancer diagnosis', 'Gender', 'Stage at cancer diagnosis'],
      dtype=object)

cancerT_df['Levels of factor'].unique()

array(['50-59', '60-69', '70-79', '80+', '0-49', 'female', 'male',
      'Stage 1', 'Stage 2', 'Stage 3', 'Stage 4'], dtype=object)
```

Figure 4.74: check unique value in cancerT\_df dataframe

Figure 4.74 shows the unique values of each column in the cancer treatment DataFrame. The results show that the values in the DataFrame are consistent, and there is no duplicate data. However, one issue arises when the data in the 'Factor of Interest' column is mixed. The data contains information about the cancer stage, age, and gender, all influencing treatment outcomes in the dataframe. To solve this problem, the dataframe is divided into three files according to its factor of interest. Figure below show the split data process. This also will help in the treatment recommendation system.

## Split data based on factor of interest

```
# Create separate columns for each factor of interest
factorAge_df = cancerT_df[cancerT_df['Factor of interest'] == 'Age at cancer diagnosis']
factorGender_df = cancerT_df[cancerT_df['Factor of interest'] == 'Gender']
factorStage_df = cancerT_df[cancerT_df['Factor of interest'] == 'Stage at cancer diagnosis']
```

Figure 4.75: split dataset

| factorAge_df |             |              |                         |                  |                           |              |                  |                   |
|--------------|-------------|--------------|-------------------------|------------------|---------------------------|--------------|------------------|-------------------|
|              | Cancer type | Chemotherapy | Factor of interest      | Levels of factor | Number of tumours treated | Radiotherapy | Tumour Resection | Year of Diagnosis |
| 0            | Bladder     | 0            | Age at cancer diagnosis | 50-59            | 90                        | 0            | 0                | 2013              |
| 1            | Bladder     | 0            | Age at cancer diagnosis | 50-59            | 179                       | 0            | 1                | 2013              |
| 2            | Bladder     | 0            | Age at cancer diagnosis | 50-59            | 14                        | 1            | 0                | 2013              |
| 3            | Bladder     | 0            | Age at cancer diagnosis | 50-59            | 4                         | 1            | 1                | 2013              |
| 4            | Bladder     | 1            | Age at cancer diagnosis | 50-59            | 48                        | 0            | 0                | 2013              |
| ...          | ...         | ...          | ...                     | ...              | ...                       | ...          | ...              | ...               |
| 2851         | Lung        | 0            | Age at cancer diagnosis | U50              | 1                         | 1            | 0                | 2019              |
| 2852         | Lung        | 1            | Age at cancer diagnosis | U50              | 25                        | 0            | 0                | 2019              |
| 2853         | Lung        | 1            | Age at cancer diagnosis | U50              | 1                         | 0            | 1                | 2019              |
| 2854         | Lung        | 1            | Age at cancer diagnosis | U50              | 42                        | 1            | 0                | 2019              |
| 2855         | Lung        | 1            | Age at cancer diagnosis | U50              | 2                         | 1            | 1                | 2019              |

1334 rows × 8 columns

```
print(factorAge_df['Levels of factor'].unique())
['50-59' '60-69' '70-79' '80+' '0-49']
```

Figure 4.76: factor of interest age dataframe.

In figure 4.76 above shown the data in factorAge\_df dataframe.

factorGender\_df

|      | Cancer type | Chemotherapy | Factor of interest | Levels of factor | Number of tumours treated | Radiotherapy | Tumour Resection | Year of Diagnosis |
|------|-------------|--------------|--------------------|------------------|---------------------------|--------------|------------------|-------------------|
| 278  | Bladder     | 0            | Gender             | female           | 840                       | 0            | 0                | 2013              |
| 279  | Bladder     | 0            | Gender             | female           | 593                       | 0            | 1                | 2013              |
| 280  | Bladder     | 0            | Gender             | female           | 271                       | 1            | 0                | 2013              |
| 281  | Bladder     | 0            | Gender             | female           | 45                        | 1            | 1                | 2013              |
| 282  | Bladder     | 1            | Gender             | female           | 129                       | 0            | 0                | 2013              |
| ...  | ...         | ...          | ...                | ...              | ...                       | ...          | ...              | ...               |
| 2955 | Lung        | 0            | Gender             | male             | 73                        | 1            | 0                | 2019              |
| 2956 | Lung        | 1            | Gender             | male             | 631                       | 0            | 0                | 2019              |
| 2957 | Lung        | 1            | Gender             | male             | 20                        | 0            | 1                | 2019              |
| 2958 | Lung        | 1            | Gender             | male             | 577                       | 1            | 0                | 2019              |
| 2959 | Lung        | 1            | Gender             | male             | 7                         | 1            | 1                | 2019              |

Figure 4.77: factor of interest Gender dataframe.

```
print(factorGender_df['Levels of factor'].unique())
['female' 'male']

factorGender_df['Levels of factor'].replace({'female': 'Female', 'male': 'Male'}, inplace=True)

print(factorGender_df['Levels of factor'].unique())
['Female' 'Male']

factorGender_df
```

|      | Cancer type | Chemotherapy | Levels of factor | Number of tumours treated | Radiotherapy | Tumour Resection | Year of Diagnosis |
|------|-------------|--------------|------------------|---------------------------|--------------|------------------|-------------------|
| 278  | Bladder     | 0            | Female           | 840                       | 0            | 0                | 2013              |
| 279  | Bladder     | 0            | Female           | 593                       | 0            | 1                | 2013              |
| 280  | Bladder     | 0            | Female           | 271                       | 1            | 0                | 2013              |
| 281  | Bladder     | 0            | Female           | 45                        | 1            | 1                | 2013              |
| 282  | Bladder     | 1            | Female           | 129                       | 0            | 0                | 2013              |
| ...  | ...         | ...          | ...              | ...                       | ...          | ...              | ...               |
| 2955 | Lung        | 0            | Male             | 73                        | 1            | 0                | 2019              |
| 2956 | Lung        | 1            | Male             | 631                       | 0            | 0                | 2019              |
| 2957 | Lung        | 1            | Male             | 20                        | 0            | 1                | 2019              |
| 2958 | Lung        | 1            | Male             | 577                       | 1            | 0                | 2019              |
| 2959 | Lung        | 1            | Male             | 7                         | 1            | 1                | 2019              |

533 rows × 7 columns

Figure 4.78: factorGender\_df change the value spelling

In figure 4.77 shows factorGender\_df data after splitting process. To make the data of gender is consistent with cancer survival rate data, the gender data spelling is change. This to ensure there is no problem in implementing system in developin process.

```
factorStage_df
```

|     | Cancer type | Chemotherapy | Factor of interest        | Levels of factor | Number of tumours treated | Radiotherapy | Tumour Resection | Year of Diagnosis |
|-----|-------------|--------------|---------------------------|------------------|---------------------------|--------------|------------------|-------------------|
| 390 | Bladder     | 0            | Stage at cancer diagnosis | Stage 1          | 75                        | 0            | 0                | 2013              |
| 391 | Bladder     | 0            | Stage at cancer diagnosis | Stage 1          | 1585                      | 0            | 1                | 2013              |
| 392 | Bladder     | 0            | Stage at cancer diagnosis | Stage 1          | 11                        | 1            | 0                | 2013              |
| 393 | Bladder     | 0            | Stage at cancer diagnosis | Stage 1          | 85                        | 1            | 1                | 2013              |
| 394 | Bladder     | 1            | Stage at cancer diagnosis | Stage 1          | 27                        | 0            | 0                | 2013              |

```
print(factorStage_df['Levels of factor'].unique())
['Stage 1' 'Stage 2' 'Stage 3' 'Stage 4']

factorStage_df['Levels of factor'] = factorStage_df['Levels of factor'].str.extract(r'(\d+)')

print(factorStage_df['Levels of factor'].unique())
['1' '2' '3' '4']
```

Figure 4.79: factor of interest Stage dataframe.

#### IV) Model Data (Machine learning)

```
# Encoding Categorical Features
Cancer_type = {'Bladder': 0, 'Breast': 1, 'Liver': 2, 'Lung': 3, 'Oesophagus': 4}
Stage = {'All stages combined': 0, '1': 1, '2': 2, '3': 3, '4': 4}

factorStage_df['Cancer type'] = factorStage_df['Cancer type'].map(Cancer_type)
factorStage_df['Levels of factor'] = factorStage_df['Levels of factor'].map(Stage)
```

Figure 4.80: machine learning data encoded

The data in Figure 4.80 is encoded into numerical form because machine learning algorithms can only read and process numerical data as input.

```
# Define the features and target variables
features = ['Cancer type', 'Levels of factor']
X = factorStage_df[features]
y = factorStage_df[['Tumour Resection', 'Chemotherapy', 'Radiotherapy']]
```

Figure 4.81: Data preparation for machine learning cancer treatment data

```
print(X)
```

|      | Cancer type | Levels of factor |
|------|-------------|------------------|
| 390  | 0           | 1                |
| 391  | 0           | 1                |
| 392  | 0           | 1                |
| 393  | 0           | 1                |
| 394  | 0           | 1                |
| ...  | ...         | ...              |
| 3158 | 3           | 4                |
| 3159 | 3           | 4                |
| 3160 | 3           | 4                |
| 3161 | 3           | 4                |
| 3162 | 3           | 4                |

[1075 rows x 2 columns]

```
print(y)
```

|      | Tumour Resection | Chemotherapy | Radiotherapy |
|------|------------------|--------------|--------------|
| 390  | 0                | 0            | 0            |
| 391  | 1                | 0            | 0            |
| 392  | 0                | 0            | 1            |
| 393  | 1                | 0            | 1            |
| 394  | 0                | 1            | 0            |
| ...  | ...              | ...          | ...          |
| 3158 | 0                | 0            | 1            |
| 3159 | 0                | 1            | 0            |
| 3160 | 1                | 1            | 0            |
| 3161 | 0                | 1            | 1            |
| 3162 | 1                | 1            | 1            |

[1075 rows x 3 columns]

Figure 4.82: print X and Y data

The data preparation process for machine learning is shown in Figure 4.82. The dataset has been split down into two data structures, X and Y. The input features are contained in DataFrame X, while the target variable will be predicted in DataFrame Y.

```
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1)
```

```
print( X.shape, X_train.shape, X_test.shape)
```

(1075, 2) (860, 2) (215, 2)

Figure 4.83: train treatment data

The treatment dataframe is divided into training and testing sets in the figure above. This data separation into training and testing sets enables us to effectively train and evaluate our machine learning model, ensuring its predictability on new, unseen data.

```

from sklearn.metrics import classification_report, confusion_matrix

def evaluate_model(model, train_features, y_train, test_features, y_test):
    """
    Function: train and evaluate a machine learning classifier
    Args:
        model: machine learning classifier
        train_features: training data extracted features
        y_train: training data labels
        test_features: test data extracted features
        y_test: test data labels
    Returns:
        results (dictionary): a dictionary of classification report and confusion matrix
    """
    results = {}

    # Train the model
    model.fit(train_features, y_train)

    # Test the model
    train_predicted = model.predict(train_features)
    test_predicted = model.predict(test_features)

    # Classification report and Confusion Matrix
    results['classification_report'] = classification_report(y_test, test_predicted)
    results['confusion_matrix'] = confusion_matrix(y_test, test_predicted)

    return results

```

```

# Initialize the models
rf = RandomForestClassifier(random_state=1)
dtree = DecisionTreeClassifier(random_state=1)
# Fit and evaluate models
results = {}
for cls in [dtree, rf]:
    cls_name = cls.__class__.__name__

    # Fit the model
    cls.fit(X_train, y_train)

    # Make predictions
    train_predictions = cls.predict(X_train)
    test_predictions = cls.predict(X_test)

    # Evaluate the model
    train_accuracy = accuracy_score(y_train, train_predictions)
    test_accuracy = accuracy_score(y_test, test_predictions)
    precision = precision_score(y_test, test_predictions, average='weighted')
    recall = recall_score(y_test, test_predictions, average='weighted')
    f1 = f1_score(y_test, test_predictions, average='weighted')

    results[cls_name] = {
        'Train Accuracy': train_accuracy,
        'Test Accuracy': test_accuracy,
        'Precision': precision,
        'Recall': recall,
        'F1-score': f1
    }

# Print the results
for result in results:
    print(result)
    print()
    for metric, value in results[result].items():
        print(metric, ":", value)
    print()
    print("-----")
    print()

```

Figure 4.84: Find the best machine learning model for treatment.

In figure 4.84, shows the process to find the best model for predicting and recommending the treatment. It shows that two model were initialize which are random forest Classifier and decision tree Classifier. The result are shown below.



```
DecisionTreeClassifier
Train Accuracy : 0.14534883720930233
Test Accuracy : 0.06976744186046512
Precision : 0.42871679413766867
Recall : 0.3476923076923077
F1-score : 0.3799467845828428
```

```
-----
RandomForestClassifier
Train Accuracy : 0.14651162790697675
Test Accuracy : 0.06046511627906977
Precision : 0.43184570449926435
Recall : 0.40923076923076923
F1-score : 0.4192076311098103
-----
```

Figure 4.85: Report Results

The evaluation metrics include Train Accuracy (accuracy of the model on the training data), Test Accuracy (accuracy of the model on the test data), Precision (weighted average precision score), Recall (weighted average recall score), and F1-score (weighted average F1-score), which are used to assess the performance of the models. Random Forest Classifier is chosen as the best model for treatment recommendation. Random Forest Classifier performs slightly better with a higher accuracy of approximately 14.65% on the training data and approximately 6.05% on the test data, compared to the Decision Tree Classifier accuracy of around 14.53% on the training data and approximately 6.98% on the test data.

#### 4.4 Developing CANCEVIVE: Data Visualisation of Cancer Survival and Treatment Recommendation.

##### 4.4.1 Front End Development (Web Application Dashboard)

The dashboard's front-end development is based on the use case and flowchart diagrams. The diagrams are used to design every flow of the system. Tableau is used to design and visualize the interface. The CANCEVIVE dashboard has three main pages: the homepage, the dashboard page, the personalize page, and the information page. This section explains the design.



Figure 4.86: CANERVIVE Homepage

Figure 4.86 displays the CANERVIVE system homepage, which includes the system's logo and name. The homepage also includes three buttons at the bottom of the page that will navigate to their respective pages which are dashboard, personalize, and information. When the Dashboard button click, the user be taken to a page with cancer survival rate visualisations. Users can predict patient's cancer survival rate and get treatment recommendation via the Personalise page. Finally, the Information page displays information on cancer survival rates as well as infographics about recommended treatments.

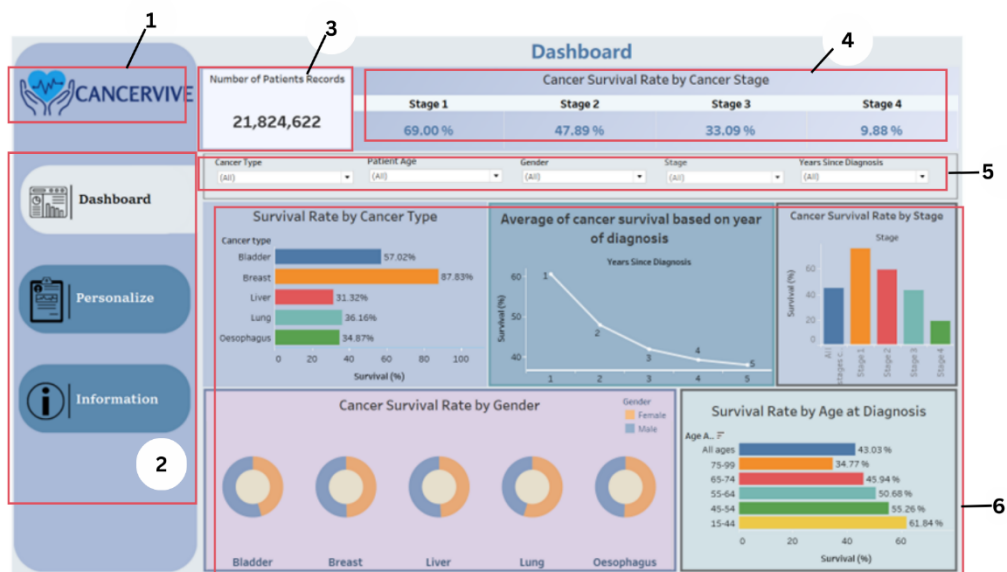


Figure 4.87: Dashboard Page

Table 4.17: Dashboard page front end.

| No. | Description                                                                                                                                                                                                         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | When user click the logo of CANCERVIVE, the system will navigate to the homepage.                                                                                                                                   |
| 2   | This section shows the navigation button at the left of the page, when user click the button, the system will navigate to the respective page that is click.                                                        |
| 3   | The number of patients that is in the records of cancer survival rate data                                                                                                                                          |
| 4   | The average survival rate by cancer stage (stages 1–4).                                                                                                                                                             |
| 5   | Users can filter the visualization of the cancer survival rate. Users can filter the visualization to make it more personalized by filtering the cancer type, patient age, gender, stage, and year since diagnosis. |
| 6   | This section shows the visualization of the average cancer survival rate by its cancer factor.                                                                                                                      |

**Personalize**

**CANCERVIVE: Cancer Survival Rate Prediction System**

Disclaimer: The Cancer Survival Rate prediction and Treatment Recommendation results is based on Machine Learning Data

**Patients' information input field**

Cancer Type  
Bladder

Cancer Stage  
1

Gender  
Female

Patient Age  
15-44

Years Since Diagnosis  
1

**Predict Cancer Survival**

**Predict cancer survival Button**

Figure 4.88: Personalize Page

**Personalize**

**CANCERVIVE: Cancer Survival Rate Prediction System**

Disclaimer: The Cancer Survival Rate prediction and Treatment Recommendation results is based on Machine Learning Data

Cancer Type  
Bladder

Patient Age  
15-44

Years Since Diagnosis  
1

Predict Cancer Survival

Figure 4.89: Fill in the information (Dropdown)

Figure 4.88 shows a webpage where a user can predict a patient's survival rate and recommended treatment, which can be used as an oncologist's guideline to plan treatment. The user can fill out the user information by selecting a drop-down option based on the patient's information (Figure 4.89). After that, complete all of the fields. The user can obtain the prediction results by clicking the "Predict Cancer Survival" button, and the prediction results will take 3 seconds to process and generate it. The results are displayed below, and users must scroll down to view the results (Figure 4.90).

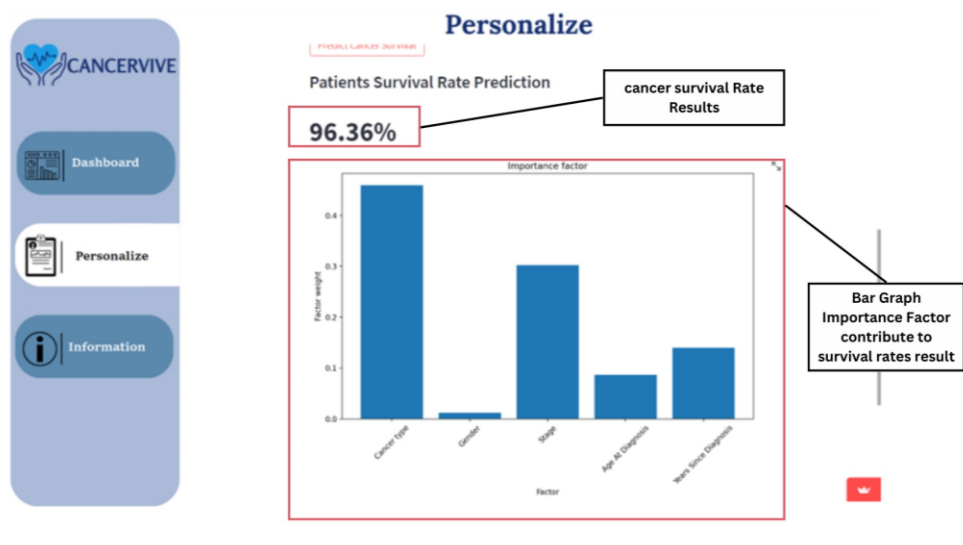


Figure 4.90: Survival rate prediction result.

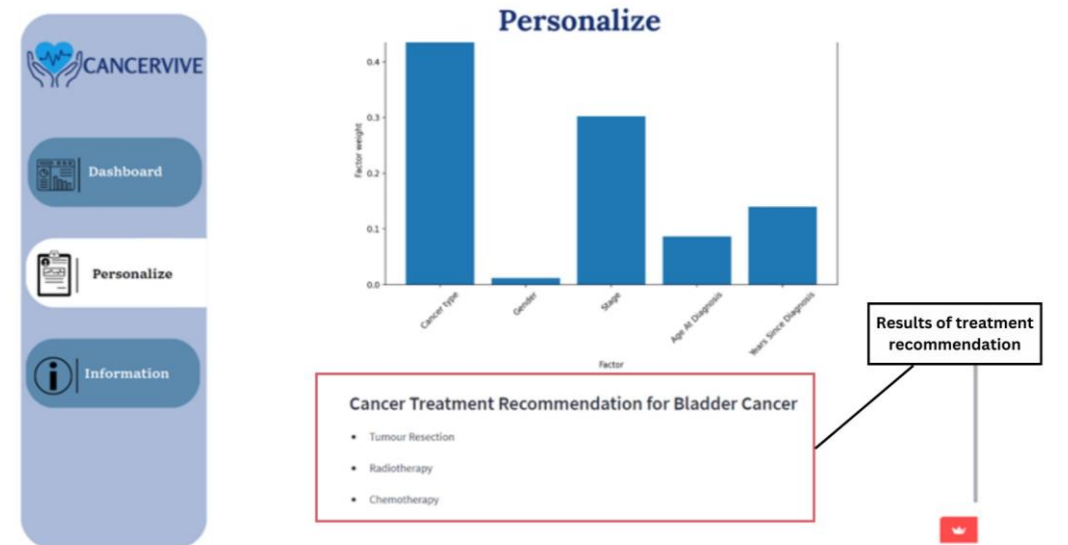


Figure 4.91: Recommended Treatment

After clicking the "Predict Cancer Survival" button, Figure 4.90 shows an example of the prediction result. Figure 4.90 shows the cancer survival rate percentage based on the previous input. Below the percentage of cancer survival rate, shows a bar graph of factors contributing to the percentage of survival rate prediction results. The bar graph is based on the weightage generated based on the factors. The prediction results also generate the patients' recommended treatment, as shown in figure 4.91.

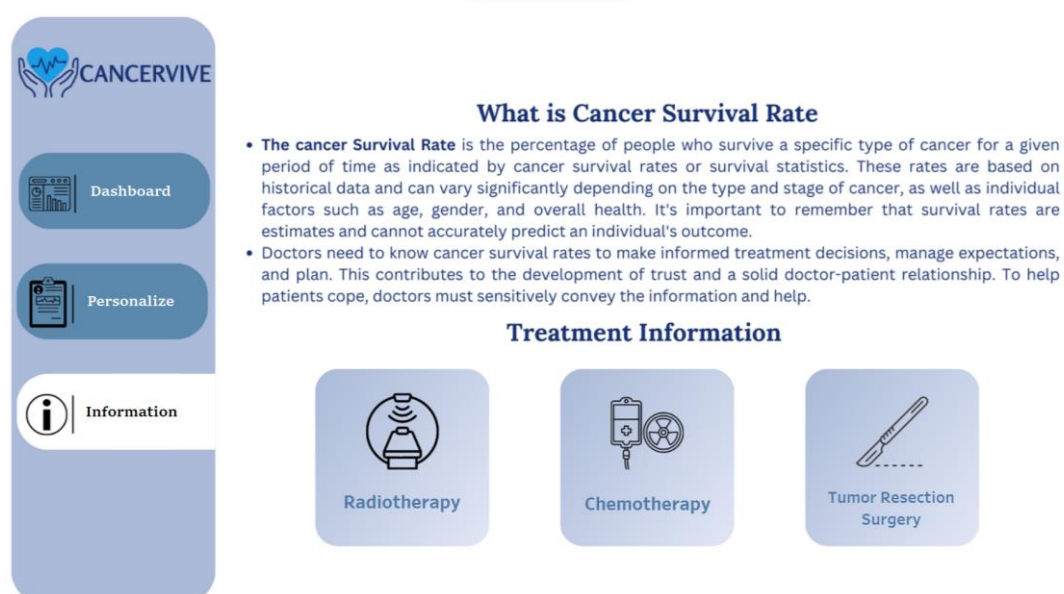


Figure 4.92: Information Page

Figure 4.92 shows the “Information” page. The page contains detailed information about the cancer survival rate. At the bottom of the page, three buttons provide treatment information. When users click on one of these buttons, the page will navigate to the respective treatment infographic page, where they can access detailed information about each treatment option.

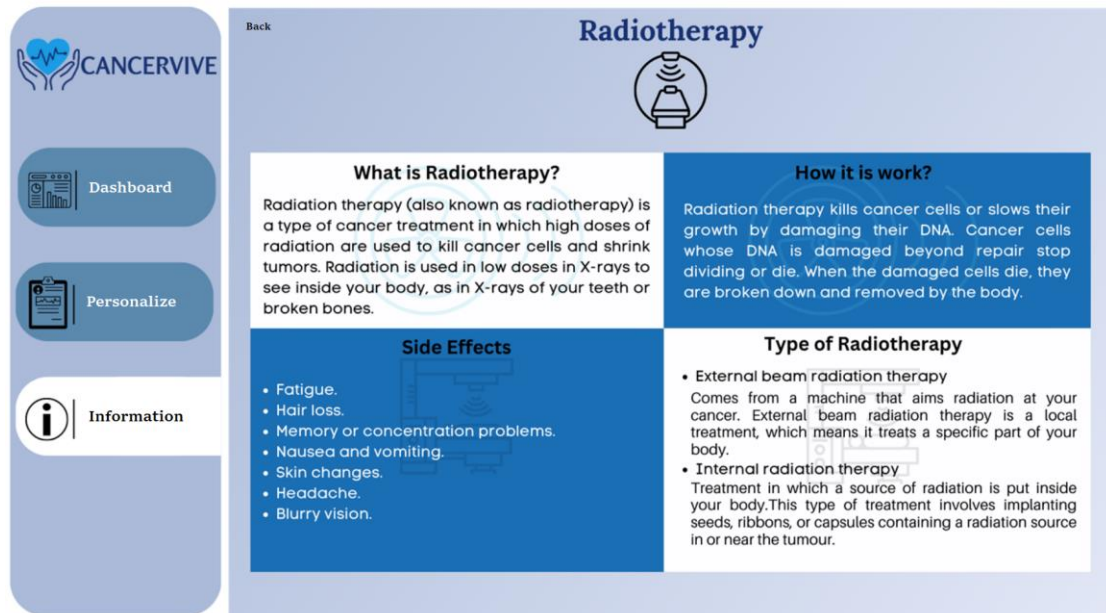


Figure 4.93: Radiotherapy treatment Infographic

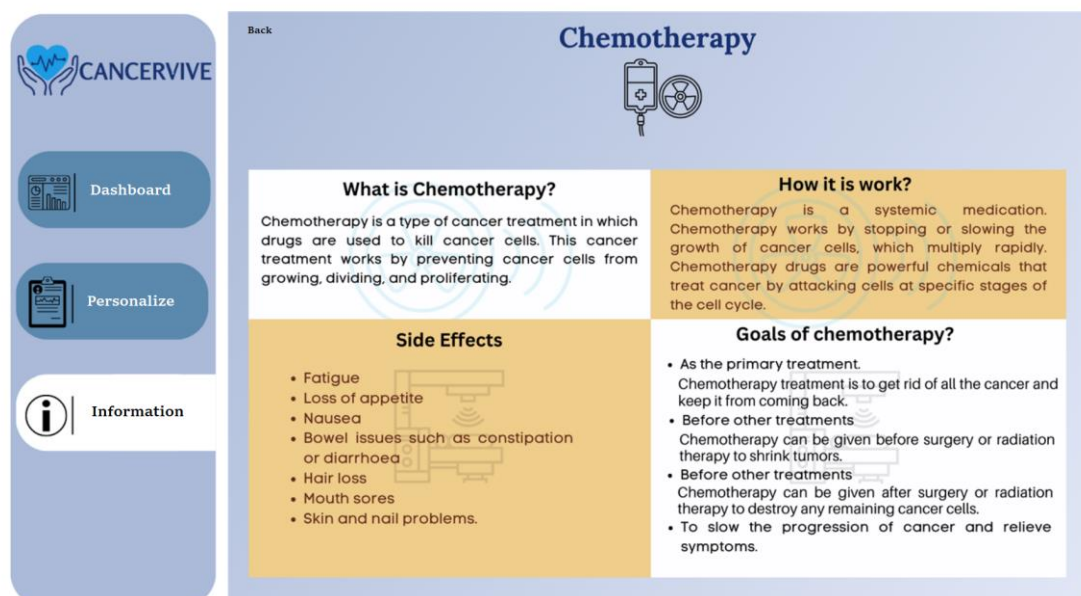


Figure 4.94: Chemotherapy treatment Infographic

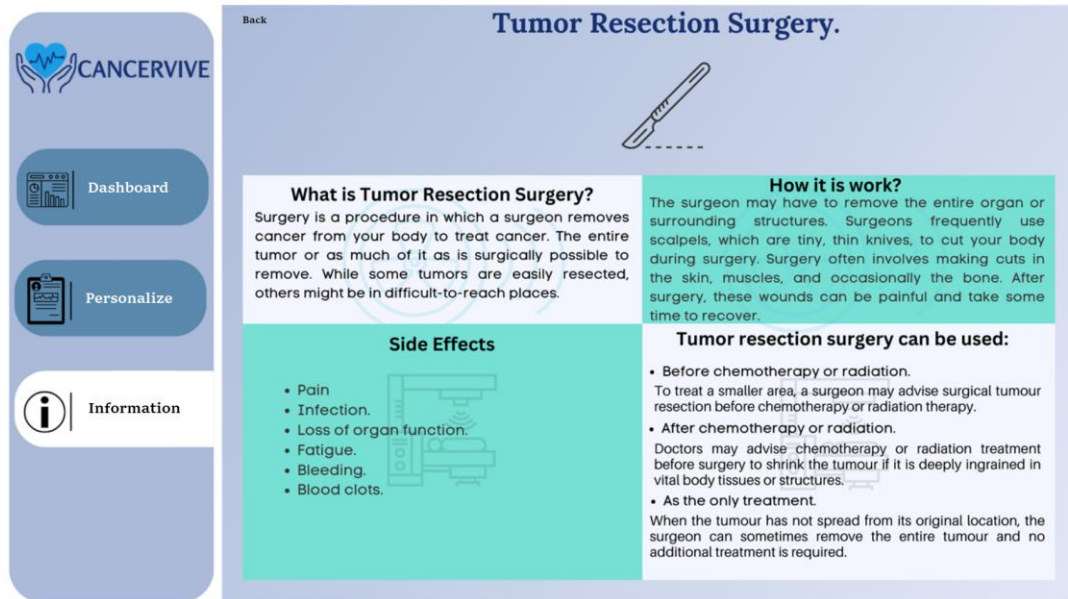


Figure 4.95: Tumour resection treatment Infographic

Lastly, in Figure 4.93 shows infographic radiotherapy treatment, figure 4.94 shows infographic chemotherapy treatment, and figure 4.95 shows infographic tumour resection surgery. The infographic provides detailed information about each treatment as well as extra materials to help users understand the treatments better. These infographics can also be shared to patients in order to provide them with clear and comprehensive explanations of the various treatment options available.

#### 4.4.2 Back End Development (Data Management)

Back-end development, also known as the development phase, focuses on the non-client-visible aspects of a platform. In this section, we will discuss the back-end development process of the user interface, as seen in 4.4.1. We will explain how the data is visualized by using Tableau and how the predictive model is developed using machine learning in the web application.



i) Developing Visualisation Dashboard (Tableau)

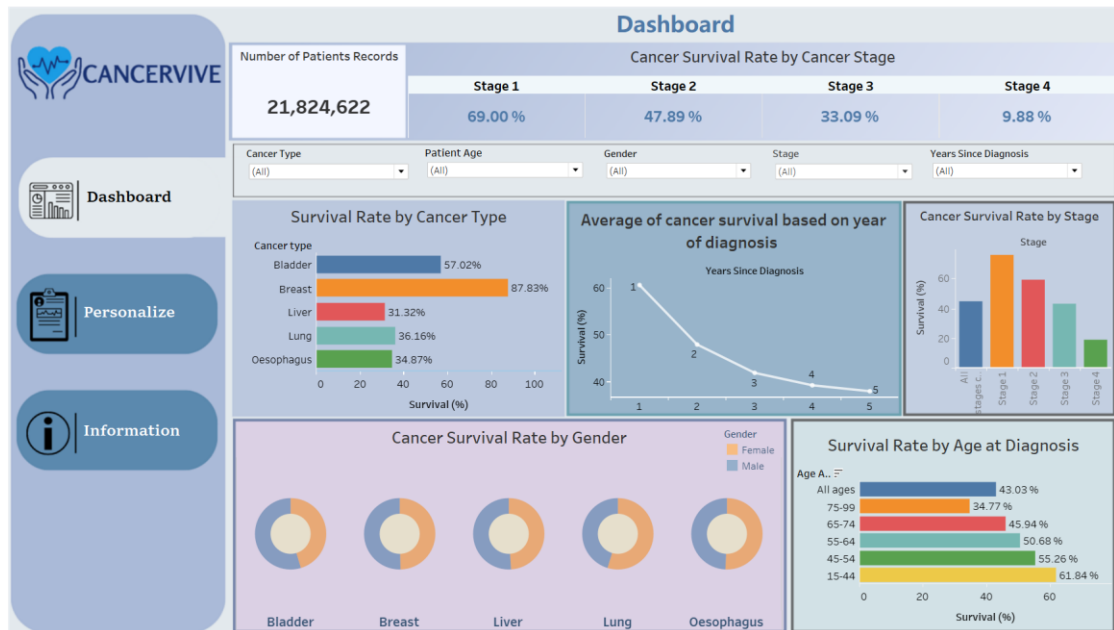


Figure 4.96: Front end visualisation Dashboard.

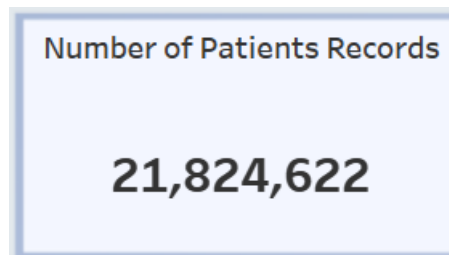


Figure 4.97: Front End total number of patients records.



Figure 4.98: Back End total number of patients record.

Figure 4.97 show the front end of total number of patients record. This section shows the total number of patients in the cancer survival data. To display that the number of



patients is SUM (Figure 4.98). The goal of displaying the total number of patients is to inform users about the amount of cancer patient data available in the system. This count is dynamic and subject to change depending on the data filtering features embedded in the dashboard. For example, if a user applies filters to narrow the data down to a specific cancer type, such as "breast," the count of patient records will adjust to reflect the data relevant to the selected criteria.

| Cancer Survival Rate by Cancer Stage |         |         |         |
|--------------------------------------|---------|---------|---------|
| Stage 1                              | Stage 2 | Stage 3 | Stage 4 |
| 69.00 %                              | 47.89 % | 33.09 % | 9.88 %  |

Figure 4.99: average of cancer survival rate by stage.

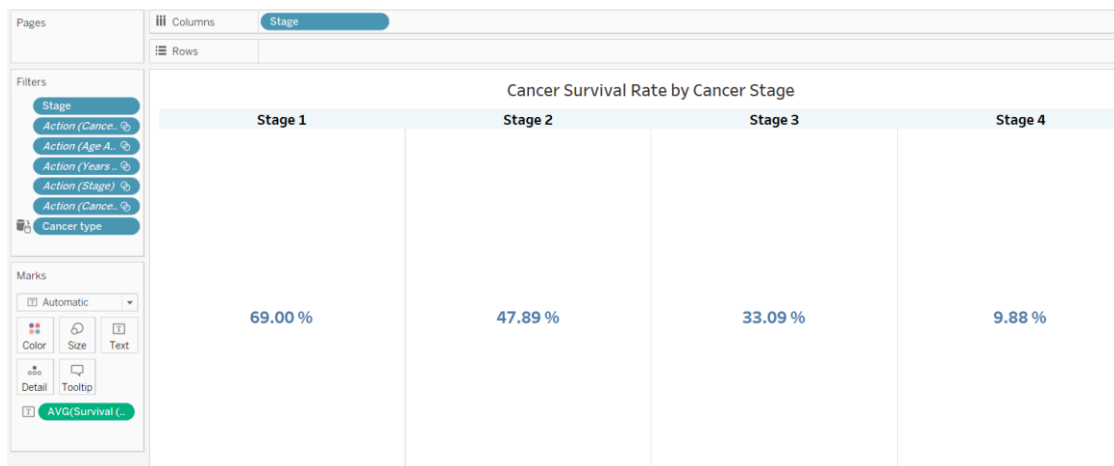


Figure 4.100: Back-end average of cancer survival rate by stage.

Figure 4.99 displays the text box average of cancer rates by stage in the dashboard. This provides a summary of the survival rate for overall cases based on different stages. Figure 4.100 shows how the back-end process to display the information using Tableau. The decision to place the average cancer survival rates by stage at the top of the dashboard page was considered, as cancer stage is one of the most important factors influencing cancer patients' overall survival rate. The specific survival rates are as follows, as shown in the text box of Figure 4.99: stage 1 has an average survival rate of 69%, stage 2 has a rate of 47.89%, stage 3 has a rate of 33.09%, and stage 4 has the lowest average survival rate of 9.88%.

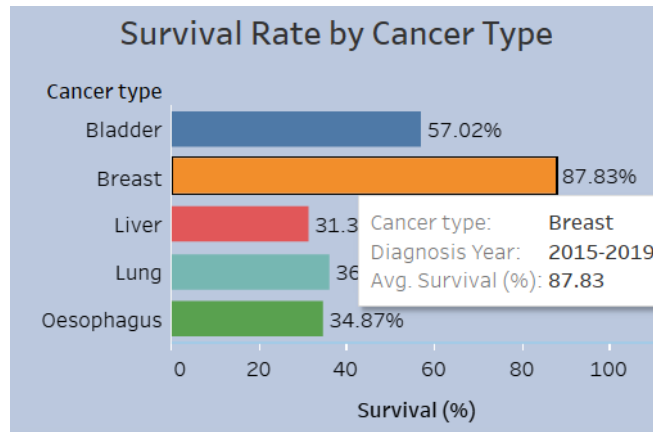


Figure 4.100: Front-End Visualize Survival Rate by Cancer Type



Figure 4.101: Back end Visualize Survival Rate by Cancer Type

Figure 4.101 shows the back-end process of visualising the bar chart of Survival Rate by Cancer Type. This visualization is created using survival rate data containing information on the average survival rate and the cancer types. Figure 4.100 shows the difference in survival rate based on cancer type in a horizontal bar graph. Each bar in the graph represents a different type of cancer, such as the blue bar displaying the average of bladder cancer, the orange bar displaying breast cancer, the red bar displaying liver cancer, the turquoise bar displaying lung cancer, and the green bar displaying oesophagus cancer. Additionally, the percentage of the average survival rate for each cancer type is displayed at the end left of each bar. The graph helps users understand the differences in cancer survival rates based on the type of cancer. It explains how cancer survival rates differ depending on the type of cancer. According to the bar chart in Figure 4.100, breast cancer has the highest average survival rate with 87.38%, followed by bladder cancer (57.02%), lung cancer (36.16%), oesophagus

cancer (34.87%), and liver cancer has the lowest average survival rate compared to other types of cancer with 31.32%.

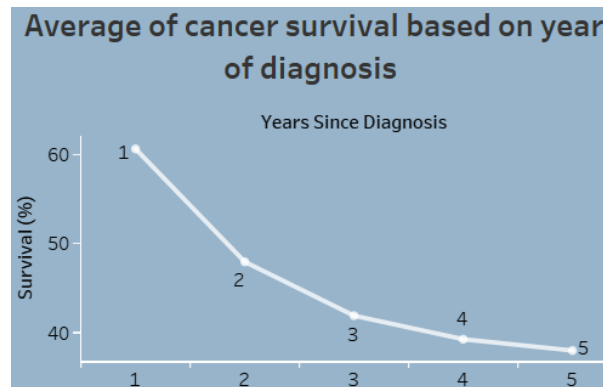


Figure 4.102: Visualisation survival rate based on year diagnosis.

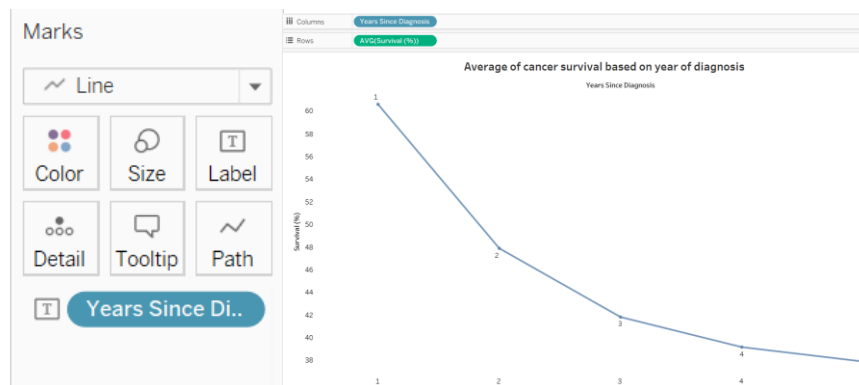


Figure 4.103: back-end visualisation survival rate based on year diagnosis.

Figure 4.102 shows the line graph average `survival rate based on year diagnosis. In the back-end process tableau, the graph uses `year since diagnosis` and `average survival rate` data from the cancer survival rate dataset (Figure 4.103). In Figure 4.102, the user can observe the trend of average survival rate and year since diagnosis. The term "year since diagnosis" can also be interpreted as the "year of survival". Notably, the analysis indicates that the 1-year survival rate (first year since diagnosis) stands as the highest at 60.59%, followed by the 2-year survival rate (47.88%), the 3-year survival rate (47.88%), the 4-year survival rate (39.16%), and the lowest survival rate being 37.89% for the 5-year survival period. The user can conclude from this graph that the survival rate decreases as the year of diagnosis increases. As a result, doctors or oncologists as system users can use this graph to recommend early treatment to mitigate cancer progression over time. Furthermore, this visualisation helps patients understand how cancer's prognosis changes over time.

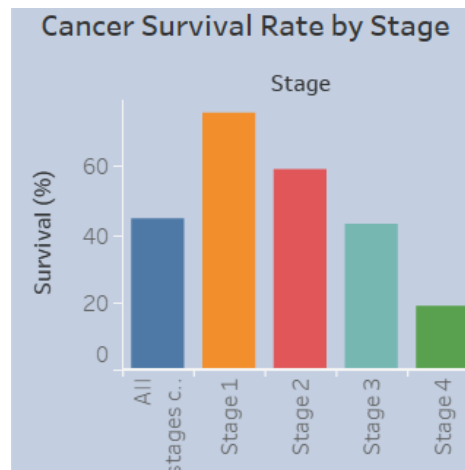


Figure 4.104: Visualisation survival rate stage.

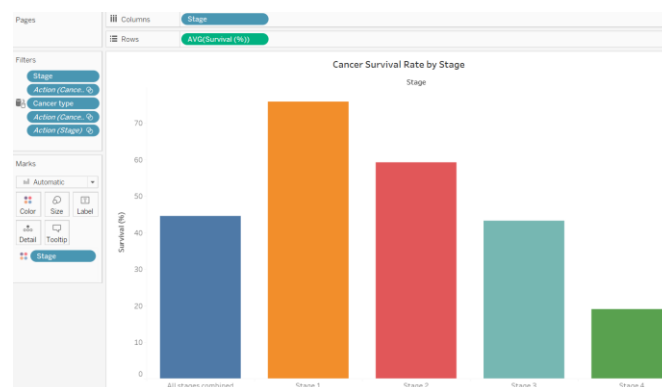


Figure 4.105: Back-end visualisation survival rate-based stage.

Figure 4.104 shows a bar graph displaying the average survival rate by stage. To create the graph, the back-end process in Tableau uses data from the cancer survival rate dataset on stage and average survival rate as shown in Figure 4.105. Each bar in the graph represents a different stage of cancer, such as the blue bar displaying the average cancer survival rate of all stages of cancer, the orange bar displaying stage 1, the red bar displaying stage 2, the turquoise bar displaying stage 3, and the green bar displaying stage 4. This visualisation graph gives users insight into the differences in survival rates among different cancer stages. The graph shows a consistent decrease in average survival rates as cancer stages progress. According to the graph in Figure 4.105, stage 1 has the highest average survival rate at 75.85%, followed by stage 2 at 59.4%, stage 3 at 43.2%, and stage 4 with the lowest average survival rate at 19.01%. Furthermore, the cancer stage is important in determining prognosis and treatment plans for patients. This analysis assists users and their patients in making informed treatment decisions and emphasises the importance of preventing cancer growth to prevent an increase in stage, which would then lead to a decrease in cancer survival rates.

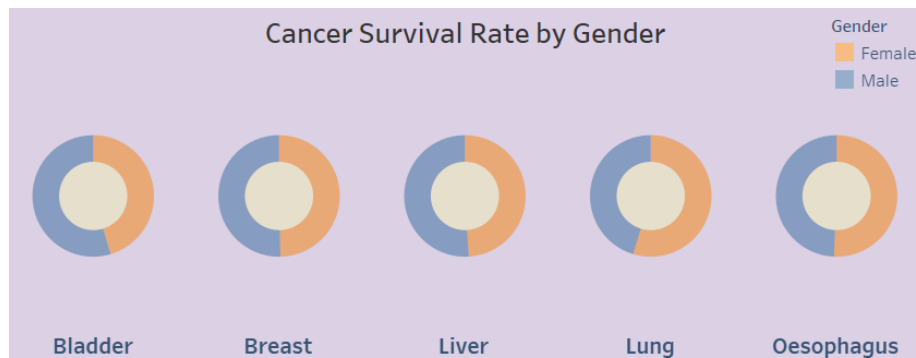


Figure 4.106: Visualisation survival rate by gender.

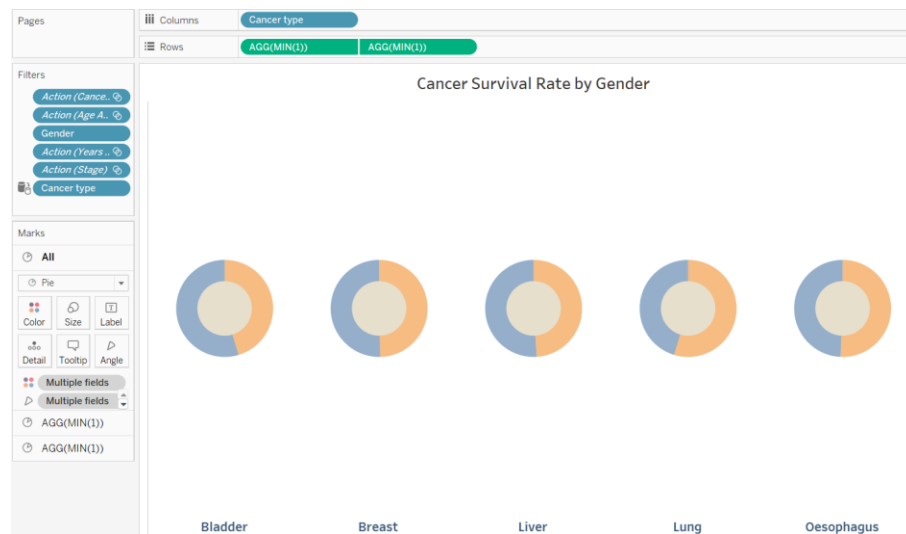


Figure 4.107: Back-end visualisation survival rate by gender.

Figure 4.106 depicts a pie chart displaying the average survival rates for each cancer type categorised by gender. This pie chart was created using data from the cancer survival rate dataset, which included gender, cancer type, and corresponding average survival rates, as shown in Figure 4.107. The colour blue represents the male gender in the pie chart, while the colour orange represents the female gender. The primary goal of this pie chart is to demonstrate the difference in survival rates between genders across various cancer types. It aims to highlight the differences in average survival rates by gender and to identify the gender with the highest average survival rate for each cancer type. According to the findings in Figure 4.106, males have a higher survival rate (62.01%) than females (51.50%) in the bladder cancer type. Regarding breast cancer, males have a higher survival rate (88.94%) than females (87.84%). Similarly, males have a higher survival rate (31.88%) than females (30.63%) in liver cancer. In contrast, females have a higher survival rate (39.55%) than males (32.76%) in lung cancer. Finally, in the context of oesophageal cancer, females have a higher survival rate (35.58%) than males (34.41%).

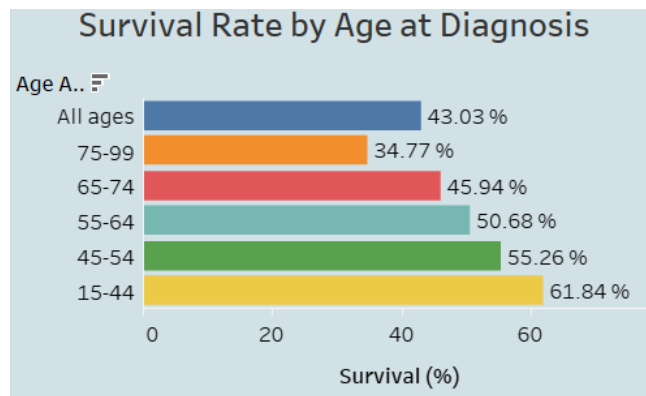


Figure 4.108: Visualisation survival rate by age.

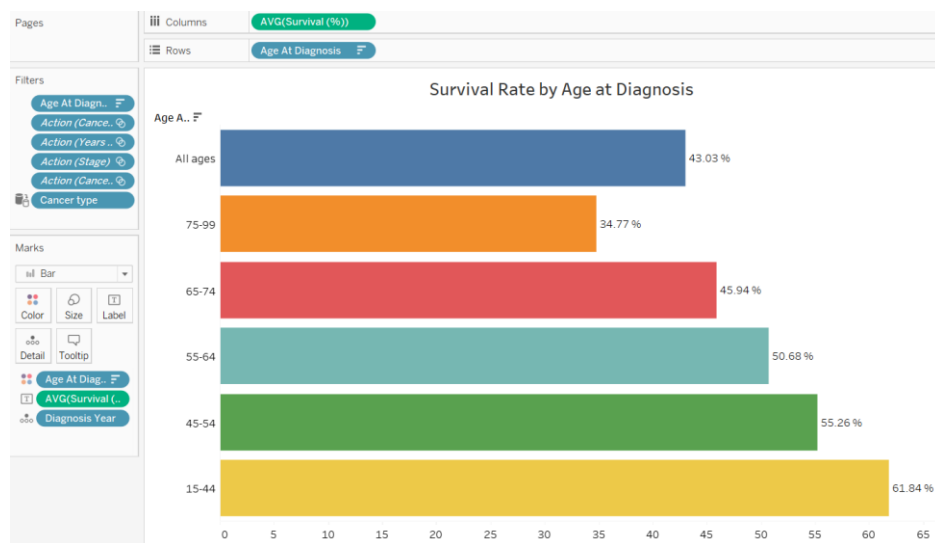


Figure 4.109: Back-end visualisation survival rate by age.

Figure 4.108 shows a bar graph displaying the average survival rate. To create the graph, the back-end process in Tableau uses data from the cancer survival rate dataset on age at diagnosis and average survival rate, as shown in Figure 4.109. A blue bar represents bar displaying the average cancer survival rate of all ages, an orange bar represents ages 75-99, a red bar represents ages 65-74, a turquoise bar represents ages 55-64, a green bar represents ages 45-54, and a yellow bar represents ages 15-44. Additionally, At the left end of each bar, the percentage of the average survival rate for each age group is also displayed. The main goal of this graph is to show how age affects survival rates visually. In general, the average survival rate decreases with age. Notably, in Figure 4.108, the age group with the highest survival rate is 61.84%, while the age group with the lowest survival rate is 75-99 years old. This graph effectively highlights the relationship between age and survival rates, demonstrating how survival chances change with age.

## ii) Prediction Cancer Survival Rate and Treatment Recommendation system (Machine Learning)

We used the Streamlit python library to create this website, writing the code in Python with VSCode. Streamlit, an open-source Python app framework, Streamlit speeds up the development of web applications for data science and machine learning projects. It integrates seamlessly with major Python libraries such as scikit-learn, Keras, PyTorch, SymPy (LaTeX), NumPy, pandas, and matplotlib, allowing for the efficient and rapid development of interactive web apps.

### CANCERVIVE: Cancer Survival Rate Prediction System

Disclaimer: The Cancer Survival Rate prediction and Treatment Recommendation results is based on Machine Learning Data

Cancer Type

Bladder

Cancer Stage

1

Gender

Female

Patient Age

15-44

Years Since Diagnosis

1

Predict Cancer Survival

Figure 4.110: prediction input

```

st.header('CANCERVIVE: Cancer Survival Rate Prediction System')

st.write('Disclaimer: The Cancer Survival Rate prediction and Treatment Recommendation results is based on Machine Learning Data')

def user_input_features():
    cancer_types = [
        "Bladder",
        "Lung",
        "Breast",
        "Liver",
        "Oesophagus"
    ]
    genders = [
        "Female",
        "Male"
    ]
    ages = [
        "15-44",
        "45-54",
        "55-64",
        "65-74",
        "75-99",
        "All ages"
    ]
    stages = [
        "1",
        "2",
        "3",
        "4"
    ]
    years_options = [
        "1",
        "2",
        "3",
        "4",
        "5"
    ]

    cancer_type = st.selectbox('**Cancer Type**', cancer_types)
    gender = st.selectbox('**Gender**', genders)
    age = st.selectbox('**Patient Age**', ages)
    stage = st.selectbox('**Cancer Stage**', stages)
    selected_years = st.selectbox('**Years Since Diagnosis**', years_options)

```

Figure 4.111: Back-end input

Figure 4.110 shows the prediction input, in which users enter cancer information about patients before performing the prediction. For data selection, the input makes use of dropdown menus. Figure 4.111 shows the back-end process of the input.

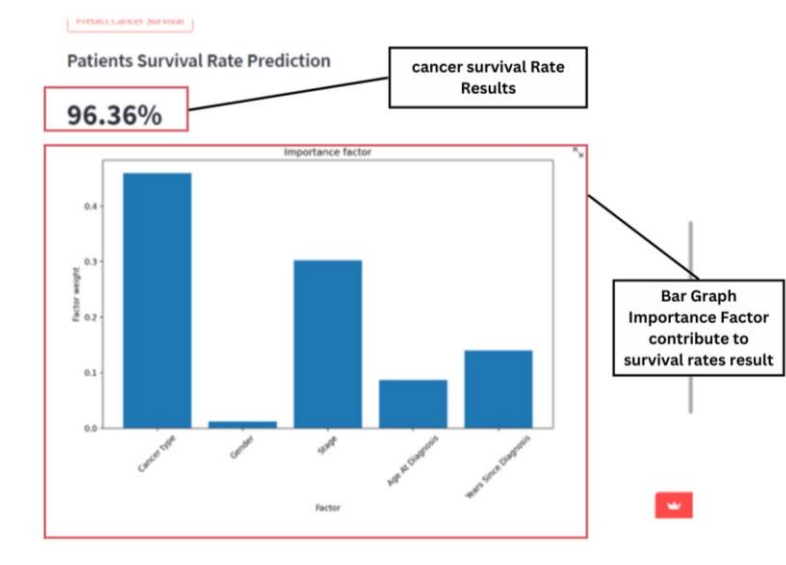


Figure 4.112: prediction result



```

# Load the Cancer survival dataset from CSV
url = 'https://raw.githubusercontent.com/sarah-liya/survival-web/main/FYP%20Cancer%20Survival%20new.csv'
cancerS_df = pd.read_csv(url, low_memory=False)

# Encode the categorical features in the original dataset
encode = ['Cancer type', 'Gender', 'Stage', 'Age At Diagnosis']
for col in encode:
    label_encoder = LabelEncoder()
    cancerS_df[col] = label_encoder.fit_transform(cancerS_df[col])
    input_features[col] = label_encoder.transform(input_features[col])

features = ['Cancer type', 'Gender', 'Stage', 'Age At Diagnosis', 'Years Since Diagnosis']
X = cancerS_df[features]
y = cancerS_df['Survival (%)']

# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)

# Build Decision Forest Regressor
rf = RandomForestRegressor()
rf.fit(X_train, y_train)

if input_features is not None:
    # Add a button to trigger the prediction
    if st.button('Predict Cancer Survival'):
        # Predict survival
        prediction = rf.predict(input_features)[0]

        # Display the prediction
        st.write('### Patients Survival Rate Prediction')
        st.title(f'{prediction:.2f}%')

```

Figure 4.113: cancer survival rate result back-end

```

# Extract the attribute weights (feature importances)
attribute_weights = rf.feature_importances_

# Create a DataFrame to store the attribute weights
attribute_weights_df = pd.DataFrame({'Attribute': X_train.columns, 'Weight': attribute_weights})

# Plot the attribute weights
plt.figure(figsize=(10, 6))
plt.bar(attribute_weights_df['Attribute'], attribute_weights_df['Weight'])
plt.xlabel('Factor')
plt.ylabel('Factor weight')
plt.title('Importance factor')
plt.xticks(rotation=45)
st.pyplot(plt)

```

Figure 4.114: importance bar graph back end.

The prediction result is displayed after the user clicks the 'Predict cancer survival' button, as shown in Figure 4.112. Figure 4.113 shows the back-end code for predicting cancer survival percentage using a random forest regressor model. The model also calculates the factors that influence the survival rate in addition to the percentage of survival rates. The significance of these factors is depicted in a bar graph in Figure 4.114, which represents the back-end code responsible for generating the graph displaying the weights of cancer survival factors.

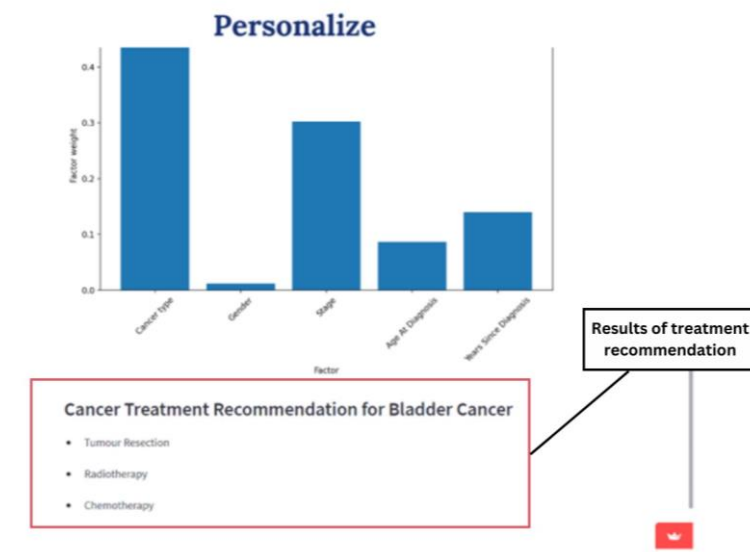


Figure 4.115: treatment recommendation results

```
selected_attributes = ['Stage', 'Gender', 'Age At Diagnosis']

# Filter the attribute weights based on selected attributes
filtered_attribute_weights_df = attribute_weights_df[attribute_weights_df['Attribute'].isin(selected_attributes)]

# Find the attribute with the highest weight
highest_weight_attribute = filtered_attribute_weights_df.loc[filtered_attribute_weights_df['Weight'].idxmax(), 'Attribute']
highest_weight = filtered_attribute_weights_df[highest_weight_attribute]['Weight'].max()

def treatment_recommendation(input_features, highest_weight_attribute):
    # Retrieve the selected cancer type from the input features
    cancer_type = input_features['Cancer type'].iloc[0]

    # Perform treatment recommendation based on conditions
    if highest_weight_attribute == 'Stage':
        if input_features['Cancer type'].iloc[0] == 0: #Bladder
            st.write('### Cancer Treatment Recommendation for Bladder Cancer')
            if input_features['Stage'].iloc[0] == 0:
                st.write('- Tumour Resection')
                st.write('- Radiotherapy')
                st.write('- Chemotherapy')
            elif input_features['Stage'].iloc[0] == 1:
                st.write('- Chemotherapy')
            elif input_features['Stage'].iloc[0] == 2:
                st.write('- Radiotherapy')
                st.write('- Chemotherapy')
            elif input_features['Stage'].iloc[0] == 3:
                st.write('- Tumour Resection')
                st.write('- Chemotherapy')

        elif input_features['Cancer type'].iloc[0] == 1: #Breast
            st.write('### Cancer Treatment Recommendation for Breast Cancer')
            if input_features['Stage'].iloc[0] == 0:
                st.write('- Tumour Resection')
                st.write('- Chemotherapy')
            elif input_features['Stage'].iloc[0] == 1:
                st.write('- Tumour Resection')
                st.write('- Radiotherapy')
                st.write('- Chemotherapy')
            elif input_features['Stage'].iloc[0] == 2:
                st.write('- Tumour Resection')
```

**treatment recommendation**

Figure 4.116: back-end recommendation results

Figure 4.115 shows the example results of recommended treatment. Figure 4.116 shows the back-end code of the recommendation treatments. The code above uses the if else condition to print the recommended treatment. So how the treatment recommendation results is process? The process of treatment recommendation machine learning is shown below.

```
# Define the features and target variables
features = ['Cancer type', 'Levels of factor']
X = factorStage_df[features]
y = factorStage_df[['Tumour Resection', 'Chemotherapy', 'Radiotherapy']]
```

```
from sklearn.ensemble import RandomForestClassifier
# Create and train the Random Forest classifier
rf = RandomForestClassifier()
rf.fit(X_train, y_train)

RandomForestClassifier()
```

```
# Predict on new data
X_new= np.array(['Breast', '1'])
X_new[:, 0] = [Cancer_type.get(value, value) for value in X_new[:, 0]]
X_new[:, 1] = [Stage.get(value, value) for value in X_new[:, 1]]
X_new = X_new.astype(float)

# Make predictions on new data
new_predictions = rf.predict(X_new)
print("Predictions for new data:", new_predictions)
```

Predictions for new data: [[0 1 0]]

```
# Predict on new data
X_new = np.array(['Breast', '2'])
X_new[:, 0] = [Cancer_type.get(value, value) for value in X_new[:, 0]]
X_new[:, 1] = [Stage.get(value, value) for value in X_new[:, 1]]
X_new = X_new.astype(float)

# Make predictions on new data
new_predictions = rf.predict(X_new)
print("Predictions for New Data:", new_predictions)
```

Predictions for New Data: [[1 1 1]]

```
# Predict on new data
X_new = np.array(['Breast', 3])
X_new[:, 0] = [Cancer_type.get(value, value) for value in X_new[:, 0]]
X_new[:, 1] = [Stage.get(value, value) for value in X_new[:, 1]]
X_new = X_new.astype(float)

# Make predictions on new data
new_predictions = rf.predict(X_new)
print("Predictions for New Data:", new_predictions)
```

Predictions for New Data: [[0 1 0]]

Figure 4.117: Treatment recommendation process

The back-end process is implemented in Jupyter Notebook. The predictive model used to determine whether a treatment is recommended based on the cancer type and stage is the Random Forest Classifier. The treatment options tested in this model are 'tumour resection', 'chemotherapy', and 'radiotherapy'. The model outputs '1' to indicate 'yes' when the treatment is recommended, and '0' to indicate 'no' when the treatment is not recommended for the patients. Figure 4.117 shows the process.

## Deploy an app

Repository

[Paste GitHub URL](#)

sarah-liya/survival-rate-and-treatment

Branch

main

Main file path

main2.py

App URL (Optional)

survival-rate-and-treatment-a9u24o342kp

.streamlit.app

Domain is available

[Advanced settings...](#)

Deploy!

Figure 4.118 (a): website deployment.

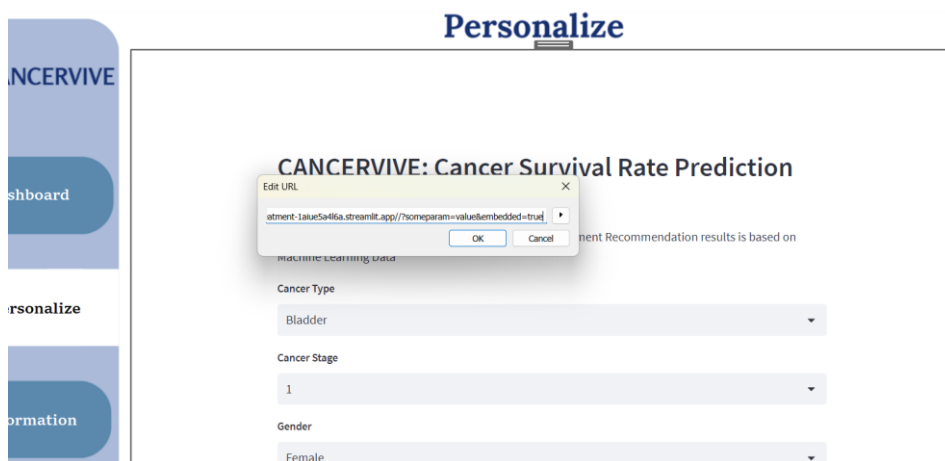


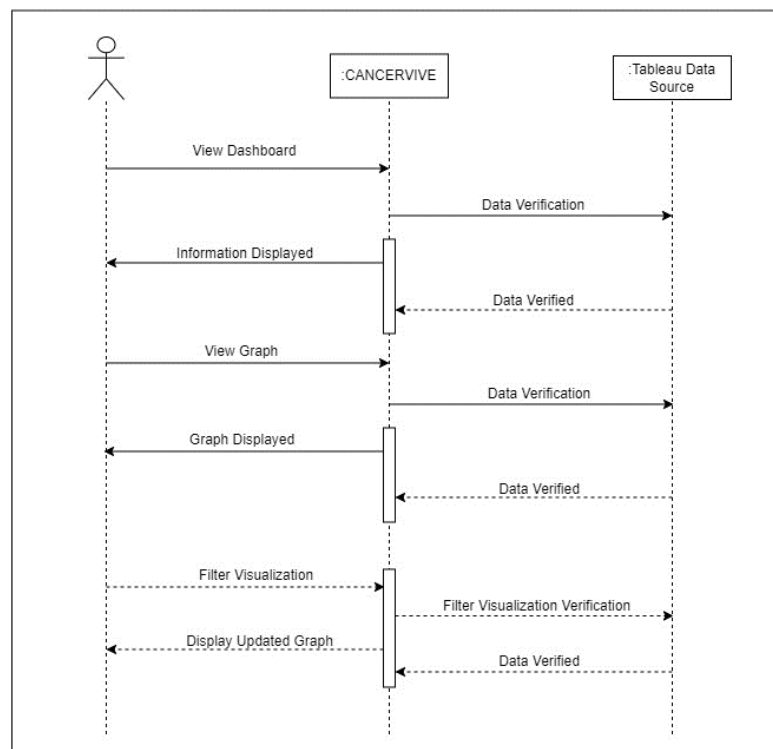
Figure 4.118 (b): web app in CANCERVIVE Tableau system.

After successfully building the website, the system deploys it to Streamlit Cloud using a Streamlit domain. The process of deploying a web app in Streamlit Cloud is illustrated in Figures 4.118 (a). To begin the deployment, the web app's code must first be uploaded to a GitHub account. This step is required because the deployment process, as shown in Figure 4.118 (a), requires the GitHub repository link to make the web app available for

public use. Following that, the deployment process produces a shareable link for the deployable website system. This link serves as a portal to the web application. The generated link is integrated into Tableau to ensure seamless integration and provide access to the web application within the CANCEVIVE system's features (Figure 4.118(b)). Users can access the web application from within the Tableau environment, ensuring a seamless and interconnected user experience.

#### 4.4.3 System Integration.

This section explained the dashboard system's functionality from both the front end and the back end. It also displays the expected user interaction with the dashboard and the expected outcome. The Sequence Diagram explains in detail how the operation is carried out. The sequence diagram visually represents the order of system interaction by using the vertical axis of the diagram to represent the sequence of interaction. Figure 4.119 shows the system's sequence diagram.



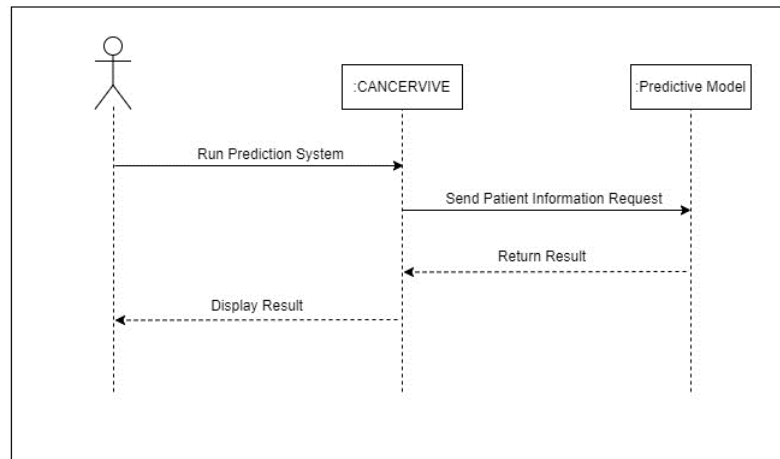
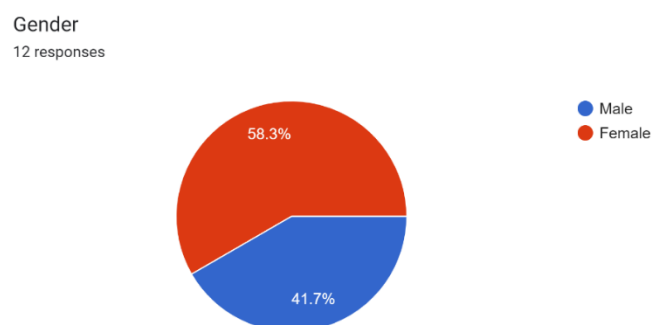


Figure 4.119: Sequence Diagram

#### 4.4.4 System Testing

In this section, users' feedback on the CANCERVIVE dashboard after testing is presented. As the final phase of dashboard development, testing is crucial to collect valuable user feedback. An online survey was conducted using Google Forms, consisting of two sections: the user background and their feedback on the dashboard's functionality. The survey testing also includes the survey functionality, the amount of information conveyed to the user, and the user experience while using the system.

##### I. User Information.



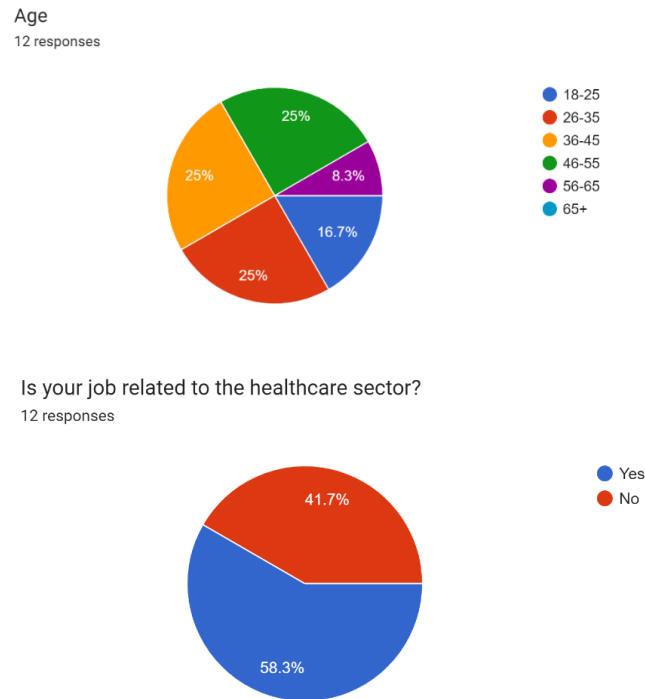


Figure 4.120: User Information

Figure 4.120 presents demographic information of users participating in the system testing. The data shows that 58.3% of the participants are female, while 41.7% are male. Next is participant age distribution, age groups 26-35 and 36-45 all have 25%, while 18.5% belong to the 18-25 age group, and 16.7% are aged 65 or older. Next, 58.7% of participant job is related to healthcare sector and the other 41.7% is not related. The participant job that related to the healthcare sector is nurse and doctor.

## II) System Usability Scale (SUS) Questionnaire

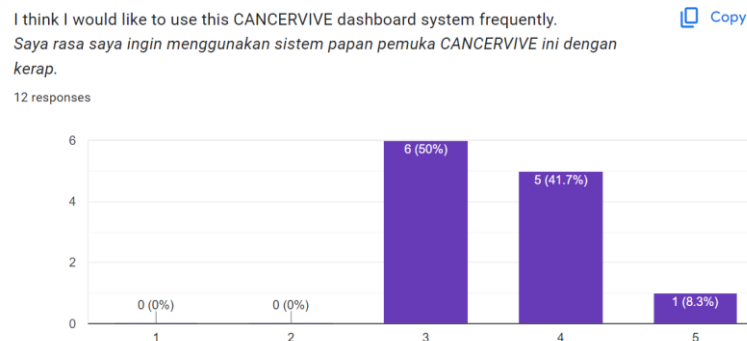


Figure 4.121: CANCEVIVE frequent usage feedback.

50% (8 respondents) of respondent is having neutral opinion on either they will use the CANCEVIVE dashboard system frequently or not. The other 41.7% (5 respondents)

is agree and 8.3% (1 respondent) is strongly agree with the question. It can be concluded that most of the respondents are neutral to use the system frequently.

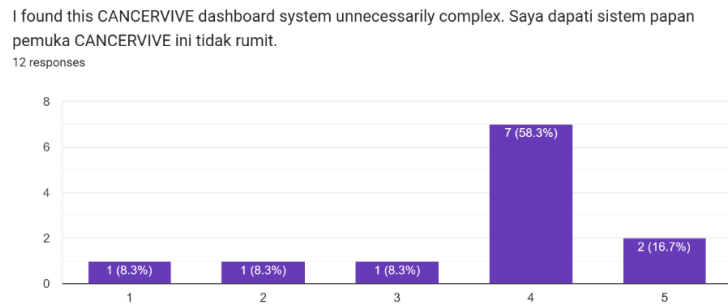


Figure 4.122: CANCEVIVE complexity feedback.

Next question is the CANCEVIVE system is not complex. 7 respondents agree that the system is not complex. 16.7% (2 respondents) strongly agree that the system is not complex and the other 8.3% respondent respectively thinks the system is complex. It can be concluded that most of the respondents agree the CANCEVIVE system is not complex to use.

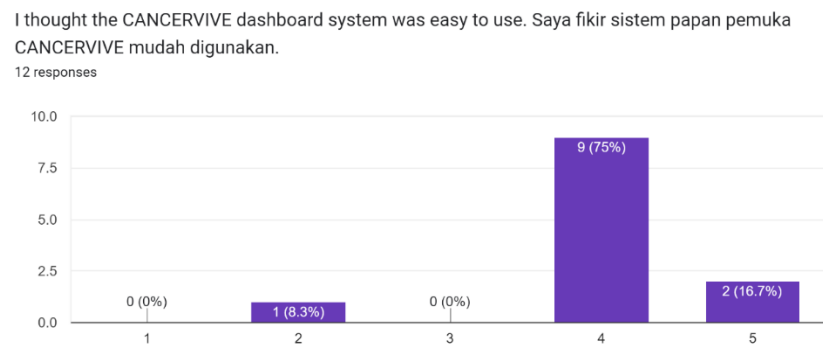


Figure 4.123: CANCEVIVE easy to use feedback.

9 respondents (75%) agree that the CANCEVIVE system is easy to use. 16.7% (2 respondents) strongly agree that the system is easy to use. However, 1 respondent (8.3%) not agree that system is easy to use. From the Figure 4.123 can conclude that most of the respondents feel that the system is easy to use.



I think that I would need the support of a technical person to be able to use CANCEVIVE dashboard system. Saya berpendapat bahawa saya memerlukan menggunakan sistem papan muka CANCEVIVE.  
12 responses

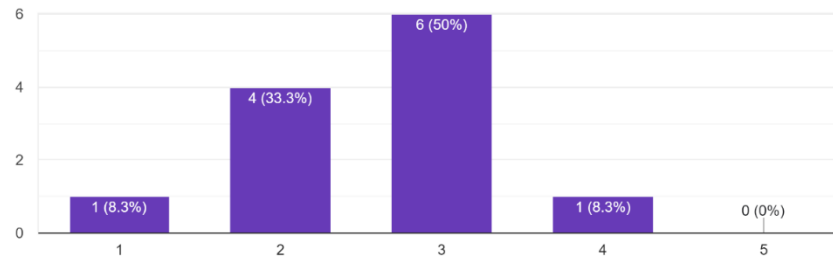


Figure 4.124: CANCEVIVE needs technical support feedback.

50% (6 respondents) have the neutral opinion that the CANCEVIVE system need the support of technical person for them to use the system. 4 respondents (33.3%) disagree that the system requires assistance to use the system. The remaining 8.3% (1 respondent) strongly disagree that a technical person is required for the system. Finally, 1 respondent (8.3%) agrees that the system requires technical assistance. The results in figure 4.124 show that most of the respondents feel they did not need a technical person to use the CANCEVIVE system.

I found the various functions in the CANCEVIVE dashboard system were well integrated. Saya dapati pelbagai fungsi dalam sistem papan muka CANCEVIVE disepadukan dengan baik.  
12 responses

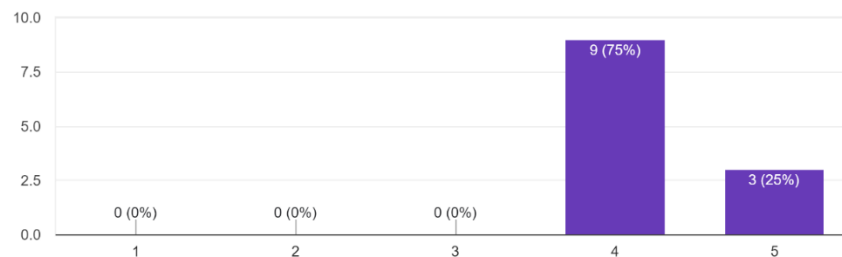


Figure 4.125: CANCEVIVE system integration feedback.

Figure 4.125 shows that 75% (9 respondents) agree that the system is well integrated. The other 25% (3 respondents) strongly agree that the function of the system is well integrated. From the results, all respondents agree that the CANCEVIVE system is well integrated.

I thought there was too much inconsistency in CANCEVIVE dashboard system. Saya pikir terdapat banyak ketidak konsistenan dalam sistem papan muka CANCEVIVE.  
12 responses

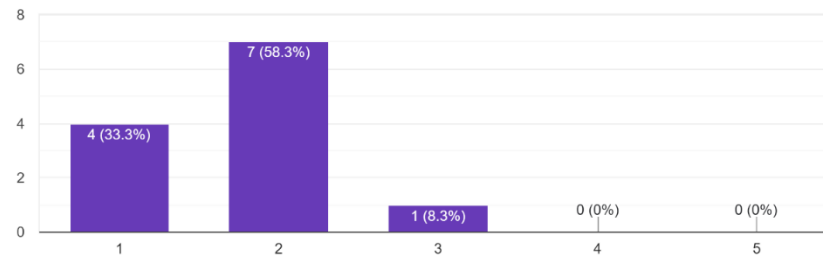


Figure 4.126: CANCEVIVE system inconsistency feedback.

From the figure 4.126 shows 7 respondents (58.3%) not agree that the CANCEVIVE system is not consistence. 4 respondents (33.3%) is strongly not agree and the other 8.3% (1 respondent) is neutral with the statement.

I would imagine that most people would learn to use CANCEVIVE dashboard system very quickly. Saya membayangkan bahawa kebanyakan...istem papan muka CANCEVIVE dengan cepat.  
12 responses

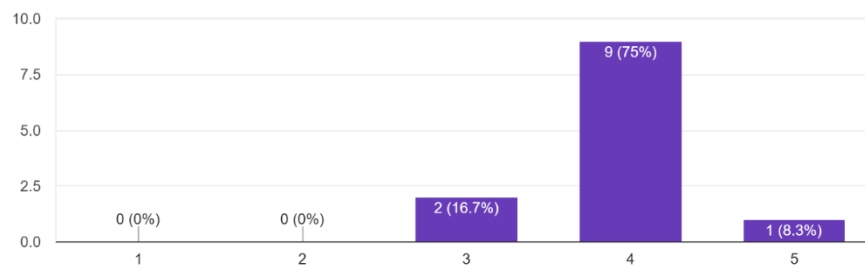


Figure 4.127: CANCEVIVE system quickly feedback.

Figure 4.127 shows that the majority of the respondents (75%) agree that people will use the system quickly, 16.7% (2 respondents) are neutral and the other 8.3% (1 respondent) is strongly agree. From the question feedback, the conclusion is the respondent thinks other people will learn how to use the CANCEVIVE system quickly.

I found the CANCEVIVE dashboard system very cumbersome to use. Saya dapati sistem papan pemuka CANCEVIVE sangat sukar digunakan.

12 responses

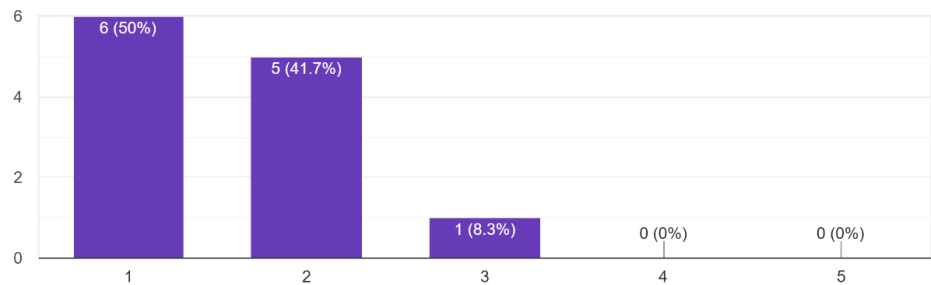


Figure 4.128: Use CANCEVIVE system is not easy to use feedback.

The majority of the respondents (50%) strongly did not agree that the CANCEVIVE system is very cumbersome to use. 5 respondents (41.7%) disagree and the other 8.3% is neutral with the statement. It can be concluded from figure 4.128 is the system is easy to use and user-friendly.

I felt very confident using the CANCEVIVE dashboard system. Saya berasa sangat yakin menggunakan sistem papan pemuka CANCEVIVE.

12 responses

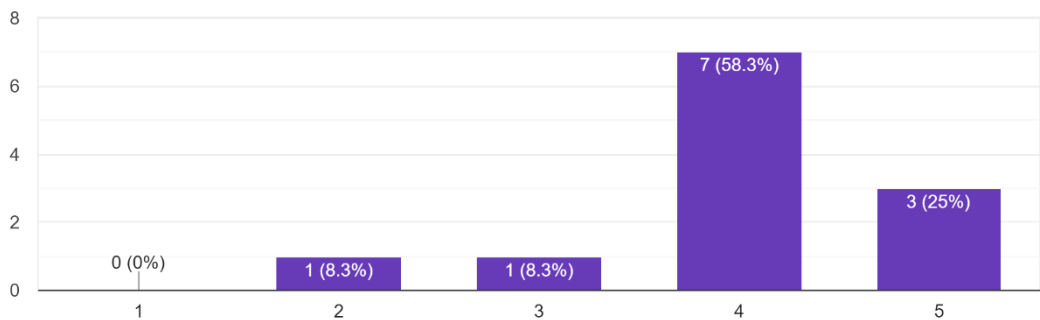


Figure 4.129: CANCEVIVE system confident feedback.

Figure 4.129 shows that 58.3% (7 respondents) agree that they are confident to iuse the CANCEVIVE system. 25% (3 respondents) strongly agree, 1 respondent is neutral and the other 1 respondent (8.3%) does not agree. Overall, most of the respondents are agree that they are really confident in using the system.

I needed to learn a lot of things before I could get going with CANCEVIVE dashboard system. Saya perlu belajar banyak perkara sebelum saya boleh menggunakan sistem papan pemuka CANCEVIVE.

12 responses

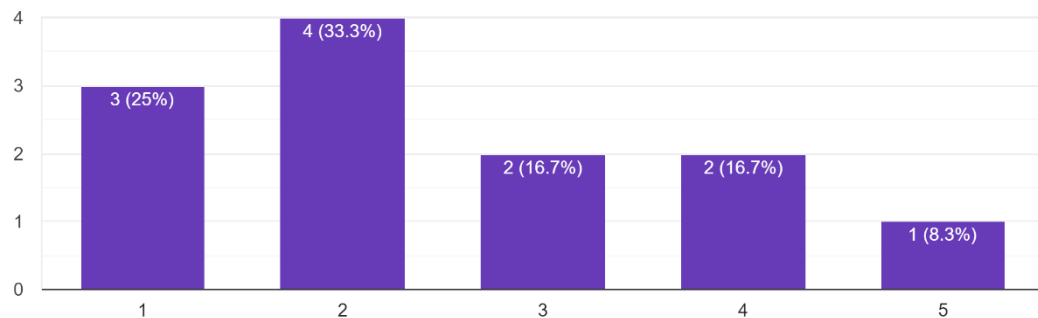


Figure 4.130: CANCEVIVE system knowledge feedback.

Figure 4.130 shows that most of the respondents (33.3%) do not agree and 25% (3 respondents) strongly did not agree that they need to learn a lot of things before use CANCEVIVE system. 16.7% (2 respondents) are neutral with the statement. Lastly, the other 16.7% and 8.3% agree and strongly agree with the statement. Overall, they are mix thoughts in the system knowledge feedback. Below is how the system usability score by the respondent is calculated.

The SUS survey contains ten statements and employs a Likert Scale. The user can decide at what level the statement is acceptable. As a result, the user responds to each question with a value between 1 and 5. The value indicates that you strongly disagree, while 5 indicates that you strongly agree. Then the SUS score has to calculate using formula. Below are the SUS score calculations using the formula:

$$X = (\text{Sum of points for all odd-numbered questions}) - 5$$

$$Y = 25 - (\text{Sum of the points for all even-numbered questions})$$

$$\text{SUS Score} = (X+Y) \times 2.5$$

Table 4.18: System Usability Scale Adjective Rating

| SUS Score   | Grade | Adjective Rating |
|-------------|-------|------------------|
| > 80.1      | A     | Excellent        |
| > 68 – 80   | B     | Good             |
| > 67 – 67.9 | C     | Okay             |
| > 51 – 66   | D     | Poor             |
| < 51        | E     | Awful            |

The adjective rating is explained in Table 4.18 above. The system is considered excellent if the SUS score is greater than 80.1, good if the score is between 68 and 80, okay if the score is between 67 and 67.9, poor if the score is between 51 and 66, and awful if the score is less than 51.

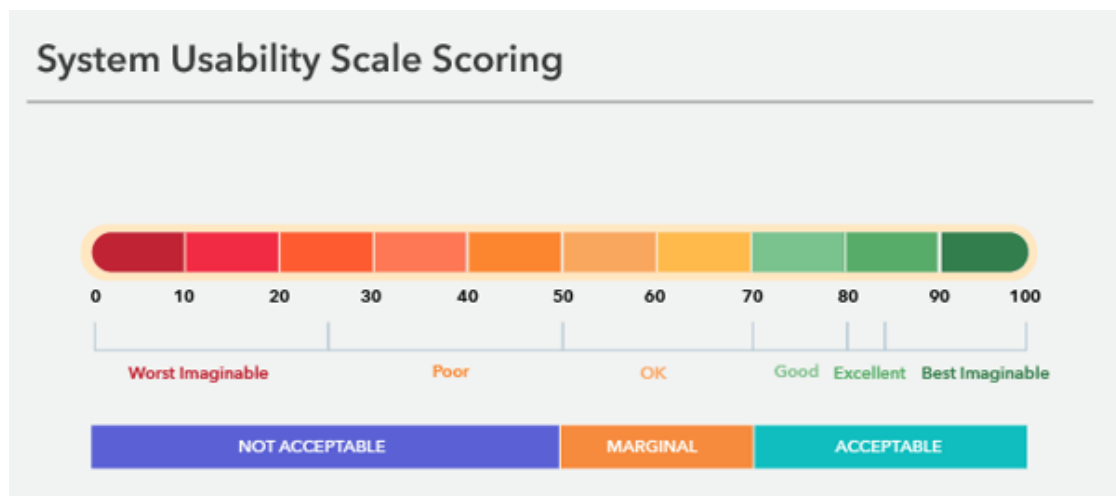


Figure 4.131: System usability score

Table 4.19: System Usability Score by Respondents

| Respondent | Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 | SUS Raw Score (X + Y) | SUS Final Score (X + Y) x 2.5 |
|------------|----|----|----|----|----|----|----|----|----|-----|-----------------------|-------------------------------|
| 1          | 4  | 5  | 5  | 3  | 5  | 2  | 4  | 1  | 5  | 2   | 30                    | 75                            |
| 2          | 4  | 4  | 3  | 3  | 4  | 2  | 3  | 2  | 3  | 4   | 22                    | 55                            |
| 3          | 4  | 2  | 4  | 4  | 4  | 3  | 5  | 1  | 4  | 1   | 30                    | 75                            |
| 4          | 5  | 1  | 4  | 1  | 5  | 1  | 4  | 1  | 5  | 1   | 38                    | 95                            |
| 5          | 3  | 3  | 2  | 3  | 5  | 1  | 3  | 3  | 2  | 5   | 20                    | 50                            |
| 6          | 4  | 4  | 4  | 2  | 4  | 2  | 4  | 2  | 4  | 3   | 27                    | 67.5                          |
| 7          | 3  | 4  | 4  | 3  | 4  | 2  | 4  | 2  | 4  | 2   | 26                    | 65                            |
| 8          | 4  | 4  | 4  | 3  | 4  | 2  | 4  | 2  | 4  | 2   | 27                    | 67.5                          |
| 9          | 3  | 5  | 5  | 2  | 4  | 1  | 4  | 2  | 4  | 3   | 27                    | 67.5                          |
| 10         | 4  | 4  | 4  | 2  | 4  | 1  | 4  | 1  | 5  | 1   | 32                    | 80                            |
| 11         | 3  | 4  | 4  | 2  | 4  | 2  | 4  | 1  | 4  | 2   | 26                    | 65                            |
| 12         | 3  | 4  | 4  | 3  | 4  | 2  | 4  | 1  | 4  | 4   | 25                    | 62.5                          |
|            |    |    |    |    |    |    |    |    |    |     | <b>Average Score</b>  | 74.32                         |

Table 4.19 shows the SUS questionnaire scores from the respondents, which are 75, 55, 75, 95, 50, 67.5, 65, 67.5, 67.5, 80, 65, and 62.5 for each respondent. Based on the responses, the average score for the CANCERVIVE system is 74.32%, indicating that the system is in the good rating category.

## **4.5 Summary**

Finally, this chapter discusses data and system requirements. This chapter's system requirement explanation includes both functional and non-functional system requirements. Following that, the chapter explains the system's data requirements as well as the data cleaning process. This also explains the design process of the CANCEVIVE system, which includes the creation of a storyboard, user interface layout, system flowchart, use case diagram, and data design. All the processes of the development are also explained in this chapter.

## **CHAPTER FIVE**

### **CONCLUSION AND FUTURE RECOMMENDATION**

#### **5.1 Introduction**

This chapter explains the overall outcome and concludes the project, including a summary of the project's findings. Furthermore, this chapter emphasized the project's limitations to identify weaknesses in the system and discuss future work recommendations for potential improvement and enhancement.

#### **5.2 Conclusion**

This project focuses on visualizing cancer survival rates, predicting patient survival rates, and recommending the best treatment based on previous survival cases. This project aims to create a CANCERVIVE dashboard that will help doctors and oncologists analyse previous cancer cases' survival rates based on factors such as cancer type, stage, age, gender, and years since diagnosis. Doctors can more easily explain the expected outcomes to patients using the CANCERVIVE system to visualize the cancer survival rate and trends. In addition, the CANCERVIVE system assists doctors in personalizing a patient's cancer survival rate and recommending suitable treatments. This feature significantly reduces the time required to analyse previous records and solves the problem of limited data access. Doctors no longer need to analyse raw record data that may contain sensitive personal information with CANCERVIVE. The CANCERVIVE dashboard lets doctors and oncologists analyse past cancer cases, visualize survival rates and trends, and provide personalized treatment recommendations. This effective method saves time and overcomes challenges in accessing sensitive personal data. As a result, it improves patient care by providing more insightful information and recommendations.

The first objective of this project is to determine the system requirement for visualizing that need to be used for the dashboard development to ensure it is achieved successfully. The objective is achieved through doing a literature review in chapter two. The literature review was done to understand better the related project topic and



what is needed and to develop the system by finding related articles, journals, and publications. In addition, the datasets used in the project are obtained from the National Cancer Registration and Analysis Service (NCRAS). They are two datasets used in this project: the data on adult cancer survival rate from year 2015 - 2019 and Treatment raw data; both are from the United Kingdom.

The second objective of this project is designing an interactive dashboard for CANCEVIVE: Data visualization of cancer survival and treatment recommendation. This was accomplished by first determining the best methodology to be used in this project. We chose Agile methodology and the OSEMN framework for this project. To begin, before we develop the CANCEVIVE dashboard, we perform a data pipeline, which is a data transformation process, using a jupyter notebook and the Python programming language. During this process, we also explore and model data. This design process is furthered by designing the dashboard user interface, including low-fidelity and high-fidelity prototypes. The dashboard wireframe design is sketched in the low-fidelity prototype, whereas the detailed interface design is included in the high-fidelity prototype. Furthermore, the system's functionality and non-functionality are identified. The goal is also met by designing the flowchart, use case diagram, and sequence diagram to view the system flow on the dashboard.

Lastly, the final objective is developed CANCEVIVE dashboard. Finally, the goal is to create the CANCEVIVE dashboard. In this process, the system is developed by following the system requirements and design discussed in the previous chapter to guide dashboard development. This goal is met when the CANCEVIVE dashboard is successfully developed with predictive features. The selection of software and hardware is critical to ensuring the smooth operation of the CANCEVIVE dashboard. Additionally, system testing, and usability testing are performed to ensure that the system runs successfully and obtain user feedback on how well the system is functioning.

### **5.3 Project Limitation**

Even though the project is successfully developed, there are some limitations in this system identified. The limitation in CANCERVIVE system will be used as an opportunity to make improvements to the system in the future.

#### **A. Data obtain.**

Both datasets that used in this project were obtained from the National Cancer Registration and Analysis Service (NCRAS) in the United Kingdom. The data is not current because the datasets that were obtained were collected between 2015 and 2019. The datasets used in this project are not from Malaysia because obtaining cancer data from Malaysia requires a lengthy and complex process. According to my research, datasets related to healthcare, particularly in Malaysia, are very difficult to obtain and must go through a lengthy process because the Malaysian Minister of Health must approve the data.

#### **B. Data Content**

The cancer survival rate datasets did not include the year in which the data was collected, the system was unable to perform a survival rate analysis based on the year. Next, for cancer treatment data, it only has three types of treatment in dataset: chemotherapy, radiotherapy, and tumour resection, because they are more treatments to treat cancer patients, according to my research. Furthermore, the column labeled "Factor of interest" in the cancer treatment data indicates that the treatment decisions in the dataset are solely based on one factor, potentially ignoring the influence of other important factors. In the context of this project, the system's analysis is limited to five cancer types: bladder, breast, lung, oesophageal, and liver cancers. This limitation is attributed to hardware and data availability constraints, which limit the number of cancer types that can be included in the analysis.

#### **C. CANCERVIVE predictive model is based on machine learning data only.**

Predictive model accuracy levels can vary depending on the machine learning algorithm used. According to the classification report, the accuracy levels reported in this case are 58.27% for the cancer survival rate prediction model

and 14.65% for the cancer treatment recommendation model. It is important to note that the predictions should not be used in place of medical advice from a qualified healthcare professional.

**D. CANCERVIVE system can only support and function in desktop mode.**

The dashboard system can only properly with desktop mode, and the dashboard interface is not supported on the mobile device.

#### **5.4. Recommendation for Future Works**

**A. Data Content**

In the future, we will improve the CANCERVIVE dashboard by obtaining datasets from Malaysia, providing more comprehensive and relevant information for analysis. We also intend to improve the presentation of data in the dashboard, making it easier to understand and interpret. In addition, we will include informative infographics that provide additional information about various types of cancer and treatment options. These modifications are intended to make the CANCERVIVE dashboard more effective and user-friendly. Additionally, for now, this project only includes five types of cancer. In the future, we will add more cancer type and prognosis factors to make the prediction various and more accurate.

**B. Compatibility**

In the future, we will make the CANCERVIVE system available on various devices. It is currently only accessible via desktop computers, but we hope to expand its compatibility to mobile devices and tablets.

**C. Collaboration**

To ensure accuracy and establish trust in the system, collaboration with experts in the field of oncology is crucial. By working closely with doctors, oncologists, and healthcare professionals, valuable insights can be gathered to determine the specific needs and requirements for the dashboard. This collaboration allows for validating the accuracy of predictions and treatment recommendations, ensuring

alignment with current medical practices and guidelines. Furthermore, a collaboration with the Malaysia Minister of Health is sought to facilitate access to data on previous cancer cases, enabling hospitals in Malaysia to utilise the dashboard effectively. Such collaboration strengthens the overall reliability and usefulness of the system in providing comprehensive cancer analysis and treatment recommendations.

#### **D. Model Enhancement**

In order to improve the accuracy of predictive models in the future, we will explore alternative machine learning algorithms and refine the feature selection process. Additionally, we plan to incorporate more extensive and diverse datasets and consider the influence of additional relevant factors in hopes of improving prediction performance for cancer survival rates and treatment recommendations.

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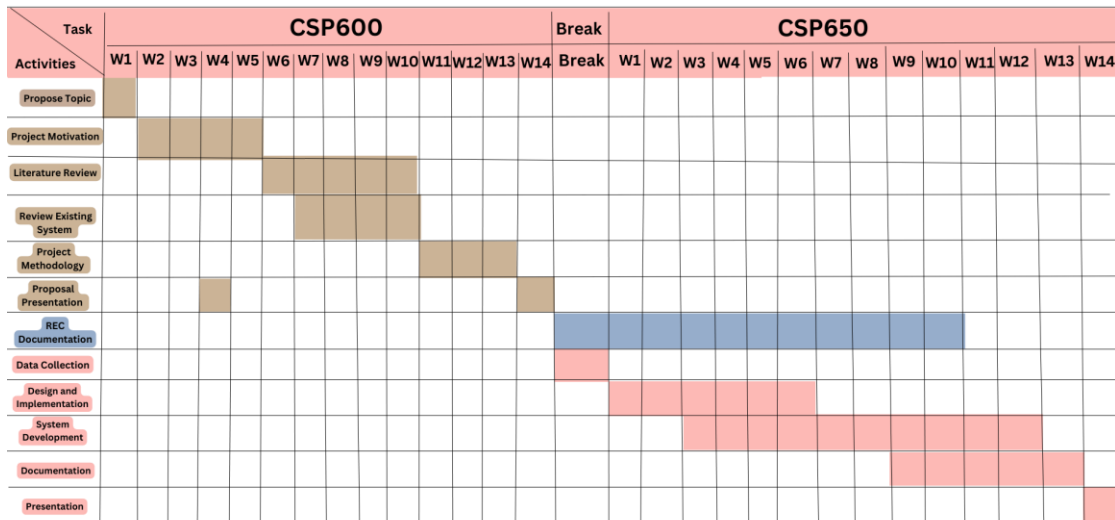
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# APPENDIX

## APPENDIX 1

### Gantt Chart



## APPENDIX 2

### System Review Overview

| System                   |  |  |  |  |  |  |
|--------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Features                 | Pennsylvania Department of Health Dashboard                                         | National Cancer Institute SEER*Explorer Application                                 | United States Cancer Statistics Website                                             | Virginia Department of Health Dashboard                                              | BC Cancer Dashboard                                                                   | WHO Global Cancer observatory dashboard                                               |
| Ease of use              | ✓                                                                                   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                    | ✓                                                                                     | ✓                                                                                     |
| customization / filter   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                    | ✓                                                                                     | ✓                                                                                     |
| Custom Chart Tooltips    | ✓                                                                                   | ✓                                                                                   | ✓                                                                                   | ✓                                                                                    | ✓                                                                                     | ✓                                                                                     |
| Dashboard Widget Linking | ✓                                                                                   | ✓                                                                                   |                                                                                     | ✓                                                                                    |                                                                                       | ✓                                                                                     |
| Drill-Through            |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |
| Dynamic Text Boxes       | ✓                                                                                   |                                                                                     |                                                                                     |                                                                                      |                                                                                       | ✓                                                                                     |

## APPENDIX 3

### Similarity index receipt

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